

Digital Circuits

DPP-1

**For the complete Solutions
follow me**

<https://unacademy.com/@bvreddy>

Use the Code : BVREDDY , to get Maximum Discount

S. No	Topics	Problems Nos	Solution link
1	Boolean algebra	1 -60	Digital DPP-1
2	Boolean algebra	61- 80	Digital DPP-2
3	Boolean algebra	81- 115	Digital DPP-3
	Logic gates	1- 10	
4	Logic gates	11-50	Digital DPP-4
5	Logic gates	51- 100	Digital DPP-5
6	Logic gates	101-121	Digital DPP-6
7	Logic gates	122-142	Digital DPP-7
8	Logic gates	143-163	Digital DPP-8
9	k- map	1- 14	Digital DPP-9
10	k- map	15- 34	Digital DPP-10
11	k- map	35- 39	Digital DPP-11
12	k- map	40-51	Digital DPP-12

Use the Code : BVREDDY , to get Maximum Discount

1. A switching function $f(A,B,C,D) = A'B'CD + A'BC'D + A'BCD + AB'C'D + AB'CD$ can also be written as

(a) $\Sigma m (1,3,5,7,9)$

(b) $\Sigma m (3,5,7,9,11)$

(c) $\Sigma m (3,5,9,11,13)$

(d) $\Sigma m (5,7,9,11,13)$

Use the Code : BVREDDY , to get Maximum Discount

2. The switching function $f(A,B,C,D) = \sum m(5,9,11,14)$ can be written as

(a) $A' B C' D + A B' C' D + A B' C D + A B C D'$

(b) $A' B' C' D + A B' C' D + A' B' C D + A B C D'$

(c) $A' B C' D + A' B C' D' + A B' C D' + A B C D$

(d) None

Use the Code : BVREDDY , to get Maximum Discount

3. The switching function $f(A,B,C) = (A+B'+C)(A'+B'+C)(A+B'+C')$ can also be written as

- | | |
|---------------------------|---------------------------|
| (a) $\Sigma m(2,3,6)$ | (b) $\Sigma m(0,1,4,5,7)$ |
| (c) $\Sigma m(1,2,5,6,7)$ | (d) $\Sigma m(0,2,4,6)$ |

Use the Code : BVREDDY , to get Maximum Discount

4. The other canonical form of $f(A,B,C) = \sum m(0,1,5,7)$ is

(a) $\prod M(2,3,4,6)$

(b) $\prod M(2,4,6,8)$

(c) $\prod M(2,5,6,7)$

(d) $\prod M(1,3,5,7)$

Use the Code : BVREDDY , to get Maximum Discount

5. If a three variable switching function is expressed as the product of maxterms by $f(A,B,C) = \prod M(0,3,5,6)$ then it can also be expressed as the sum of minterms by

(a) $\sum m(0,3,5,6)$

(b) $\prod M(1,2,4,7)$

(c) $\sum m(1,2,4,7)$

(d) $\prod M(1,2,4,7)$

Use the Code : BVREDDY , to get Maximum Discount

6. The dual of Boolean theorem $x(y+z) = xy+xz$ is

(a) $x + yz = xy + xz$

(b) $x(y+z) = (x+y)(x+z)$

(c) $x+yz = (x+y)(x+z)$

(d) None

Use the Code : BVREDDY , to get Maximum Discount

7. Given Boolean theorem $AB + A'C + BC = AB + A'C$ which of the following is true?

(a) $(A+B)(A'+C)(B+C) = (A+B)(A'+C)$

(b) $AB + A' C + BC = AB + BC$

(c) $AB + A' C + BC = (A+B)(A'+C)(B+C)$

(d) $(A+B)(A'+C)(B+C) = AB + A' C$

Use the Code : BVREDDY , to get Maximum Discount

8. $(A' + B' + C)'$ is equal to

(a) $A' B' C'$

(b) ABC

(c) $A+B+C$

(d) $A'+B'+C'$

Use the Code : BVREDDY , to get Maximum Discount

9. $AB + A' C + BC$ is equivalent to

(a) $AB + BC$

(b) $AB + A' C$

(c) $A' C + BC$

(d) AC

Use the Code : BVREDDY , to get Maximum Discount

10. The dual of a Boolean theorem is obtained by
- (a) interchanging all zeros and ones only
 - (b) changing all zeros to ones only
 - (c) changing all ones to zeros only
 - (d) interchanging operators and identity elements

Use the Code : BVREDDY , to get Maximum Discount

11. The number of min-terms after minimizing the following Boolean expression is _____.

$$[D' + AB' + A'C + AC'D + A'C'D]'$$

Use the Code : BVREDDY , to get Maximum Discount

12. The simplified SOP (Sum of Product) form of the Boolean expression.

$$(P + \bar{Q} + \bar{R}).(P + \bar{Q} + R).(P + Q + \bar{R})$$

(a) $(\bar{P}Q + \bar{R})$ (b) $(P + \bar{Q} \bar{R})$

(c) $(\bar{P}Q + R)$ (d) $(PQ + R)$

Use the Code : BVREDDY , to get Maximum Discount

13. The logic expression $F = XY + XZ' + YZ$ is known as

- (a) SSOP form
- (b) SOP form
- (c) POS form
- (d) SPOS form

Use the Code : BVREDDY , to get Maximum Discount

14. The logic expression $F = (x+y+z)(x+y')(y+z')(x+z)$ is known as

- (a) SOP form
- (b) SSOP form
- (c) SPOS form
- (d) POS form

Use the Code : BVREDDY , to get Maximum Discount

15. The logic expression $F = \sum m(0,3,6,7,10,12,15)$ is equivalent to

(a) $F = \prod M(0,3,6,7,10,12,15)$

(b) $F = \prod M(1,2,4,5,8,9,11,13,14)$

(c) $F = \sum m(0,1,5,6,7,12,15)$

(d) $F = \sum m(1,2,4,5,8,9,11,13,14)$

Use the Code : BVREDDY , to get Maximum Discount

16. In Boolean Algebra '1' is called

- (a) Additive identity
- (b) Multiplicative identity
- (c) Either 1 or 2
- (d) None

Use the Code : BVREDDY , to get Maximum Discount

17. In Boolean Algebra '0' is called

- (a) Additive identity
- (b) Multiplicative identity
- (c) Both 1 and 2
- (d) None

Use the Code : BVREDDY , to get Maximum Discount

18. A literal in Boolean algebra is a

- (a) Primed or unprimed Boolean variable
- (b) Primed Boolean variable only
- (c) Unprimed Boolean variable only
- (d) None

Use the Code : BVREDDY , to get Maximum Discount

19. The Boolean expression $x + x' y$ is equal to

(a) x

(b) $x+y$

(c) y

(d) $x + y'$

Use the Code : BVREDDY , to get Maximum Discount

20. The Boolean expression $(x+y)(x+z)$ is equal to

- (a) $x+z$ (b) $x+y$ (c) $x+yz$ (d) $y+xz$

Use the Code : BVREDDY , to get Maximum Discount

21. Identify number of literals in the given Boolean function $F = x'yz + xyz + xy'z$

(a) 5

(b) 4

(c) 3

(d) 6

Use the Code : BVREDDY , to get Maximum Discount

22. A minterm is nothing but

- | | |
|--|---------------------------|
| (a) Standard sum term | (b) Standard product term |
| (c) May be standard sum term or product term | (d) None |

Use the Code : BVREDDY , to get Maximum Discount

23. A maxterm is nothing but a

- | | |
|--|---------------------------|
| (a) Standard sum term | (b) Standard product term |
| (c) May be standard sum term or product term | (d) None |

Use the Code : BVREDDY , to get Maximum Discount

24. The Boolean function expressed in standard sum of products form or standard product of sums form is called

- (a) Canonical form
- (b) Conical form
- (c) Both 1 and 2
- (d) None

Use the Code : BVREDDY , to get Maximum Discount

25. If P, Q, R are Boolean variables, then $(P + \overline{Q})(P.\overline{Q} + P.R) (\overline{P}.\overline{R} + \overline{Q})$ Simplifies to

(A) $P.\overline{Q}$

(B) $P.\overline{R}$

(C) $P.\overline{Q} + R$

(D) $P.\overline{R} + Q$

Use the Code :BVREDDY, to get the Maximum discount

26. The min-term expansion of $f(P, Q, R) = PQ + Q\bar{R} + P\bar{R}$ is _____.

(A) $m_2 + m_4 + m_6 + m_7$

(B) $m_0 + m_1 + m_3 + m_5$

(C) $m_0 + m_1 + m_6 + m_7$

(D) $m_2 + m_3 + m_4 + m_5$

Use the Code :BVREDDY, to get the Maximum discount

27. Boolean algebra is based on

- (a) numbers
- (b) switch logic
- (c) truth
- (d) symbols

Use the Code : BVREDDY , to get Maximum Discount

28. For the identity $AB + \bar{A}C + BC = AB + \bar{A}C$, the dual form is

(a) $(A+B) (\bar{A}+C) (B+C) = (A+B) (\bar{A}+C)$

(b) $(\bar{A}+\bar{B})(A + \bar{C})(\bar{B} + \bar{C}) = (\bar{A}+\bar{B}) (A+\bar{C})$

(c) $(A+B) (\bar{A}+C) (B+C) = (\bar{A}+\bar{B})(A+\bar{C})$

(d) $\bar{A} \bar{B} + A \bar{C} + \bar{B} \bar{C} = \bar{A} \bar{B} + A \bar{C}$

Use the Code : BVREDDY , to get Maximum Discount

29. The minimized form of the Boolean expression $F(A, B, C) = \prod(0, 2, 3)$

(a) $A + \overline{B}C$

(b) $A + B\overline{C}$

(c) $\overline{A}C + B$

(d) $\overline{A}B\overline{C} + \overline{A}B$

Use the Code : BVREDDY , to get Maximum Discount

30. Which one of the following Boolean expressions is correct ?

(a) $\overline{\overline{x + y}} = \bar{x} \bar{y}$

(b) $\overline{\bar{x} + y} = \bar{x} \bar{y}$

(c) $\overline{\bar{x} \bar{y}} = \bar{x} + \bar{y}$

(d) $\overline{\bar{x} + \bar{y}} = \bar{x} \bar{y}$

Use the Code : BVREDDY , to get Maximum Discount

31. A Boolean function can be expressed.

- (a) as sum of maxterms or product of min-terms
- (b) as product of maxterms or sum of min-terms
- (c) partly as product of maxterms and partly as sum of min-terms
- (d) partly as sum of maxterms and partly as product of min-terms

Use the Code : BVREDDY , to get Maximum Discount

32. A function $F(A, B, C)$ defined by three Boolean variables A , B and C when expressed as sum of products is given by:

$$F = \bar{A} \cdot \bar{B} \cdot \bar{C} + \bar{A} \cdot B \cdot \bar{C} + A \cdot \bar{B} \cdot \bar{C}$$

Where, $\bar{A}$, $\bar{B}$, and $\bar{C}$ are the complements of the respective variables. The product of sums (POS) form of the function F is

(a) $F = (A + B + C) \cdot (A + \bar{B} + C) \cdot (\bar{A} + B + C)$

(b) $F(\bar{A} + \bar{B} + \bar{C}) \cdot (\bar{A} + B + \bar{C}) \cdot (A + \bar{B} + \bar{C})$

(c) $F = (A + B + \bar{C}) \cdot (A + \bar{B} + \bar{C}) \cdot (\bar{A} + B + \bar{C}) \cdot (\bar{A} + \bar{B} + C) \cdot (\bar{A} + \bar{B} + \bar{C})$

(d) $F = (\bar{A} + \bar{B} + C) \cdot (\bar{A} + B + C) \cdot (A + \bar{B} + C) \cdot (A + B + \bar{C}) \cdot (A + B + C)$

Use the Code : BVREDDY , to get Maximum Discount

33. The compliment of the Boolean expression $AB.(\overline{B}C+AC)$ is

(a) $(\overline{A} + \overline{B})+(B+\overline{C})+ (\overline{A} +\overline{C})$

(b) $(\overline{A} . \overline{B})+(B\overline{C} + \overline{A}\overline{C})$

(c) $(\overline{A} + \overline{B}) + (B+\overline{C})(\overline{A} + \overline{C})$

(d) $(A+B)(\overline{B} +C)(A+C)$

Use the Code :BVREDDY, to get the Maximum discount

34. The product of sum expression of a Boolean function $F(A, B, C)$ of three variables is given by $F(A, B, C) = (A + B + \bar{C}) \cdot (A + \bar{B} + \bar{C}) \cdot (\bar{A} + B + C) \cdot (\bar{A} + \bar{B} + \bar{C})$. The canonical sum of product expression of $F(A, B, C)$ is given by

- (a) $\bar{A}\bar{B}C + \bar{A}BC + A\bar{B}\bar{C} + ABC$
- (b) $\bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}C + AB\bar{C}$
- (c) $AB\bar{C} + A\bar{B}\bar{C} + \bar{A}BC + \bar{A}\bar{B}\bar{C}$
- (d) $\bar{A}\bar{B}\bar{C} + \bar{A}BC + AB\bar{C} + ABC$

Use the Code :BVREDDY, to get the Maximum discount

35. Which one of the following gives the simplified sum of products expression for the Boolean function $F = m_0 + m_2 + m_3 + m_5$, where m_0, m_2, m_3 and m_5 , are minterms corresponding to the inputs A, B and C with A as the MSB and C as the LSB?

(a) $\bar{A}B + \bar{A}\bar{B}\bar{C} + A\bar{B}C$

(b) $\bar{A}\bar{C} + \bar{A}B + A\bar{B}C$

(c) $\bar{A}\bar{C} + A\bar{B} + A\bar{B}C$

(d) $\bar{A}BC + \bar{A}\bar{C} + A\bar{B}C$

Use the Code :BVREDDY, to get the Maximum discount

36. The Boolean expression $AB + A\bar{C} + BC$ simplifies to

(a) $BC + A\bar{C}$

(b) $AB + A\bar{C} + B$

(c) $AB + A\bar{C}$

(d) $AB + BC$

Use the Code :BVREDDY, to get the Maximum discount

37. The Boolean expression $\overline{(a + \bar{b} + c + \bar{d})} + (b + \bar{c})$ simplifies to

(a) 1

(b) $\overline{a.b}$

(c) $a.b$

(d) 0

Use the Code :BVREDDY, to get the Maximum discount

38. The Boolean expression $XY + (X' + Y')Z$ is equivalent to

(a) $XYZ' + X'Y'Z$

(b) $X'Y'Z' + XYZ$

(c) $(X + Z)(Y + Z)$

(d) $(X' + Z)(Y' + Z)$

Use the Code :BVREDDY, to get the Maximum discount

39. The Boolean expression

$$F(X, Y, Z) = \bar{X}Y\bar{Z} + X\bar{Y}\bar{Z} + XY\bar{Z} + XYZ$$

converted into the canonical product of sum (POS) form is

(a) $(X + Y + Z)(X + Y + \bar{Z})(X + \bar{Y} + \bar{Z})(\bar{X} + Y + \bar{Z})$

(b) $(X + \bar{Y} + Z)(\bar{X} + Y + \bar{Z})(\bar{X} + \bar{Y} + Z)(\bar{X} + \bar{Y} + \bar{Z})$

(c) $(X + Y + Z)(\bar{X} + Y + \bar{Z})(X + \bar{Y} + Z)(\bar{X} + \bar{Y} + \bar{Z})$

(d) $(X + \bar{Y} + \bar{Z})(\bar{X} + Y + Z)(\bar{X} + \bar{Y} + Z)(X + Y + Z)$

Use the Code :BVREDDY, to get the Maximum discount

40. A function of Boolean variables X, Y and Z is expressed in terms of the min-terms as

$$F(X, Y, Z) = \sum(1, 2, 5, 6, 7)$$

Which one of the product of sums given below is equal to the function $F(X, Y, Z)$?

- (a) $(\bar{X} + \bar{Y} + \bar{Z}). (\bar{X} + Y + Z). (X + \bar{Y} + \bar{Z})$
- (b) $(X + Y + Z). (X + \bar{Y} + \bar{Z}). (\bar{X} + Y + Z)$
- (c) $(\bar{X} + \bar{Y} + Z). (\bar{X} + Y + \bar{Z}). (X + \bar{Y} + Z). (X + Y + \bar{Z}). (X + Y + Z)$
- (d) $(X + Y + \bar{Z}). (\bar{X} + Y + Z). (\bar{X} + Y + \bar{Z}). (\bar{X} + \bar{Y} + Z). (\bar{X} + \bar{Y} + \bar{Z})$

Use the Code :BVREDDY, to get the Maximum discount

41. Consider the following Sum of Products expression, F.

$$F = ABC + \overline{A}\overline{B}C + A\overline{B}C + \overline{A}BC + \overline{A}\overline{B}\overline{C}$$

The equivalent Product of Sums expression is

(a) $F = (A + \overline{B} + C)(\overline{A} + B + C)(\overline{A} + \overline{B} + C)$

(b) $F = (A + B + \overline{C})(A + B + C)(\overline{A} + \overline{B} + \overline{C})$

(c) $F = (\overline{A} + B + \overline{C})(A + \overline{B} + \overline{C})(A + \overline{B} + C)$

(d) $F = (\overline{A} + \overline{B} + C)(A + B + \overline{C})(A + B + C)$

Use the Code :BVREDDY, to get the Maximum discount

42. The Boolean expression $(X + Y)(X + \bar{Y}) + \overline{X\bar{Y}} + \bar{X}$ simplifies to

(a) X

(b) Y

(c) XY

(d) $X + Y$

Use the Code :BVREDDY, to get the Maximum discount

43. If $X = 1$ in the logic equation

$$[X + Z\{\bar{Y} + (\bar{Z} + X\bar{Y})\}]\{\bar{X} + \bar{Z}(X + Y)\} = 1, \text{ then}$$

(a) $Y = Z$

(b) $Y = \bar{Z}$

(c) $Z = 1$

(d) $Z = 0$

Use the Code :BVREDDY, to get the Maximum discount

44. The Boolean expression $AC + B\bar{C}$ is equivalent to

(a) $\bar{A}C + B\bar{C} + AC$

(b) $\bar{B}C + AC + B\bar{C} + \bar{A}C\bar{B}$

(c) $AC + B\bar{C} + \bar{B}C + ABC$

(d) $ABC + \bar{A}B\bar{C} + A\bar{B}C + AB\bar{C}$

Use the Code :BVREDDY, to get the Maximum discount

45. The simplified form of the Boolean expression $Y = (\bar{A}.BC + D)(\bar{A}.D + \bar{B}.\bar{C})$ can be written as

(a) $\bar{A}.D + \bar{B}.\bar{C}.D$

(b) $A.D + B.\bar{C}.D$

(c) $(\bar{A} + D)(\bar{B}.C + \bar{D})$

(d) $A.\bar{D} + B.C.\bar{D}$

Use the Code :BVREDDY, to get the Maximum discount

46. If the functions W, X, Y and Z are as follows.

$$W = R + \bar{P}Q + \bar{R}S$$

$$X = PQ\bar{R}\bar{S} + \bar{P}\bar{Q}\bar{R}\bar{S} + P\bar{Q}\bar{R}\bar{S}$$

$$Y = RS + \overline{PR + P\bar{Q} + \bar{P}\bar{Q}}$$

$$Z = R + S + \overline{PQ + \bar{P}\bar{Q}\bar{R} + P\bar{Q}\bar{S}} \text{ then}$$

(a) $W = Z, X = \bar{Z}$ (b) $W = Z, X = Y$

(c) $W = Y$ (d) $W = Y = \bar{Z}$

Use the Code :BVREDDY, to get the Maximum discount

47. The Boolean expression

$\bar{X}Y\bar{Z} + \bar{X}\bar{Y}Z + XY\bar{Z} + X\bar{Y}Z + XYZ$ Can be simplified to

(a) $X\bar{Z} + \bar{X}Z + YZ$

(b) $XY + \bar{Y}Z + Y\bar{Z}$

(c) $\bar{X}Y + YZ + XZ$

(d) $\bar{X}Y + Y\bar{Z} + \bar{X}Z$

Use the Code :BVREDDY, to get the Maximum discount

48. The logic expression $y = A + \bar{A}B$ is equivalent to

(a) $y = AB$

(b) $y = \bar{A}B$

(c) $y = \bar{A} + B$

(d) $y = A + B$

Use the Code :BVREDDY, to get the Maximum discount

49. The simplest form of the Boolean expression $AB\bar{C}\bar{D} + AB\bar{C}D + ABC\bar{D} + ABCD$ is

(a) AD

(b) BC

(c) $A\bar{B}$

(d) AB

Use the Code :BVREDDY, to get the Maximum discount

50. The expression $A + \bar{A}B$ is equivalent to

(a) $A + B$

(b) $AB + A$

(c) $A + \bar{B}$

(d) AB

Use the Code :BVREDDY, to get the Maximum discount

51. From the truth table given below, choose the correct logic expression for Q

A	0	0	0	0	1	1	1	1
B	0	0	1	1	0	0	1	1
C	0	1	0	1	0	1	0	1
Q	0	0	0	1	0	1	1	1

(a) $AB + BC + CA$

(b) $A + B + C$

(c) $A\bar{B} + B\bar{C} + C\bar{A}$

(d) $\bar{A}\bar{B} + \bar{B}\bar{C} + \bar{C}\bar{A}$

Use the Code :BVREDDY, to get the Maximum discount

52. The minimal sum of products form $f = A\bar{B}CD + \bar{A}BC + \bar{A}\bar{B}C + BCD$ is

(a) $\bar{A}\bar{C} + \bar{B}\bar{D}$

(b) $\bar{A}C + CD$

(c) $AC + \bar{B}\bar{D}$

(d) $A\bar{B} + C\bar{D}$

Use the Code :BVREDDY, to get the Maximum discount

53. The term $AB + AC + B\bar{C}$ reduced to.

(a) $AB + CA$

(b) $AC + BC$

(c) $AC + B\bar{C}$

(d) $AB + \bar{B}\bar{C}$

Use the Code :BVREDDY, to get the Maximum discount

54. If X, Y and Z are Boolean variables, then the expression $X(X+XY) Z(X+Y+Z)$ is equal to

(a) $X + \bar{X} Y$

(b) $X+Y+Z$

(c) XYZ

(d) XZ

Use the Code :BVREDDY, to get the Maximum discount

55. What is dual of $A+[B+(AC)] + D$

(a) $A+[B(A+C)] + D$

(b) $A[B+AC] D$

(c) $A+[B(A+C)] D$

(d) $A[B(A+C)] D$

Use the Code :BVREDDY, to get the Maximum discount

56. The product-of-sum expressions for given truth table is:

X	Y	Z
0	0	1
0	1	0
1	0	1
1	1	0

(a) $(\bar{X} + \bar{Y})(X+Y)$

(b) $(X + \bar{Y})(\bar{X} + \bar{Y})$

(c) $(X+Y)(\bar{X} + \bar{Y})$

(d) None of the above

Use the Code :BVREDDY, to get the Maximum discount

57. What is dual of $X + \bar{X}Y = X + Y$

(a) $X + Y = XY$

(b) $\bar{X} + XY = XY$

(c) $X(\bar{X} + Y) = XY$

(d) $X(\bar{X} + Y) + X + Y$

Use the Code :BVREDDY, to get the Maximum discount

58. What is Boolean expression for a gating network that will have output 0 only, when
 $X=1, Y=1, Z=1$; $X=0, Y=0, Z=0$; $X=1, Y=0, Z=0$?

(a) $XYZ + \bar{X}\bar{Y}Z + X\bar{Y}Z$

(b) $XYZ (\bar{X} + \bar{Y} + Z) (X + \bar{Y} + Z)$

(c) $(\bar{X} + \bar{Y} + \bar{Z}) (X + Y + Z) (\bar{X} + Y + Z)$

(d) $\bar{X}\bar{Y}Z + XYZ + \bar{X}YZ$

Use the Code :BVREDDY, to get the Maximum discount

59. which one of the following is the dual form of the Boolean identity?

$$\overline{A} B + \overline{A} C = (A+C) (\overline{A} + B)?$$

(a) $AB + \overline{A} C = AC + \overline{A} B$

(b) $(A+B) + (A+C) = (A+C) (A+B)$

(c) $(\overline{A} + B) (\overline{A} + C) = AC + \overline{A} B$

(d) $AB + \overline{A} C = AB + \overline{A} C + BC$

Use the Code :BVREDDY, to get the Maximum discount

60. In a digital system there are three inputs A, B and C. The output should be high when at least two inputs are high, the Boolean expression for the output is.

(a) $AB+BC+AC$

(b) $ABC+AB\bar{C}+\bar{A}CB+A\bar{B}C$

(c) $AB\bar{C}+A\bar{B}C+\bar{A}BC$

(d) $A\bar{B}+B\bar{C}+\bar{A}C$

Use the Code :BVREDDY, to get the Maximum discount

61. The Boolean theorem:

$AB + \bar{A}C + BC = AB + \bar{A}C$ corresponds to

(a) $(A+B).(\bar{A} + C).(B+C) = (A+B).(\bar{A} + C)$

(b) $AB + \bar{A}C + BC = AB + BC$

(c) $AB + \bar{A}C + BC = AB + BC$

(d) $(A+B).(\bar{A} + C).(B+C) = (AB).(\bar{A}C)$

Use the Code :BVREDDY, to get the Maximum discount

62. Which one of the following is equivalent to Boolean expression.

$$Y = \overline{A} \overline{B} + \overline{B} \overline{C} + \overline{C} \overline{A} \text{ is?}$$

(a) $\overline{AB + BC + CA}$

(b) $(\overline{A} + \overline{B}) (B + \overline{C}) (\overline{A} + \overline{C})$

(c) $\overline{(A + B)(B + C)(C + A)}$

(d) $\overline{(A + B)} \overline{(B + C)} \overline{(C + A)}$

Use the Code :BVREDDY, to get the Maximum discount

63. Given Boolean theorem, $AB + \bar{A}C + BC = AB + \bar{A}C$. Which one of the following identities is true?

(a) $(A+B) (\bar{A}+C) (B+C) = (A+B) (\bar{A}+C)$

(b) $(AB + \bar{A}C + BC) = AB + BC$

(c) $AB + \bar{A}C + BC = (A+B) (\bar{A}+C) (B+C)$

(d) $(A+B) (\bar{A} + C) (B+C) = AB + \bar{A}C$

Use the Code :BVREDDY, to get the Maximum discount

64. Consider the Boolean expression $X = ABCD + A\bar{B}CD + \bar{A}C\bar{B}D + \bar{A}BCD$. The simplified form of X is

(a) $\bar{C} + \bar{D}$

(b) BC

(c) CD

(d) BC

Use the Code :BVREDDY, to get the Maximum discount

65. The Boolean expression $(\bar{A} + B)(A + \bar{C})(\bar{B} + \bar{C})$ simplifies to

(a) $(A + B)\bar{C}$

(b) $(A + \bar{B})\bar{C}$

(c) $(\bar{A} + B)\bar{C}$

(d) $(\bar{A} + \bar{B})\bar{C}$

Use the Code :BVREDDY, to get the Maximum discount

66. A, B and C are three Boolean variables. which one of the following Boolean expressions cannot be minimized any further?

(a) $Z = A\bar{B}\bar{C} + AB\bar{C} + ABC + \bar{A}\bar{B}\bar{C}$

(b) $Z = A\bar{B}C + AB\bar{C} + ABC + \bar{A}\bar{B}\bar{C}$

(c) $Z = A\bar{B}\bar{C} + \bar{A}\bar{B}C + A.BC + \bar{A}B\bar{C}$

(d) $Z = \bar{A}B\bar{C} + AB\bar{C} + ABC + \bar{A}\bar{B}C$

Use the Code :BVREDDY, to get the Maximum discount

67. The Boolean function $(x + y) (\bar{x} + z) (y + z)$ is equal to which one of the following expressions?

(a) $(x + y) (y + z)$

(b) $(\bar{x} + z) (y + z)$

(c) $(x + y) (\bar{x} + z)$

(d) $(x + y) (y + \bar{z})$

Use the Code :BVREDDY, to get the Maximum discount

68. $AB + \bar{A}C = (A + C)(\bar{A} + B)$ Which one of the following is the dual form of the Boolean identity given above?

(a) $AB + \bar{A}C = AC + \bar{A}B$

(b) $(A + B)(\bar{A} + C) = (A + C)(\bar{A} + B)$

(c) $(A + B)(\bar{A} + C) = AC + \bar{A}B$

(d) $AB + \bar{A}C = AB + \bar{A}C + BC$

Use the Code :BVREDDY, to get the Maximum discount

69. What does the Boolean expression

$AD + ABCD + ACD + \overline{A}B + \overline{A}\overline{B}$ on minimization result into

- (a) $A + D$
- (b) $AD + \overline{A}$
- (c) AD
- (d) $\overline{A} + D$

Use the Code :BVREDDY, to get the Maximum discount

70. If A and B are Boolean variables, then what is $(A + B).(A + \overline{B})$ equal to?

(a) B

(b) A

(c) $A + B$

(d) AB

Use the Code :BVREDDY, to get the Maximum discount

71. The Boolean functions can be expressed in canonical SOP (Sum Of Products) and POS (Product Of Sums) form. For the functions, $Y = A + \overline{B} C$, which are such two forms

(a) $Y = \sum (1,2,6,7)$ and $Y = \prod (0,2,4)$

(b) $Y = \sum (1,4,5,6,7)$ and $Y = \prod (0,2,3)$

(c) $Y = \sum (1,2,5,6,7)$ and $Y = \prod (0,1,3)$

(d) $Y = \sum (1,2,4,5,6,7)$ and $Y = \prod (0,2,3,4)$

Use the Code :BVREDDY, to get the Maximum discount

72. The Boolean function $A + BC$ is a reduced form of which one of the following.

(a) $AB + BC$

(b) $\bar{A}B + A\bar{B}C$

(c) $(A + B).(A + C)$

(d) None of the above

Use the Code :BVREDDY, to get the Maximum discount

73. Which one of the following statements is not correct.

(a) $X + \bar{X}Y = X$

(b) $X(\bar{X} + Y) = XY$

(c) $XY + X\bar{Y} = X$

(d) $ZX + Z\bar{X}Y = ZX + ZY$

Use the Code :BVREDDY, to get the Maximum discount

74. Which of the following Boolean Algebra rules is correct?

(a) $A \cdot \bar{A} = 1$

(b) $A + AB = A + B$

(c) $A + \bar{A} B = A + B$

(d) $A (A + B) = B$

Use the Code :BVREDDY, to get the Maximum discount

75. What are the ultimate purposes of Minimizing logic expressions?

1. To get a small size expression.
2. To reduce the number of variables in the given expression.
3. To implement the function of the logic expression with least hardware.
4. To reduce the expression for making it feasible for hardware implementation.

Select the correct answer from the codes given below:

- | | |
|------------|-------------|
| (a) 1 only | (b) 2 and 3 |
| (c) 3 only | (d) 3 and 4 |

Use the Code :BVREDDY, to get the Maximum discount

76. The Boolean expression $\overline{A + \overline{B} + C} + \overline{\overline{A} + \overline{B} + C} + \overline{A + \overline{B} + \overline{C}} + ABC$ reduces to:

(a) A

(b) B

(c) C

(d) $A + B + C$

Use the Code :BVREDDY, to get the Maximum discount

77. The standard SOP expression for Boolean expression $A \bar{B} + AC + BC$ is:

(a) $A\bar{B}C + \bar{A}BC + A\bar{B}\bar{C} + ABC$ (b) $AB\bar{C} + \bar{A}BC + ABC$

(c) $A\bar{B}C + AB\bar{C} + ABC$ (d) $\bar{A}\bar{B}C + AB\bar{C} + A\bar{B}\bar{C}$

Use the Code :BVREDDY, to get the Maximum discount

78. The complement of the expression $Y = ABC + AB\bar{C} + \bar{A}\bar{B}C + \bar{A}BC$ is:

(a) $(A + \bar{B})(A + \bar{C})$ (b) $(\bar{A} + B)(A + C)$

(c) $(A + \bar{B})(\bar{A} + C)$ (d) $(\bar{A} + \bar{B})(A + \bar{C})$

Use the Code :BVREDDY, to get the Maximum discount

79. The logic function $f = \overline{(x \cdot \bar{y} + \bar{x} \cdot y)}$ is the same as

(a) $f = (x + y) (\bar{x} + \bar{y})$

(b) $f = \overline{(\bar{x} + \bar{y})(x + y)}$

(c) $f = \overline{(x \cdot y)} (\bar{x} \cdot \bar{y})$

(d) None of the above

Use the Code :BVREDDY, to get the Maximum discount

80. If the Boolean expression $\overline{P}Q + QR + PR$ is Minimized, the expression becomes:

(a) $\overline{P}Q + QR$

(b) $\overline{P}Q + PR$

(c) $QR + PR$

(d) $\overline{P}Q + QR + PR$

Use the Code :BVREDDY, to get the Maximum discount

81. The Boolean equation $X = [(A + \bar{B})(B + C)] B$ can be simplified to

(a) $X = \bar{A} B$

(b) $X = A \bar{B}$

(c) $X = A B$

(d) $X = \bar{A} \bar{B}$

Use the Code :BVREDDY, to get the Maximum discount

82. The correct expression is

(a) $\bar{A} B + A \bar{B} = \overline{AB} (A + B)$

(b) $\bar{A} B + \bar{A} B = AB (\bar{A} + \bar{B})$

(c) $\bar{A} B + A \bar{B} = AB (\bar{A} + \bar{B})$

(d) $\bar{A} B + \bar{A} B = \overline{AB} (A + B)$

Use the Code :BVREDDY, to get the Maximum discount

83. Simplified form of the logic expression $(A + \overline{B} + C)(A + \overline{B} + \overline{C})(A + B + C)$ is

(a) $\overline{A}B + \overline{C}$

(b) $A + \overline{B}C$

(c) A

(d) $AB + \overline{C}$

Use the Code :BVREDDY, to get the Maximum discount

84. Logic function $(\bar{A} + B)(A + B)$ can be reduced to:

(a) B

(b) $\bar{B}$

(c) A

(d) $\bar{A}$

Use the Code :BVREDDY, to get the Maximum discount

85. Logic function $A \bar{B} D + A \bar{B} \bar{D}$ can be reduced to:

(a) $\bar{A} \bar{B}$

(b) $A \bar{B}$

(c) $\bar{B} \bar{D}$

(d) $A \bar{D}$

Use the Code :BVREDDY, to get the Maximum discount

86. The logic function $f(A, B, C, D) = (\bar{A} + BC)(B + CD)$ can be expressed to:

(a) $\bar{A}B + BC + \bar{A}CD + BCD$

(b) $AB + A\bar{B} + A\bar{C}D + BCD$

(c) $AB + \bar{A}\bar{B} + \bar{A}CD + B\bar{C}D$

(d) $A\bar{B} + \bar{A}B + \bar{A}CD + BCD$

Use the Code :BVREDDY, to get the Maximum discount

87. The logic function $(A + B)$ can be expressed in terms of min terms as:

(a) $A \bar{B} + B \bar{A}$

(b) $\bar{A}B + \bar{B}A + AB$

(c) $\bar{A} \bar{B} + \bar{A} B$

(d) $AB + \bar{B}A$

Use the Code :BVREDDY, to get the Maximum discount

88. The min terms for $AB + ACD$ are

(a) $\bar{A}\bar{B}CD + AB\bar{C}\bar{D} + A\bar{B}C\bar{D} + A\bar{B}C\bar{D} + \bar{A}BCD$

(b) $AB\bar{C}\bar{D} + AB\bar{C}\bar{D} + ABC\bar{D} + ABCD + A\bar{B}CD$

(c) $A\bar{B}CD + AB\bar{C}\bar{D} + ABC\bar{D} + \bar{A}BCD + A\bar{B}C\bar{D}$

(d) $AB\bar{C}\bar{D} + A\bar{B}CD + \bar{A}BCD + ABC\bar{D} + \bar{A}\bar{B}CD$

Use the Code :BVREDDY, to get the Maximum discount

89. On simplification of expression $Y = \overline{(A \cdot B + \bar{C})}(\bar{A} + \bar{B} + C)$ using Boolean algebra, the solution is

(a) $(A \cdot B + C)(A + B \cdot C)$

(b) $(\bar{A} + \bar{B} + \bar{C})(A + B + C)$

(c) $(A \cdot B + \bar{C})(A \cdot C + \bar{B})$

(d) $(B \cdot C + \bar{A})(A \cdot B + \bar{C})$

Use the Code :BVREDDY, to get the Maximum discount

90. The function $Y = A + \overline{B} \cdot C$ in canonical sum of product form is

(a) $Y = \sum m(1,3,5,6,7)$

(b) $Y = \sum m(1,4,5,6,7)$

(c) $Y = \sum m(2,3,5,6)$

(d) $Y = \sum m(2,3,5,7)$

Use the Code :BVREDDY, to get the Maximum discount

91. The simplified form of the Boolean expression $AB + A(B + C) + B(B + C)$ is given by

(a) $AB + AC$

(b) $B + AC$

(c) $BC + AC$

(d) $AB + C$

Use the Code :BVREDDY, to get the Maximum discount

92. Product of Max terms representation for the Boolean function $F = \overline{B}D + \overline{A}D + BD$ is

(a) $\prod M(1,3,5,7)$

(b) $\prod M(0,2,4,6)$

(c) $\prod M(0,1,2,3)$

(d) $\prod M(4,5,6,7)$

Use the Code :BVREDDY, to get the Maximum discount

93. Simplified form of the Boolean expression $Y = \overline{(A \cdot B + \bar{C})}(\bar{A} + \bar{B} + C)$ is

(a) $\bar{A}\bar{C} + A\bar{C} + \bar{B}\bar{C} + \bar{B}C$

(b) $(\bar{A} + \bar{B} + \bar{C})(A + B + C)$

(c) $(\bar{A} + \bar{B})(A + \bar{C})$

(d) $A(B + C)$

Use the Code :BVREDDY, to get the Maximum discount

94. What is the function $Y = A + \overline{B}C$ in Product-Of-Sums (POS) form?

(a) $M_6 M_5 M_4 M_3$

(b) $M_3 M_2 M_1 M_0$

(c) $M_0 M_2 M_3$

(d) $M_4 M_3 M_2 M_1$

Use the Code :BVREDDY, to get the Maximum discount

95. In the negative logic system,

- (a) The more negative of two logic levels represent a logic '1' state
- (b) The more negative of two logic levels represent a logic '0' state
- (c) All input and output voltage levels are negative
- (d) The output is always complement of the intended logic function.

Use the Code :BVREDDY, to get the Maximum discount

96. What is the Boolean expression for the truth table shown below?

A	0	0	0	0	1	1	1	1
B	0	0	1	1	0	0	1	1
C	0	1	0	1	0	1	0	1
f	0	0	0	1	0	0	1	0

(a) $B (A + C) (\bar{A} + \bar{C})$

(b) $B (A + \bar{C}) (\bar{A} + C)$

(c) $\bar{B} (A + C) (\bar{A} + C)$

(d) $\bar{B} (A + C) (\bar{A} + \bar{C})$

Use the Code :BVREDDY, to get the Maximum discount

97. A 3-variable truth table has a high output for the inputs: 010, 011 and 110. The Boolean expression for the Sum of Products (SOP) can be written as:

(a) $\overline{A}B + B\overline{C}$

(b) $A\overline{B} + \overline{B}C$

(c) $\overline{A}\overline{B} + BC$

(d) $AB + \overline{B}C$

Use the Code :BVREDDY, to get the Maximum discount

98. Four logical expressions are given below:

1. $\bar{A} . \bar{B} . \bar{C} . \bar{D} . \bar{E} . \bar{F} . \bar{G} . \bar{H}$

2. $\overline{AB} . \overline{CD} . \overline{EF} . \overline{GH}$

3. $\bar{A} + \bar{B} + \bar{C} + \bar{D} + \bar{E} + \bar{F} + \bar{G} + \bar{H}$

4. $(\bar{A} + \bar{B}) (\bar{C} + \bar{D}) (\bar{E} + \bar{F}) (\bar{G} + \bar{H})$

Two of these expressions are equal.

(a) 1 and 2

(b) 3 and 4

(c) 1 and 3

(d) 2 and 4

Use the Code :BVREDDY, to get the Maximum discount

99. The expression $(X+Y) (X+\bar{Y}) (\bar{X}+Y)$ is equivalent to

(a) $\bar{X} \bar{Y}$

(b) $\bar{X} Y$

(c) $X \bar{Y}$

(d) XY

Use the Code :BVREDDY, to get the Maximum discount

100. In Boolean algebra if $F = (A+B)(\bar{A}+C)$ then

(a) $F = AB + \bar{A}C$

(b) $F = AB + \bar{A}\bar{B}$

(c) $F = AC + \bar{A}B$

(d) $F = A\bar{A} + \bar{A}B$

Use the Code :BVREDDY, to get the Maximum discount

101. The simplified form of a logic function $Y = \overline{(\overline{A}\overline{B}) (\overline{\overline{A}} \overline{\overline{B}})}$

(a) $A+B$

(b) AB

(c) $\overline{A} + \overline{B}$

(d) $\overline{A} B + A \overline{B}$

Use the Code :BVREDDY, to get the Maximum discount

102. The reduced form of the Boolean expression $A[B+C(\overline{AB+AC})]$ is.

(a) $\overline{A} B$

(b) $A\overline{B}$

(c) AB

(d) $AB+B\overline{C}$

Use the Code :BVREDDY, to get the Maximum discount

103. Which of the following expression is not correct?

(a) $X + \bar{X}Y = X$

(b) $X.(\bar{X} + Y) = XY$

(c) $X + X\bar{Y} = X$

(d) $ZX + Z\bar{X}Y = ZX + ZY$

Use the Code :BVREDDY, to get the Maximum discount

104. The Boolean expression $\bar{Y}\bar{Z} + \bar{X}\bar{Z} + \bar{X}\bar{Y}$ is logically equal to

(a) $YZ + \bar{X}$

(b) $YZX + \bar{X}\bar{Y}\bar{Z}$

(c) $YZ + XZ + XY$

(d) $X\bar{Y}\bar{Z} + \bar{X}\bar{Y}\bar{Z} + \bar{X}Y\bar{Z} + \bar{X}\bar{Y}Z$

Use the Code :BVREDDY, to get the Maximum discount

105. The function $F = AB\bar{C} + ABC + \bar{A}BC + \bar{A}\bar{B}\bar{C}$ can be reduced to which one of the following

(a) $F = A$

(b) $F = AB$

(c) $F = ABC$

(d) $F = B$

Use the Code :BVREDDY, to get the Maximum discount

106. What is the simplified form of the Boolean expression $T = (X + Y)(X + \bar{Y})(\bar{X} + Y)$

(a) $\bar{X} \bar{Y}$

(b) $\bar{X} Y$

(c) XY

(d) $X\bar{Y}$

Use the Code :BVREDDY, to get the Maximum discount

107. Match List-I (expression 1) with List-II (expression 2) and select the correct answer using the codes given below the Lists:

List-I

A. $ABC + AB\bar{C} + A\bar{B}C$

B. $\bar{A}B\bar{C} + AB\bar{C} + B\bar{C}$

C. $\bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$

D. $\bar{A}\bar{B} + \bar{A}B + ABC$

List-II

1. $\bar{A} + BC$

2. $A(B+C)$

3. $B\bar{C}$

4. $AB+BC+AC$

Codes:

A	B	C	D
(a)2	1	4	3
(b)4	3	2	1
(c)2	3	4	1
(d)4	1	2	3

Use the Code :BVREDDY, to get the Maximum discount

108. The Boolean expression $A.B + \bar{A}\bar{B}$ is logically equivalent to which of the following?

(a) $(A + \bar{B}).(\bar{A} + B)$

(b) $(\bar{A} + \bar{B}).(A + B)$

(c) $\overline{(A + \bar{B}).(\bar{A} + B)}$

(d) $\overline{\bar{A} + B} . \overline{\bar{A} + \bar{B}}$

Use the Code :BVREDDY, to get the Maximum discount

109. The Boolean expression $F = \overline{A + \overline{B} + C} + \overline{\overline{A} + \overline{B} + C} + \overline{A + \overline{B} + \overline{C}} + ABC$ reduces to

(a) A

(b) B

(c) C

(d) A + B + C

Use the Code :BVREDDY, to get the Maximum discount

110. The logic function $A+BC$ is the simplified form of which of the following?

- (a) $AB + BC$ (b) $\overline{A}B + A\overline{B}C$
(c) $\overline{ABC}$ (d) $(A+B)(A+C)$

Use the Code :BVREDDY, to get the Maximum discount

111. The logical expression, $A B \bar{C} + A \bar{B} C + A \bar{B} \bar{C}$ is equivalent to

(a) $\bar{A} (B+C)$

(b) $\bar{A} + \bar{B} + \bar{C}$

(c) $\bar{A} \bar{B} \bar{C}$

(d) $A (\bar{C} + \bar{B})$

Use the Code :BVREDDY, to get the Maximum discount

112. Let $*$ be defined as $x*y = \overline{x} + y$. Let $z = x*y$. value of $z*x$ is

(A) $x + y$

(B) x

(C) 0

(D) 1

Use the Code :BVREDDY, to get the Maximum discount

113. Let $f(A, B) = A' + B$. Simplified expression for function $f(f(x + y, y), z)$ is

(A) $x' + z$

(B) xyz

(C) $xy' + z$

(D) None of these

Use the Code : BVREDDY , to get Maximum Discount

114. The Boolean function $x'y' + xy + x'y$ is equivalent to

(A) $x' + y'$

(B) $x + y$

(C) $x + y'$

(D) $x' + y$

Use the Code :BVREDDY, to get the Maximum discount

115. Given f_1, f_2, f_3 and $f = f_1 f_2 + f_3$ in canonical sum of products form (in decimal) for the

circuit $f_1 = \sum m(4,5,6,7,8);$

$$f_2 = \sum m(1,6,15)$$

$$f = \sum m(1,6,8,15)$$

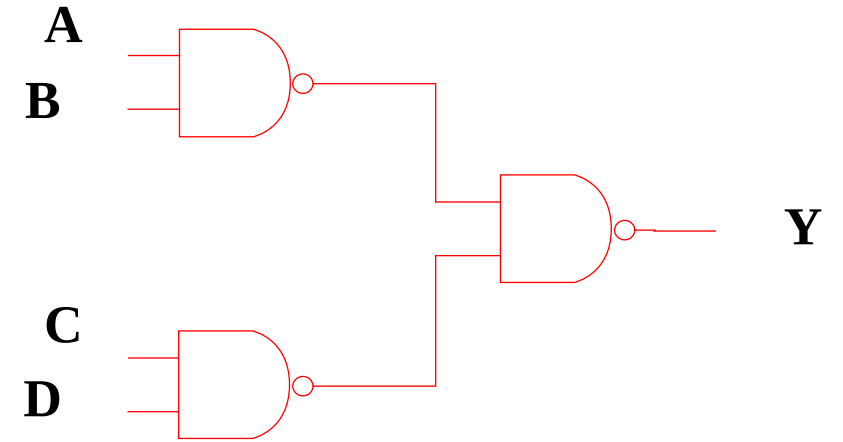
Then f_3 is

Use the Code : BVREDDY , to get Maximum Discount

Logic Gates

1. In the logic circuit shown in the figure, Y is given by.

- (a) $Y=ABCD$ (b) $Y=(A+B)(C+D)$
(c) $Y=A+B+C+D$ (d) $Y=AB+CD$



Use the Code :BVREDDY, to get the Maximum discount

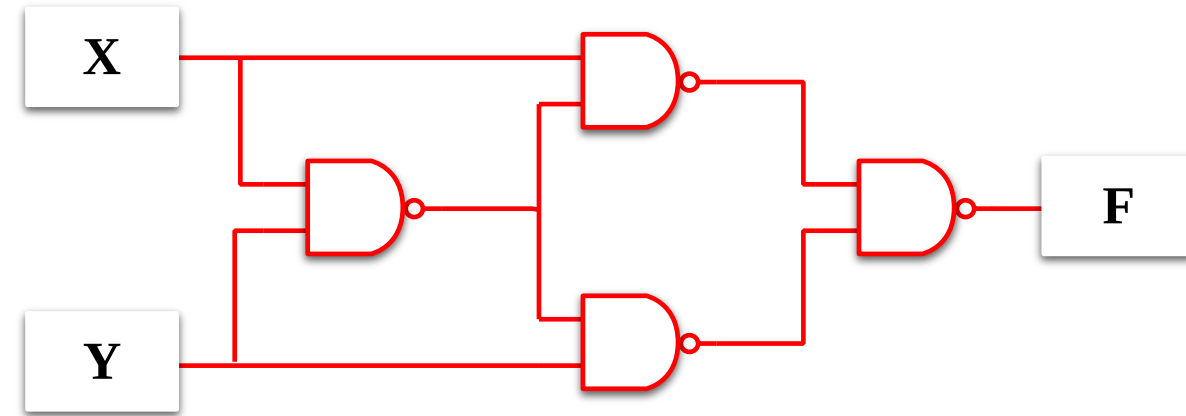
2. The Boolean function $F(X, Y)$ realized by the given circuit is:

(a) $\bar{X}Y + X\bar{Y}$

(b) $\bar{X}\bar{Y} + XY$

(c) $X + Y$

(d) $\bar{X} \cdot \bar{Y}$



Use the Code :BVREDDY, to get the Maximum discount

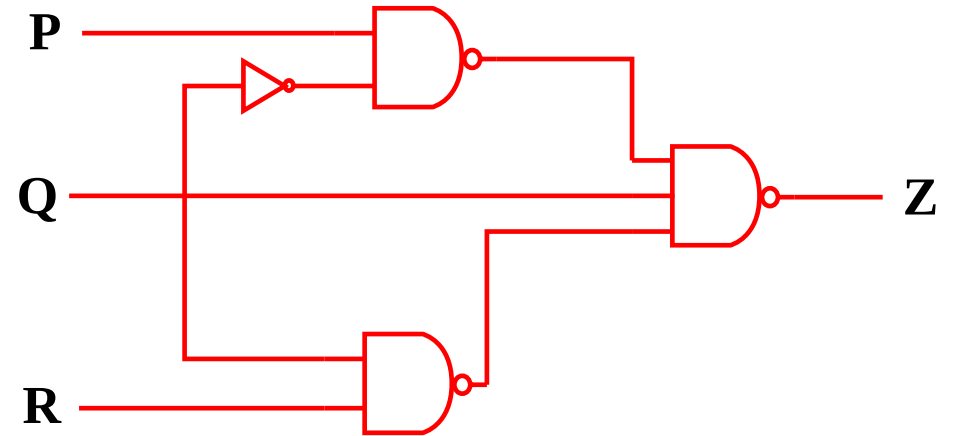
3. For a 3-input logic circuit shown below, the output Z can be expressed as

(a) $Q + \bar{R}$

(b) $P\bar{Q} + R$

(c) $\bar{Q} + R$

(d) $P + \bar{Q} + R$



Use the Code :BVREDDY, to get the Maximum discount

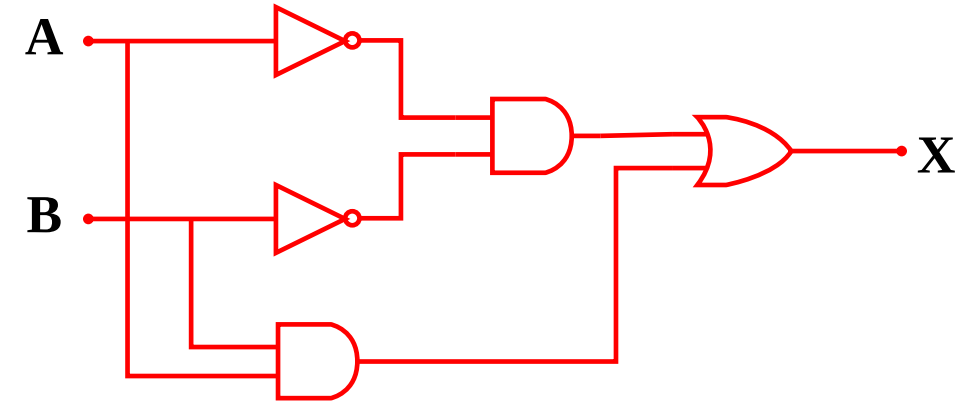
4. A and B are the logical inputs and X is the logical output shown in the figure.

(a) $X = \bar{A}B + \bar{B}A$

(b) $X = AB + \bar{B}A$

(c) $X = AB + \bar{B}\bar{A}$

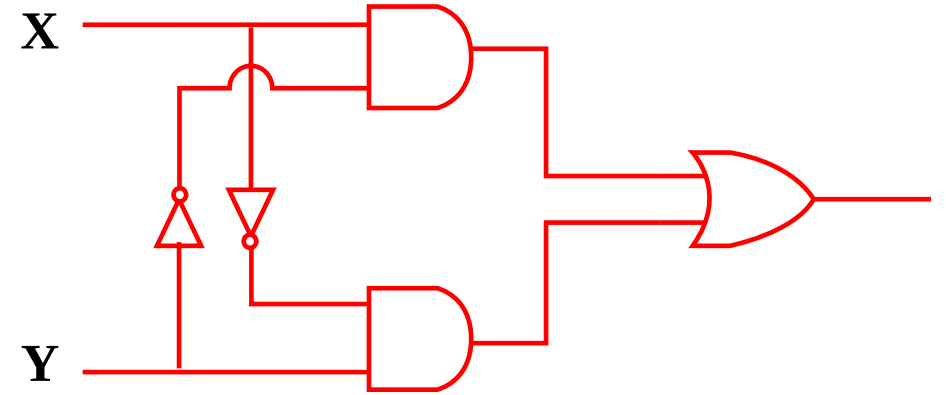
(d) $X = \bar{A}\bar{B} + \bar{B}A$



Use the Code :BVREDDY, to get the Maximum discount

5. The logic evaluated by the circuit at the output is

- (a) $X\bar{Y} + Y\bar{X}$ (b) $(\overline{X + Y})XY$
(c) $\overline{XY} + XY$ (d) $\bar{X}Y + X\bar{Y} + X + Y$



Use the Code :BVREDDY, to get the Maximum discount

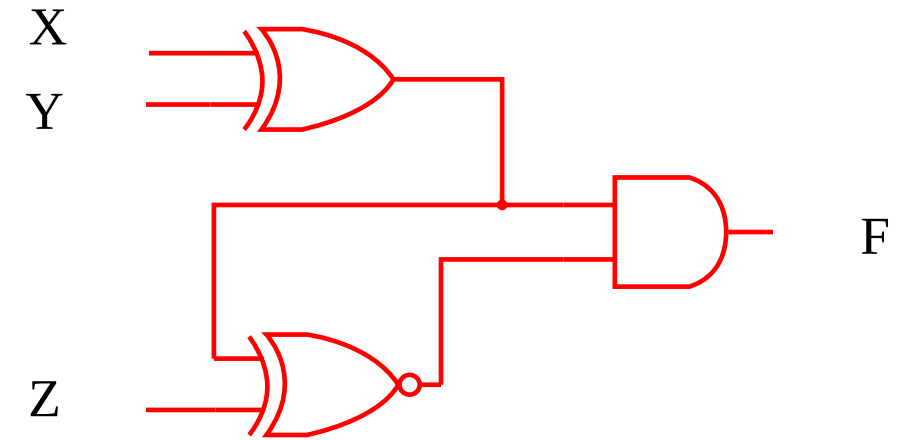
6. The output F in the digital logic circuit shown in the figure is

(a) $F = \bar{X}YZ + X\bar{Y}Z$

(b) $F = \bar{X}Y\bar{Z} + X\bar{Y}Z$

(c) $F = \overline{X\bar{Y}Z} + XYZ$

(d) $F = \overline{X\bar{Y}Z} + XYZ$



Use the Code :BVREDDY, to get the Maximum discount

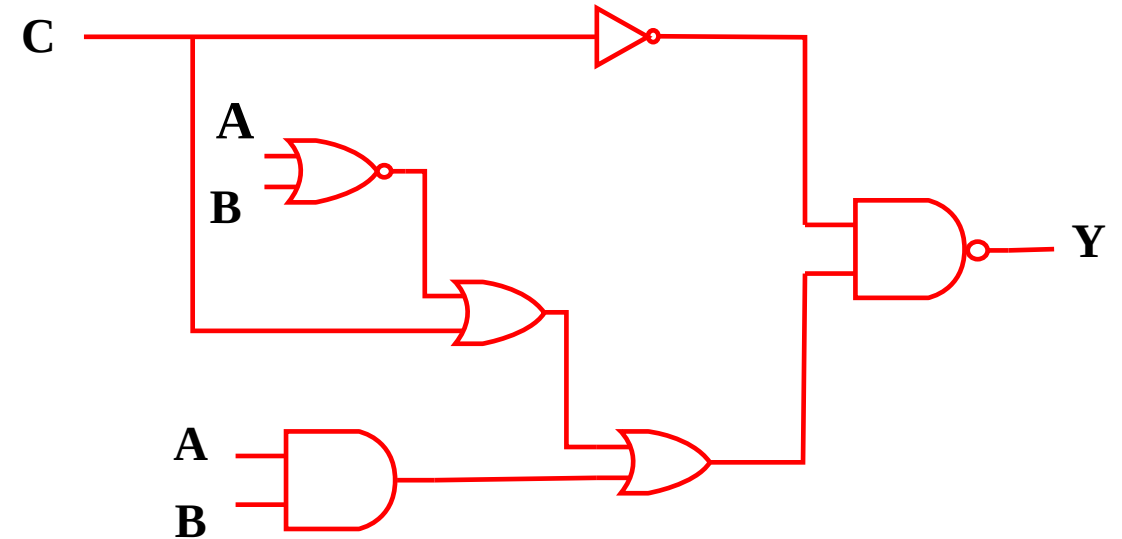
7. In the circuit shown in the figure, if $C = 0$, the expression for Y is

(a) $Y = A\bar{B} + \bar{A}B$

(b) $Y = A + B$

(c) $Y = \bar{A} + \bar{B}$

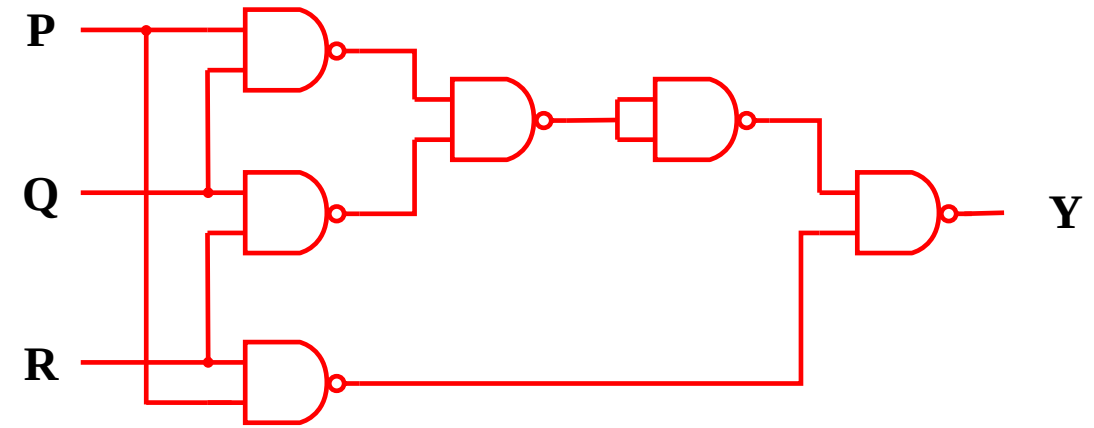
(d) $Y = AB$



Use the Code :BVREDDY, to get the Maximum discount

8. The output Y in the circuit below is always '1' when

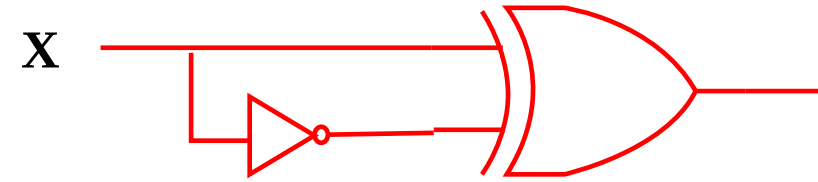
- (a) Two or more of the inputs P, Q, R are '0'
- (b) Two or more of the inputs P, Q, R are '1'
- (c) Any odd number of the inputs P, Q, R is '0'
- (d) Any odd number of the inputs P, Q, R is '1'



Use the Code :BVREDDY, to get the Maximum discount

9. The output Y of the logic circuit give below is

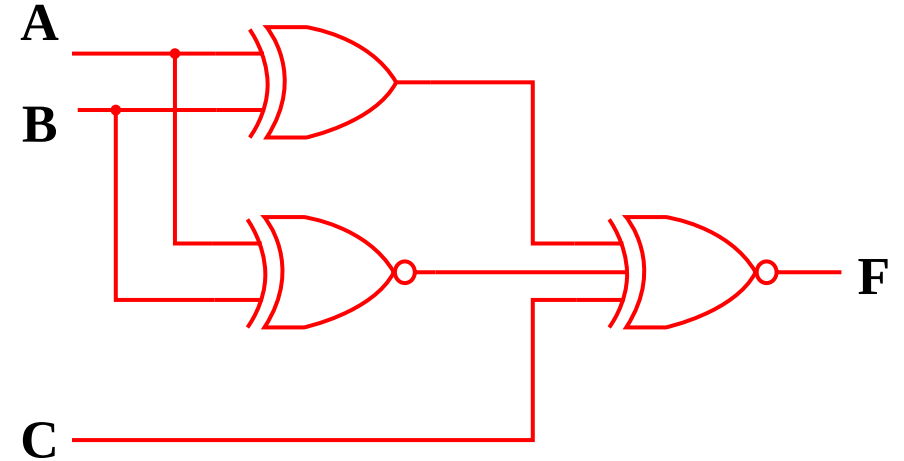
- | | |
|-------|---------------|
| (a) 1 | (b) 0 |
| (c) X | (d) $\bar{X}$ |



Use the Code :BVREDDY, to get the Maximum discount

10. For the output F to be 1 in the logic circuit shown, the input combination should be

- (a) $A = 1, B = 1, C = 0$
- (b) $A = 1, B = 0, C = 0$
- (c) $A = 0, B = 1, C = 0$
- (d) $A = 0, B = 0, C = 1$



Use the Code :BVREDDY, to get the Maximum discount

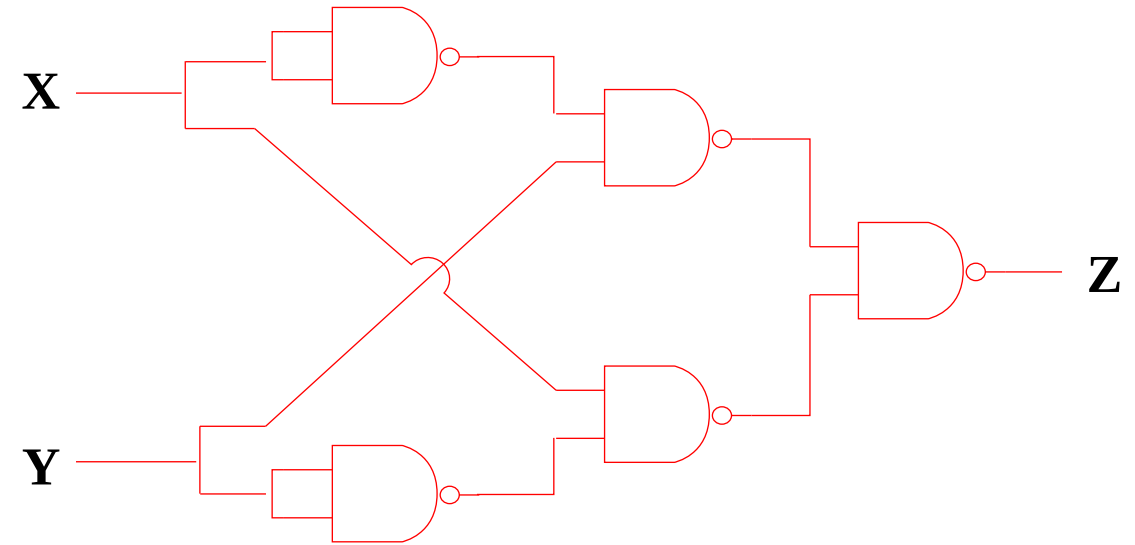
11. The logic gate circuit shown in the figure realizes the function

(a) XOR

(b) XNOR

(c) Half adder

(d) Full adder



Use the Code :BVREDDY, to get the Maximum discount

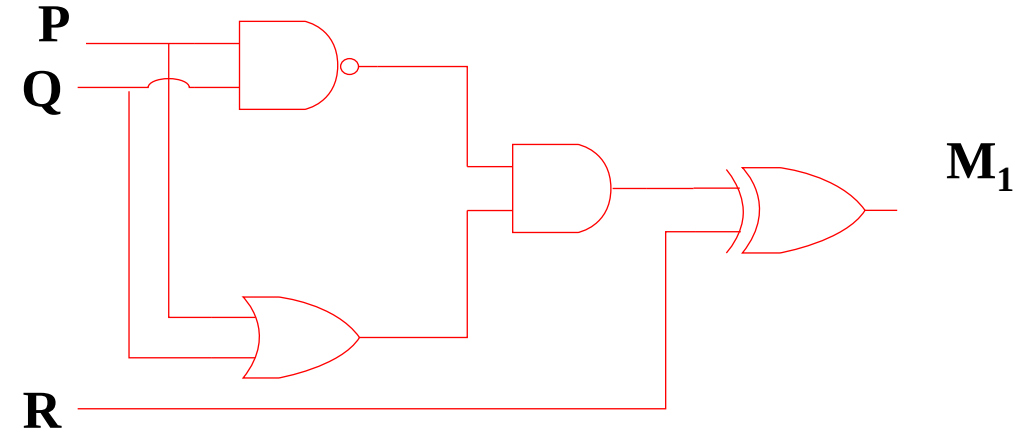
12. Which of the following Boolean Expression correctly represents the relation between P, Q, R and M_1 ?

(a) $M_1 = (P + Q) \oplus R$

(b) $M_1 = (P \cdot Q) \oplus R$

(c) $M_1 = (P \odot Q) \oplus R$

(d) $M_1 = (P \oplus Q) \oplus R$



Use the Code :BVREDDY, to get the Maximum discount

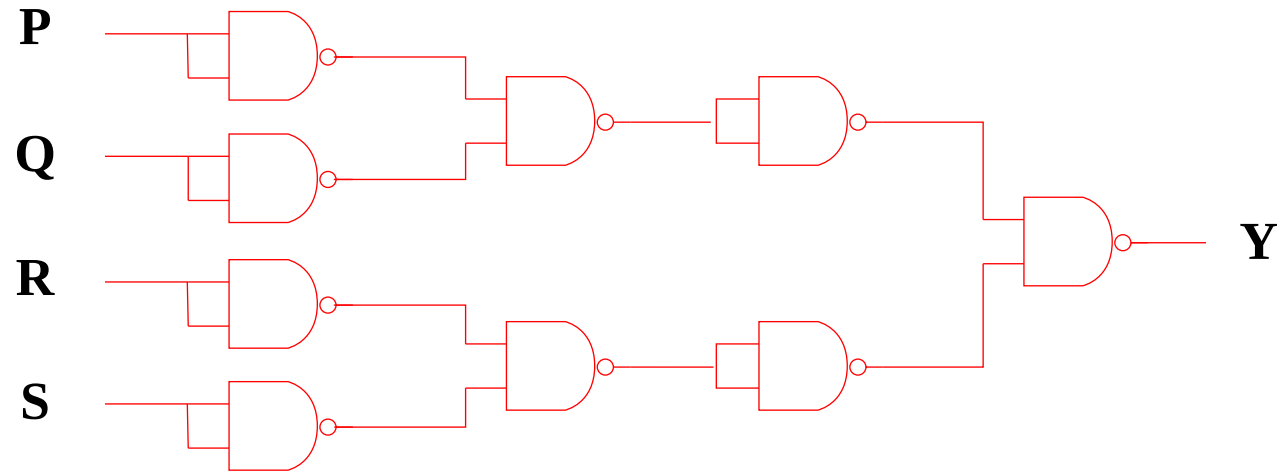
13. For the circuit shown in figure, the Boolean expression for the output Y in terms of inputs P, Q, R and S is

(a) $\bar{P} + \bar{Q} + \bar{R} + \bar{S}$

(b) $P + Q + R + S$

(c) $(\bar{P} + \bar{Q})(\bar{R} + \bar{S})$

(d) $(P + Q)(R + S)$



Use the Code :BVREDDY, to get the Maximum discount

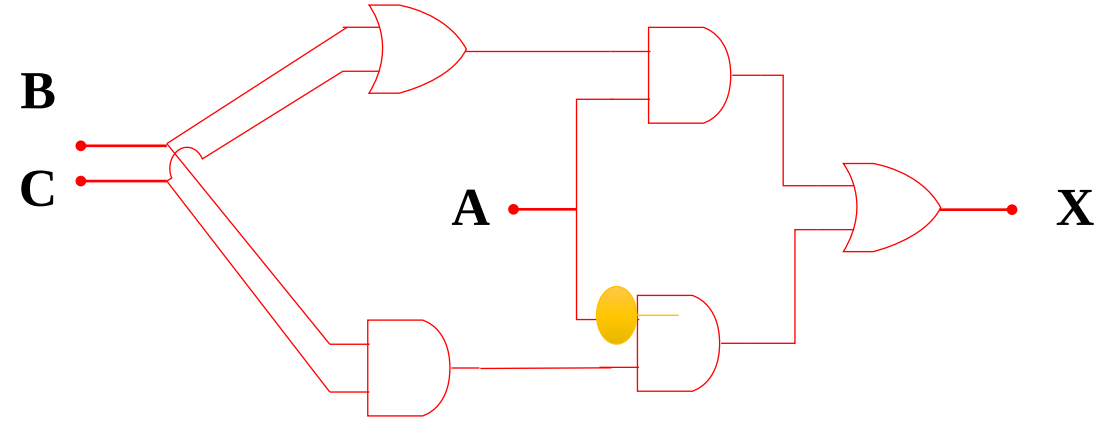
14. In the logic circuit shown in figure, The output x is.

(a) $A\bar{B} + B\bar{C} + C\bar{A}$

(b) $A + B + C$

(c) $AB + BC + CA$

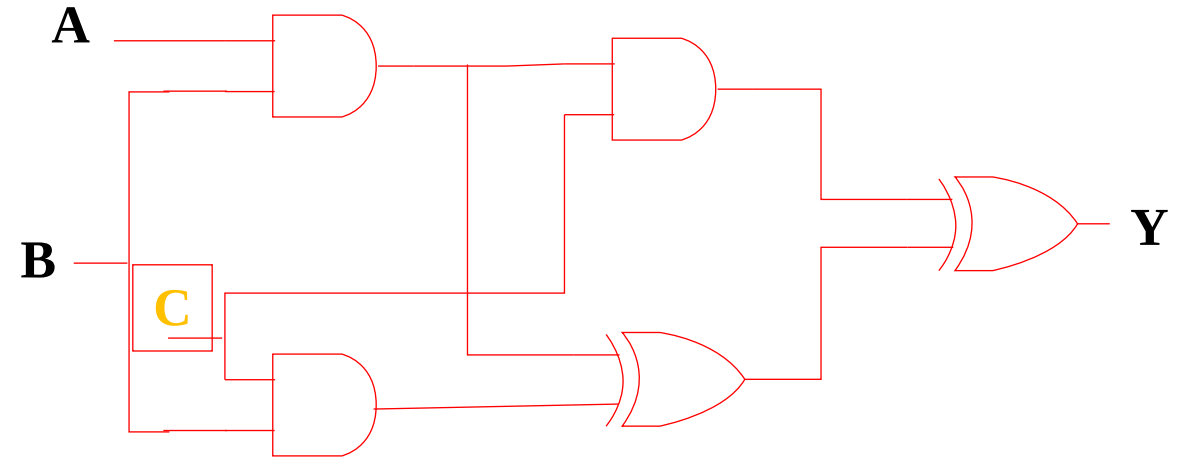
(d) $\bar{A}\bar{B} + \bar{B}\bar{C} + \bar{C}\bar{A}$



Use the Code :BVREDDY, to get the Maximum discount

15. The output of the combinational circuit given below is,

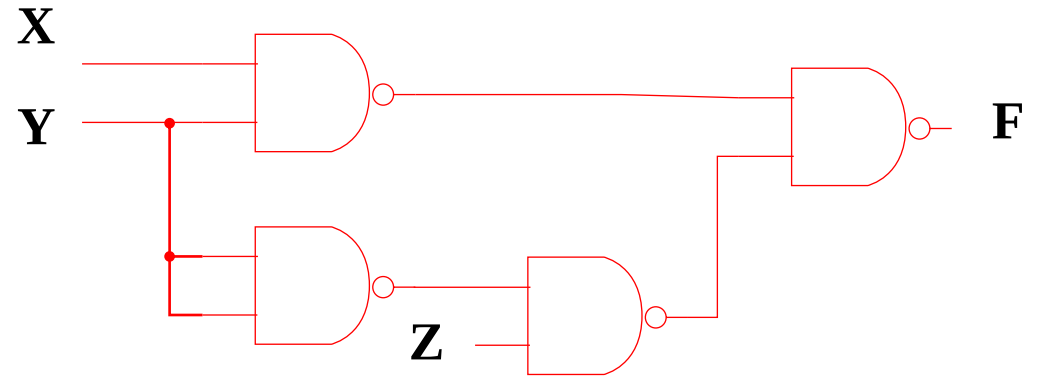
- (a) $A + B + C$ (b) $A(B + C)$
(c) $B(C + A)$ (d) $C(A + B)$



Use the Code :BVREDDY, to get the Maximum discount

16. In the digital circuit given below, F is

- (a) $XY + Y\bar{Z}$ (b) $XY + \bar{Y}Z$
(c) $\bar{X}\bar{Y} + \bar{Y}Z$ (d) $XZ + \bar{Y}$



Use the Code :BVREDDY, to get the Maximum discount

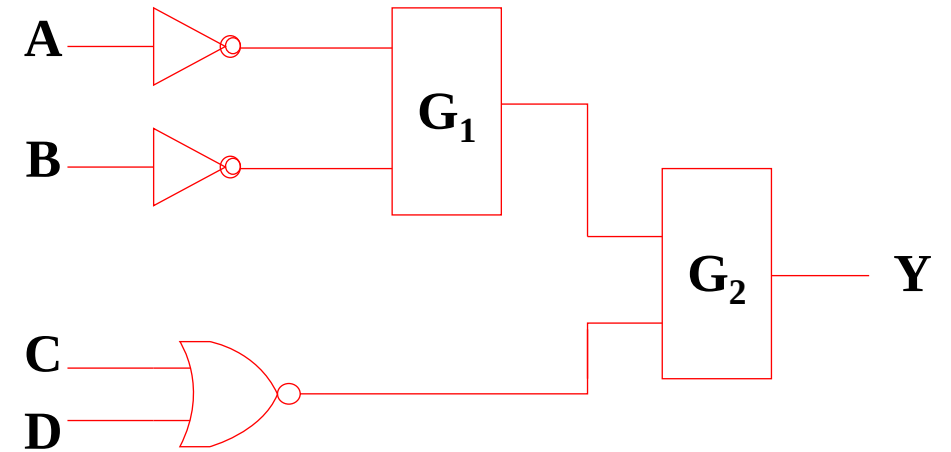
17. In the figure shown, the output Y is required to be $Y = AB + \bar{C} \bar{D}$. The gates G1 and G2 must be, respectively,

(a) NOR, OR

(b) OR, NAND

(c) OR, NAND

(d) AND, NAND



Use the Code :BVREDDY, to get the Maximum discount

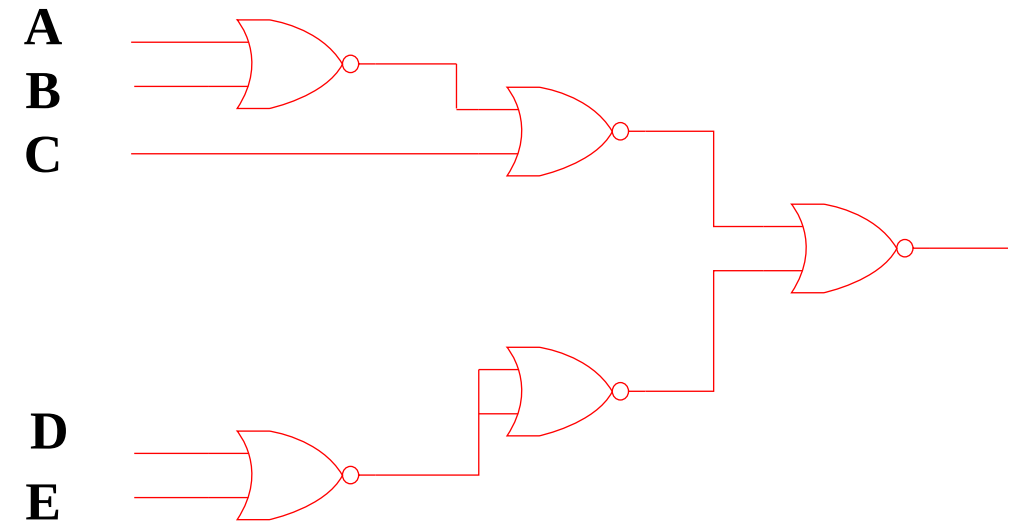
18. The circuit shown in the figure realizes the function:

(a) $(A+B+C) (\overline{D}\overline{E})$

(b) $(A+\overline{(B+C)})(\overline{D}E)$

(c) $(A+B+C) (\overline{D} \overline{E})$

(d) $(\overline{A+B+C})(\overline{D} \overline{E})$



Use the Code :BVREDDY, to get the Maximum discount

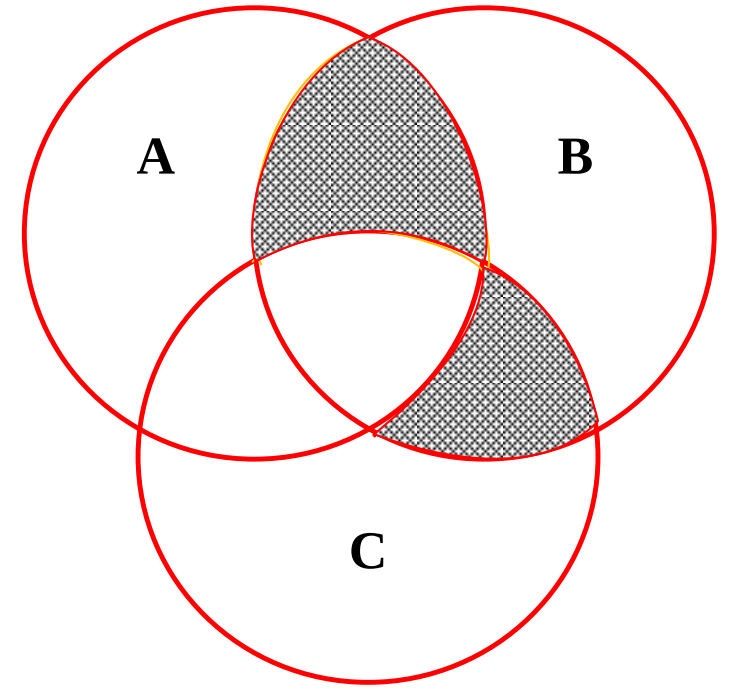
19. The expression for shaded area shown below is:

(a) $AB+BC$

(b) $\bar{A} \bar{B} C + A \bar{B} \bar{C}$

(c) $AB\bar{C}+\bar{A} BC$

(d) None of the above



Use the Code :BVREDDY, to get the Maximum discount

20. The number of Boolean functions which can be generated with four variables is?

(a) 4

(b) 16

(c) 256

(d) 65,536

Use the Code :BVREDDY, to get the Maximum discount

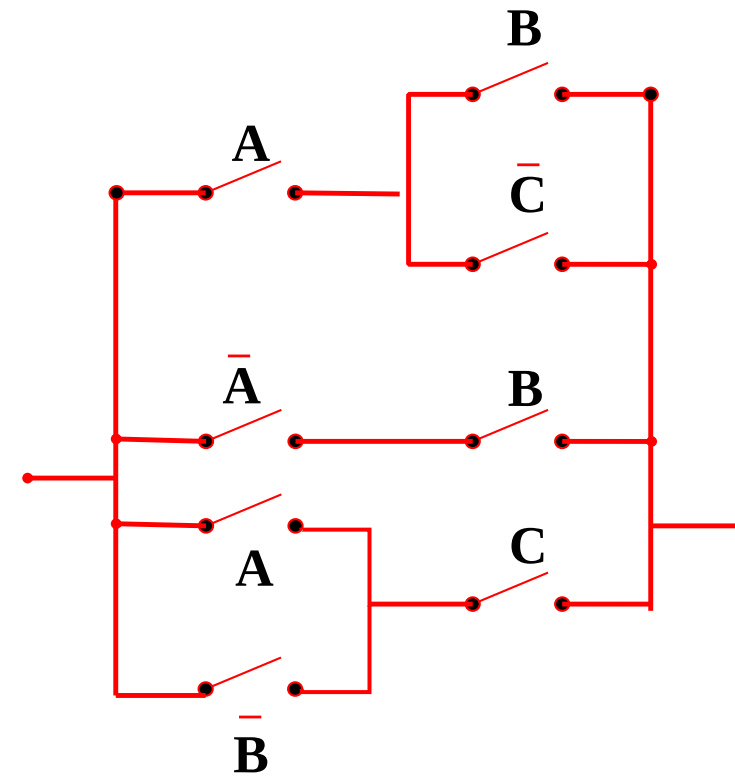
21. The minimum Boolean for the following circuit is.

(a) $AB+AC+BC$

(b) $A+BC$

(c) $A+B$

(d) $A+B+C$



Use the Code :BVREDDY, to get the Maximum discount

22. The Number of switching functions of 3 variables is

(a) 8

(b) 64

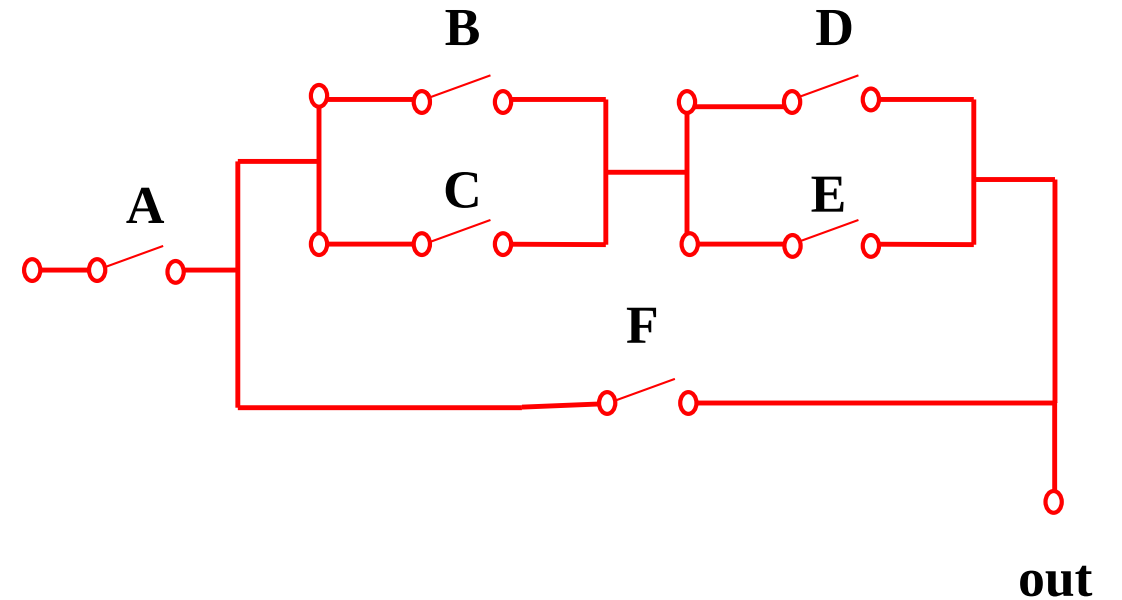
(c) 128

(d) 256

Use the Code :BVREDDY, to get the Maximum discount

23. What Boolean function does the following circuit represents:

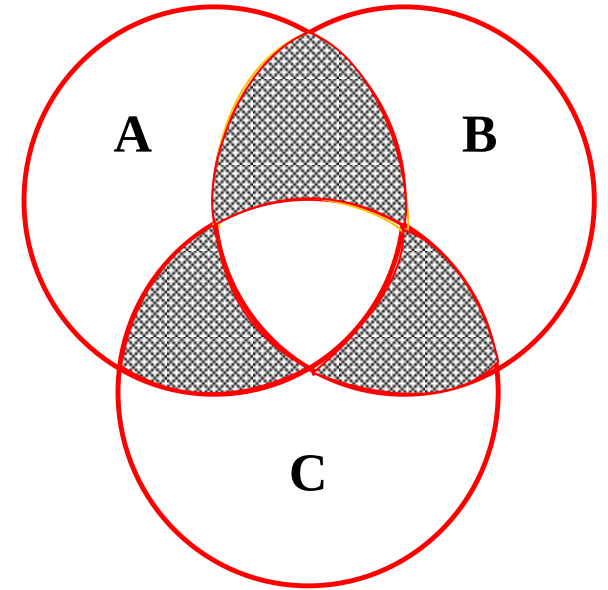
- (a) $A[F+(B+C) (D+E)]$
- (b) $A[F+(B+C) (DE)]$
- (c) $A[F+(BC) + (DE)]$
- (d) $A[F(B+C) + (D+E)]$



Use the Code :BVREDDY, to get the Maximum discount

24. The Boolean expression for the shaded area in the given Venn diagram is:

- (a) $AB+BC+CA$ (b) $AB\bar{C} + \bar{A}BC + A\bar{B}C$
(c) $ABC+\bar{A}\bar{B}\bar{C}$ (d) $A\bar{B}\bar{C} + A\bar{B}C$



Use the Code :BVREDDY, to get the Maximum discount

25. In a digital system there are three inputs A, B and C. The output should be high when at least two inputs are high, the Boolean expression for the output is.

(a) $AB+BC+AC$

(b) $ABC+AB\bar{C}+\bar{A}CB+A\bar{B}C$

(c) $AB\bar{C}+A\bar{B}C+\bar{A}BC$

(d) $A\bar{B}+B\bar{C}+\bar{A}C$

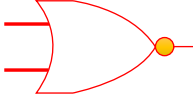
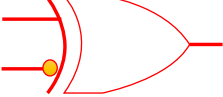
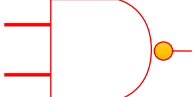
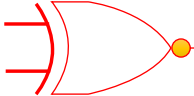
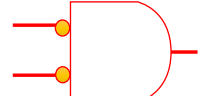
Use the Code :BVREDDY, to get the Maximum discount

26. A bulb in a staircase has two switches, one switch being at the ground floor and the other one at the first floor. The bulb can be turned ON and also can be turned OFF by any one of the switches irrespective of the state of the other switch. The logic of switching of the bulb resembles

- | | |
|-----------------|-----------------|
| (a) an AND gate | (b) an OR gate |
| (c) an XOR gate | (d) a NAND gate |

Use the Code :BVREDDY, to get the Maximum discount

27. Match the logic gates in Column A with their equivalents in Column B.

Column A	Column B
P 	1 
Q 	2 
R 	3 
S 	4 

(a) P-2, Q-4, R-1, S-3

(b) P-4, Q-2, R-1, S-3

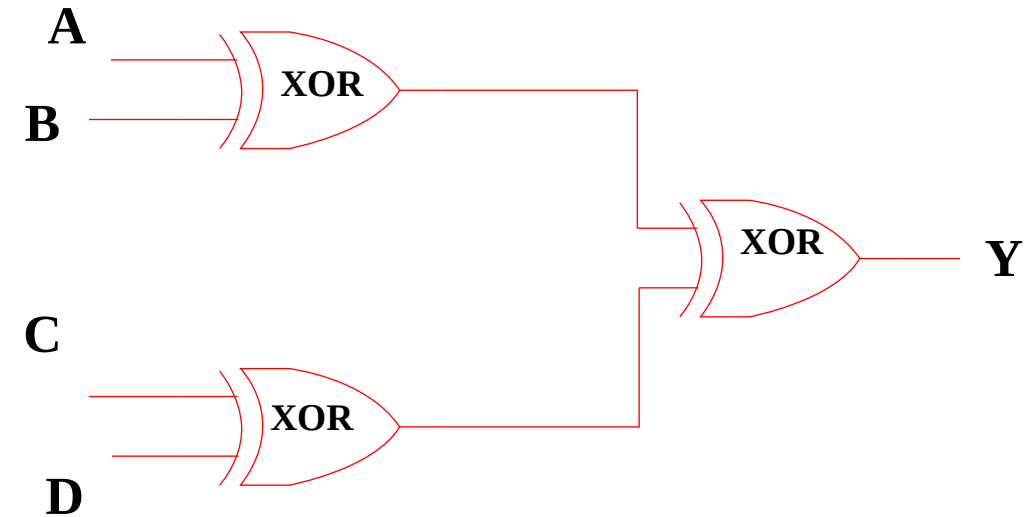
(c) P-2, Q-4, R-3, S-1

(d) P-4, Q-2, R-3, S-1

Use the Code :BVREDDY, to get the Maximum discount

28. A, B, C and D are input bits, and Y is the output bit in the XOR gate circuit of the figure below. Which of the following statements about the sum S of A, B, C, D and Y is correct?

- (a) S is always either zero or odd
- (b) S is always either zero or even
- (c) $S = 1$ only if the sum of A, B, C and D is even
- (d) $S = 1$ only if the sum of A, B, C and D is odd



Use the Code :BVREDDY, to get the Maximum discount

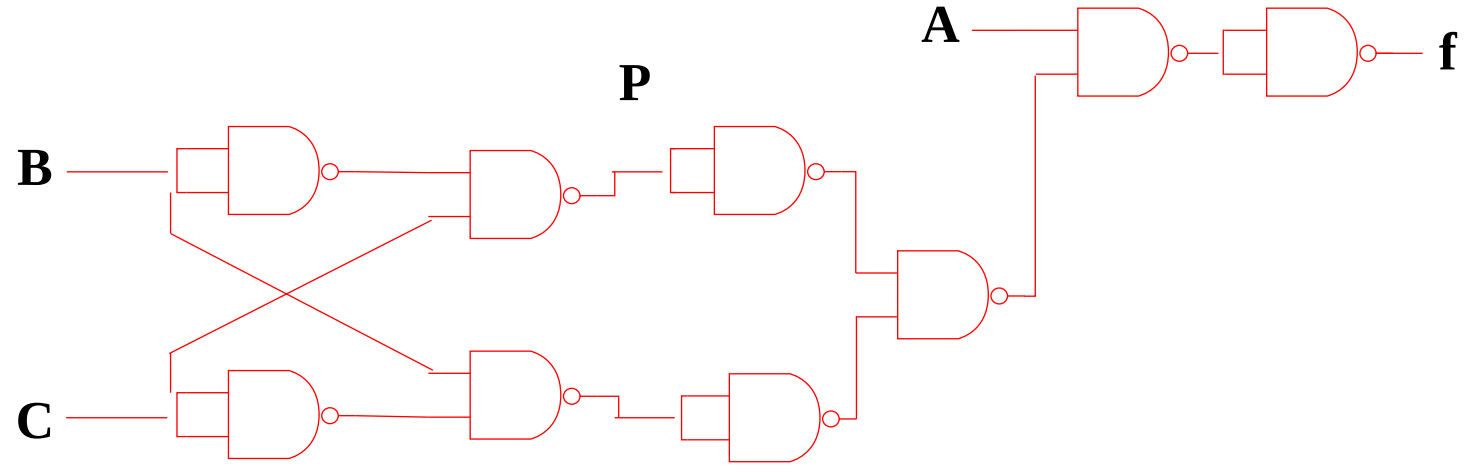
29. The point P in the following figure is stuck-at-1. The output f will be

(a) $\overline{ABC}$

(b) $\overline{A}$

(c) ABC

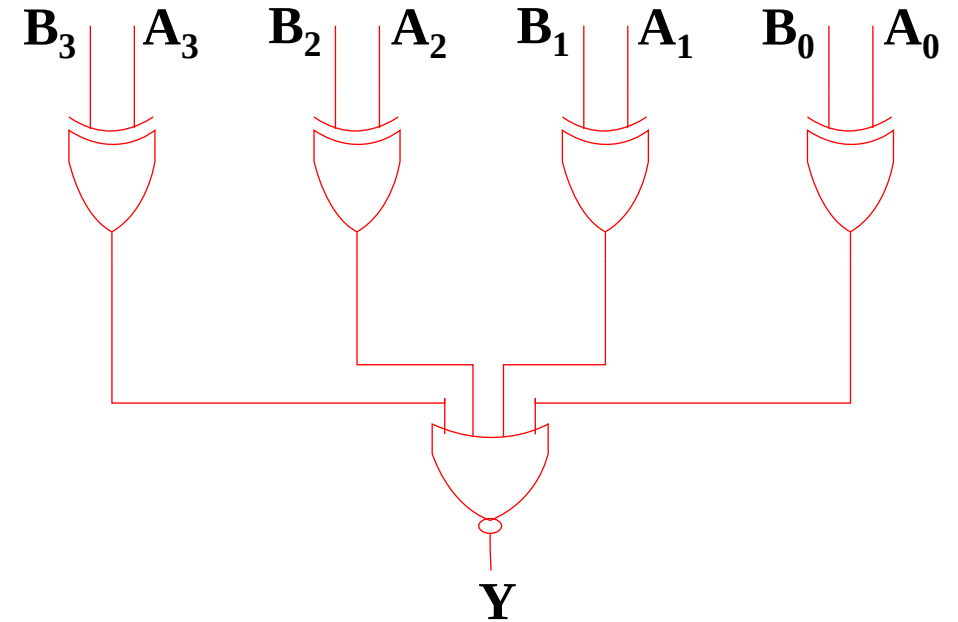
(d) A



Use the Code :BVREDDY, to get the Maximum discount

30. A digital circuit, which compares two numbers, $A_3, A_2, A_1, A_0, B_3, B_2, B_1, B_0$ is shown in figure. To get output $Y = 0$, choose one pair of correct input numbers.

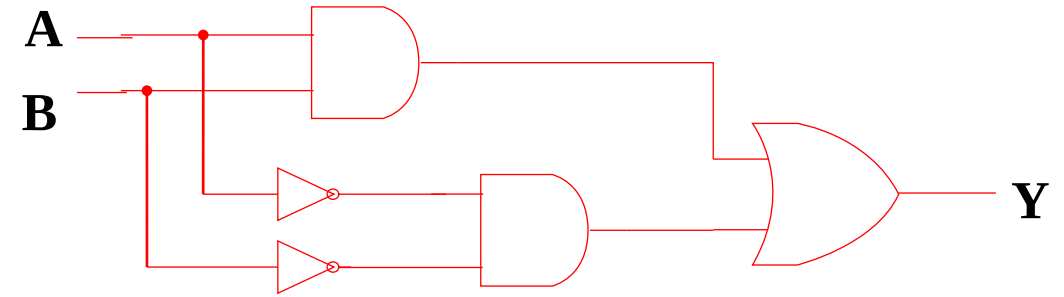
- (a) 1010, 1010 (b) 0101, 0101
(c) 0010, 0010 (d) 0010, 1011



Use the Code :BVREDDY, to get the Maximum discount

31. The logic circuit of figure is a

- | | |
|-----------------------|----------------|
| (a) Half adder | (b) XOR |
| (c) Equality detector | (d) Full adder |



Use the Code :BVREDDY, to get the Maximum discount

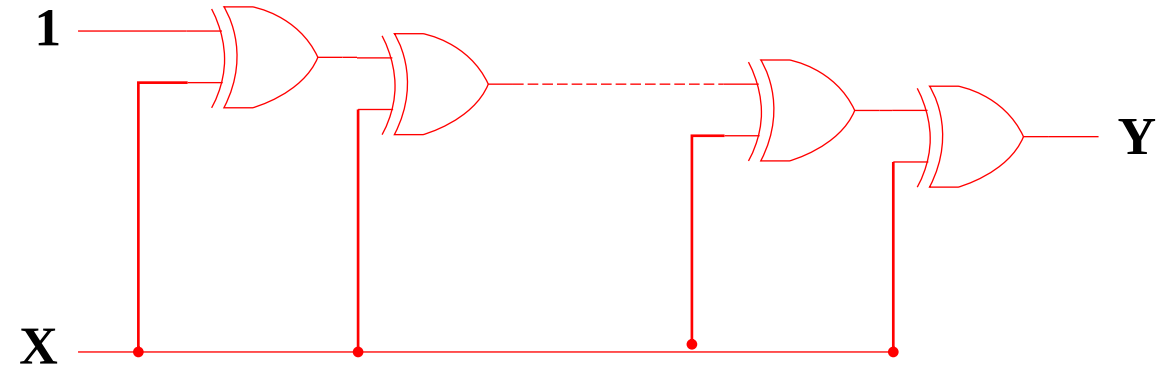
32. If the input to the digital circuit (shown in the given figure) consisting of a cascade of 20 XOR-gates is X , then the output Y is equal to

(a) 0

(b) 1

(c) $\bar{X}$

(d) X



Use the Code :BVREDDY, to get the Maximum discount

33.The output of a logic gate is “1” when all its inputs are at logic "0". The gate is either

- (a) A NAND or EX-OR gate
- (b) A NOR or an EX-OR gate
- (c) An AND or an EX-OR gate
- (d) A NOR or an EX-NOR gate

Use the Code :BVREDDY, to get the Maximum discount

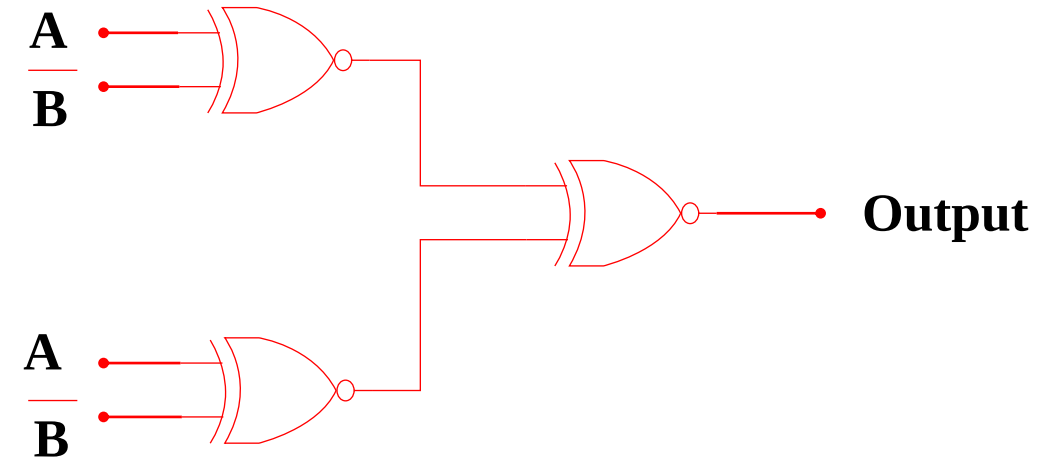
34. The output of the circuit shown (in figure) is equal to

(a) 0

(b) 1

(c) $\bar{A}B + A\bar{B}$

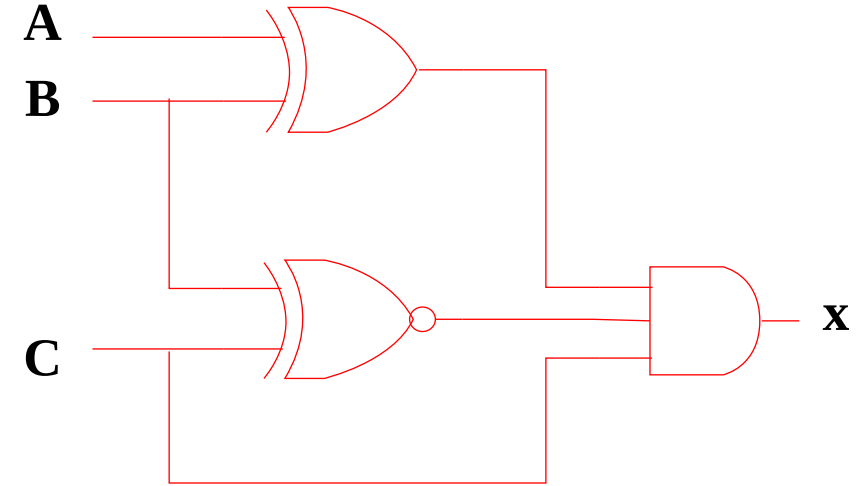
(d) $(\overline{A * B}) * (\overline{A * B})$



Use the Code :BVREDDY, to get the Maximum discount

35. For the logic circuit shown in the given figure, the required input condition (A, B, C) to make the output (X)=1 is

- | | |
|-------------|-------------|
| (a) 1, 0, 1 | (b) 0, 0, 1 |
| (c) 1, 1, 1 | (d) 0, 1, 1 |



Use the Code :BVREDDY, to get the Maximum discount

36. The operation $x \oplus \bar{y}$ represents:

(a) $x - y$

(b) $\bar{x}y + \overline{xy}$

(c) $xy + \bar{x}\bar{y}$

(d) $x - \bar{y}$

Use the Code :BVREDDY, to get the Maximum discount

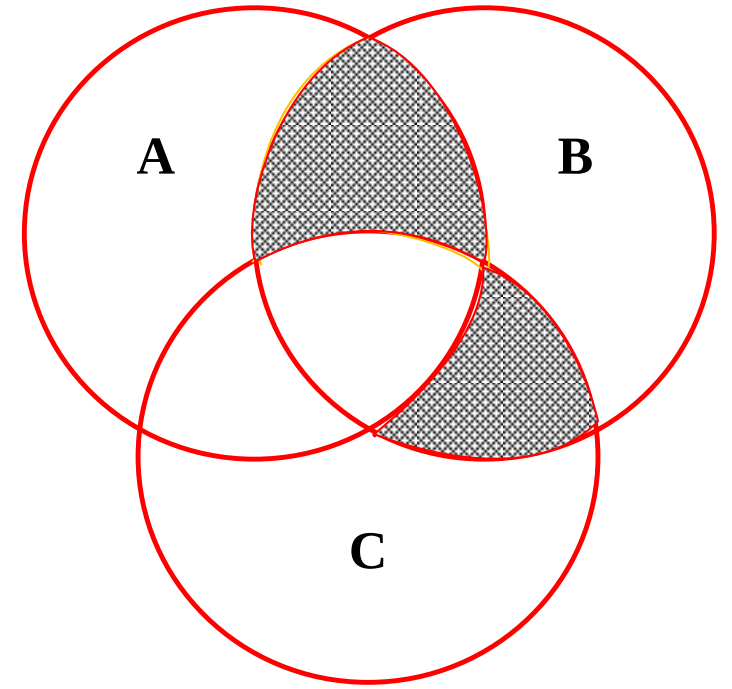
37. The expression for shaded area shown below is:

(a) $AB+BC$

(b) $\bar{A}\bar{B}C + A\bar{B}\bar{C}$

(c) $AB\bar{C}+\bar{A}BC$

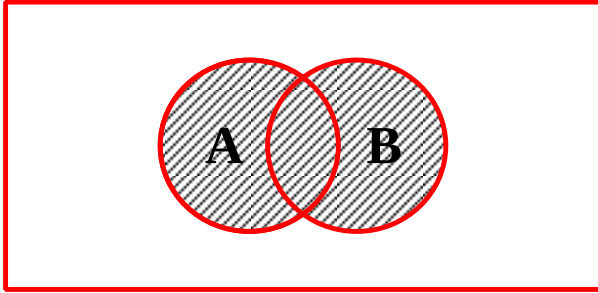
(d) None of the above



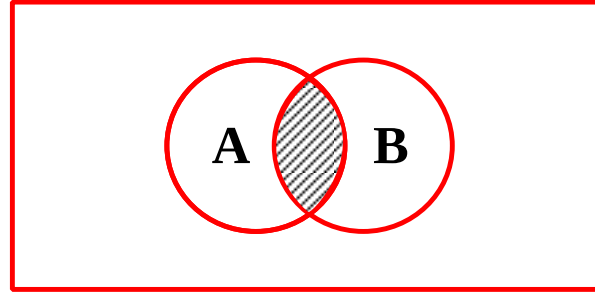
Use the Code :BVREDDY, to get the Maximum discount

38. The Venn diagram representing the Boolean expression $(A + \bar{A}.B)$ is.

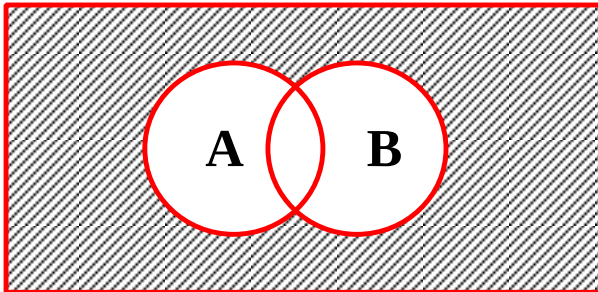
(a)



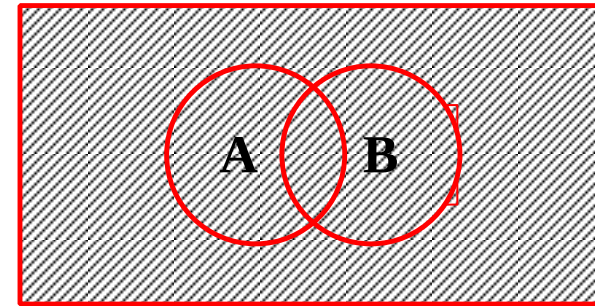
(b)



(c)



(d)



Use the Code :BVREDDY, to get the Maximum discount

39. Match List-I with List-II and select the correct answer using the codes given below the Lists:

List-I

A. $A \oplus B = 0$

B. $\overline{A + B} = 0$

C. $\overline{A} . B = 0$

D. $A \oplus B = 1$

List-II

1. $A \neq B$

2. $A = B$

3. $A=1$ or $B=1$

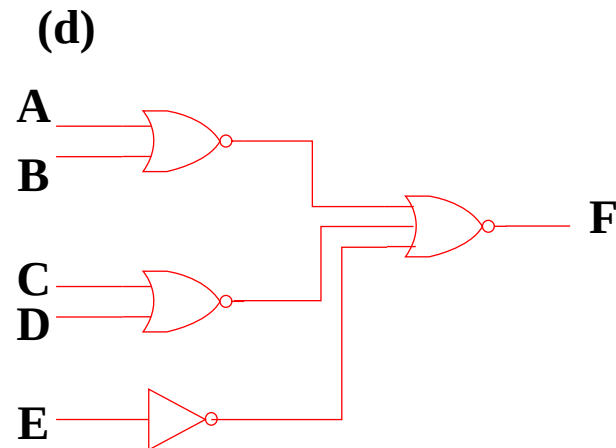
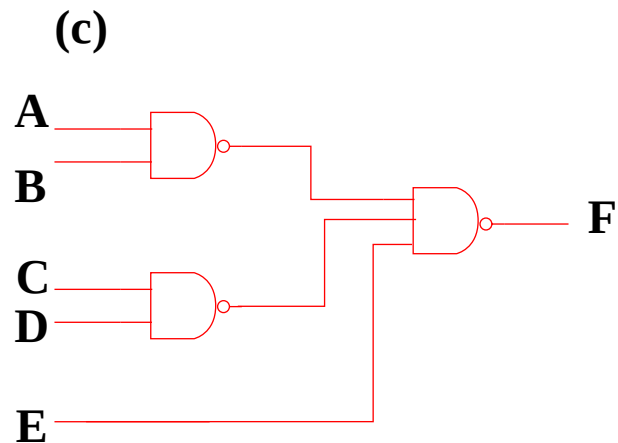
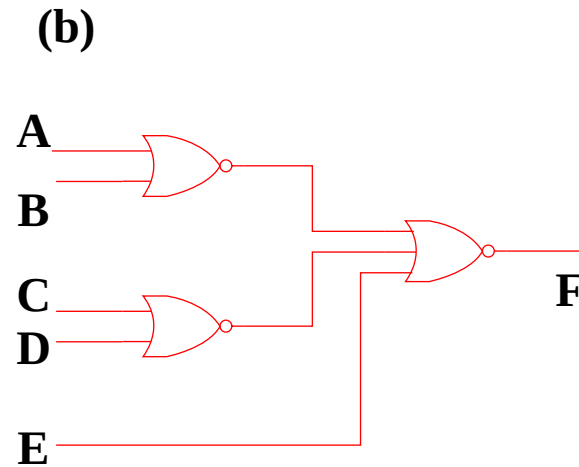
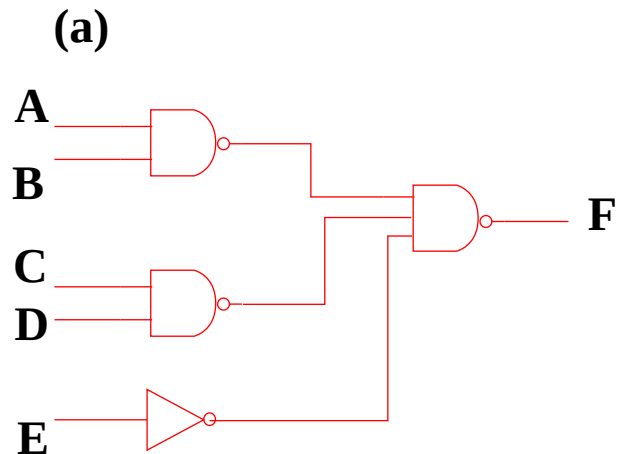
4. $A=1$ or $B=0$

Codes:

	A	B	C	D
(a)	3	2	1	4
(b)	2	3	4	1
(c)	3	2	4	1
(d)	2	3	1	4

Use the Code :BVREDDY, to get the Maximum discount

40. The Boolean function $F=AB+CD+E$ can be realized as



Use the Code :BVREDDY, to get the Maximum discount

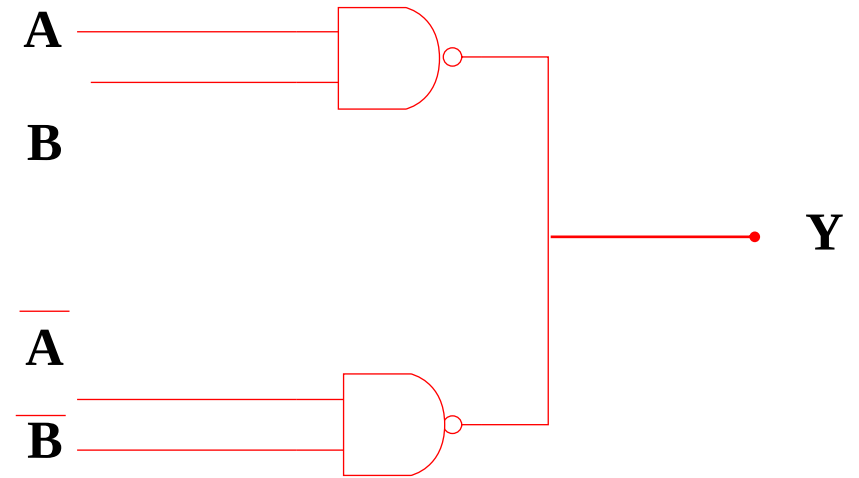
41. The open collector wired circuit shown below functions as:

(a) EX – NOR

(b) AND

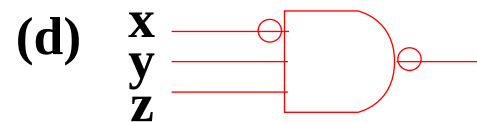
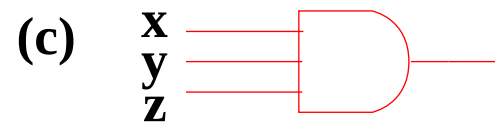
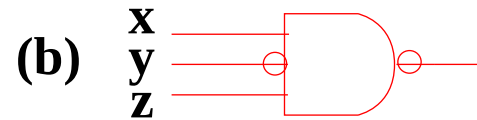
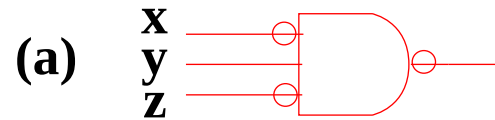
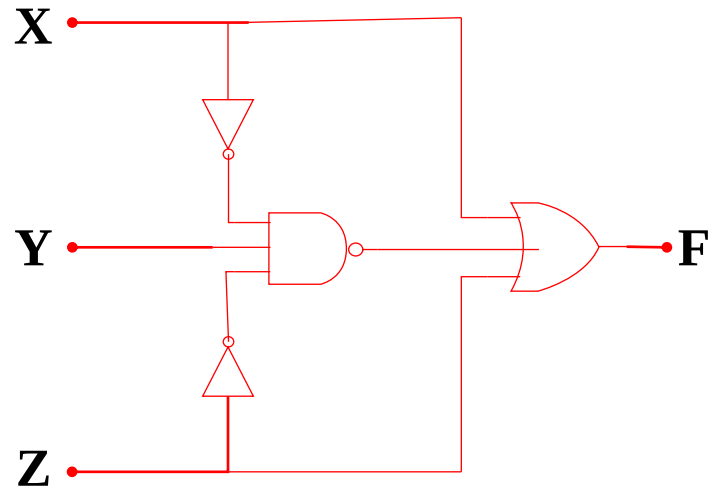
(c) EX – OR

(d) NOR



Use the Code :BVREDDY, to get the Maximum discount

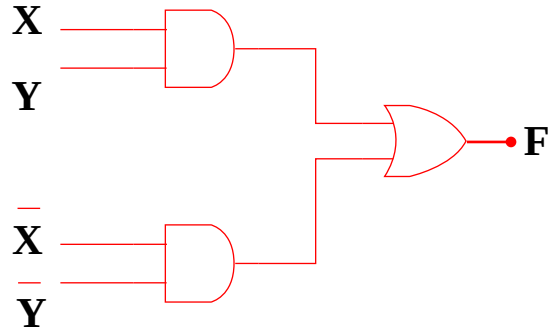
42. The Minimized version for the logic circuit shown in the figure is:



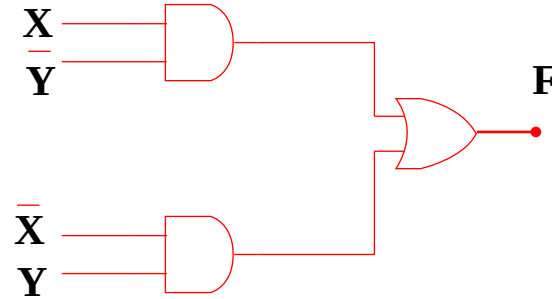
Use the Code :BVREDDY, to get the Maximum discount

43. Which of the following is a coincidence logic circuit:

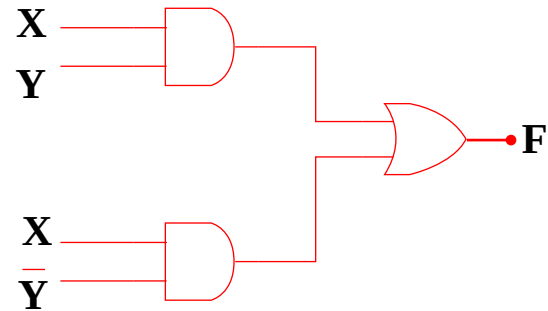
(a)



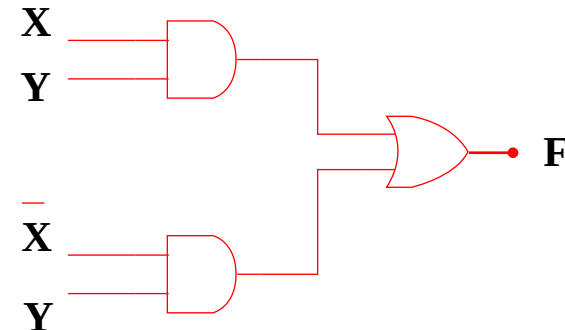
(b)



(c)



(d)



Use the Code :BVREDDY, to get the Maximum discount

44. The gate whose output is LOW if and only if all the inputs are HIGH, is

- (a) NAND
- (b) NOR
- (c) OR
- (d) AND

Use the Code :BVREDDY, to get the Maximum discount

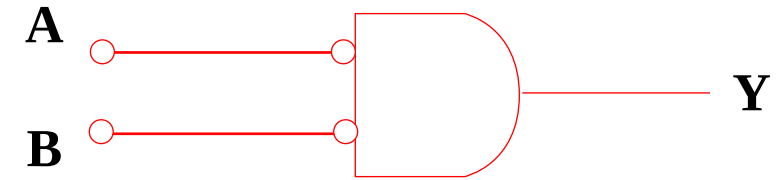
45. The negative logic AND gate shown in the given figure is equivalent to positive logic.

(a) AND gate

(b) OR gate

(c) NAND gate

(d) NOR gate



Use the Code :BVREDDY, to get the Maximum discount

46. Which one of the following sets of gates are best suited for parity checking and parity generation?

(a) AND, OR, NOT gates

(b) X-OR, X-NOR gates

(c) NAND gates

(d) NOR gates

Use the Code :BVREDDY, to get the Maximum discount

47. The output of a logic gate is '1' when all its inputs are at logic '0'. The gate is either.

(a) A NAND or an EX-OR

(b) An OR or an EX-OR

(c) An AND or an EX-OR

(d) A NOR or an EX-NOR

Use the Code :BVREDDY, to get the Maximum discount

48. A three-input NAND gate is to be used as an inverter. Which one of the following measures will achieve better results?

- (a) The two inputs not used are kept open
- (b) The two inputs not used are connected to ground (0 level)
- (c) The two inputs not used are connected to V_{cc}
- (d) None of the above

Use the Code :BVREDDY, to get the Maximum discount

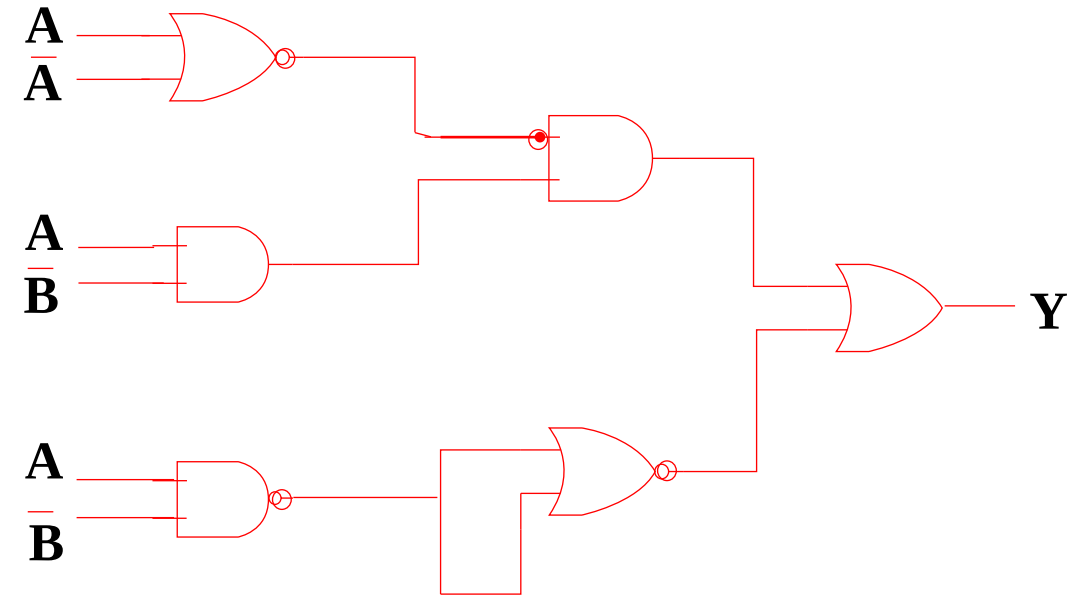
49. The output Y for the logic circuit shown in the given figure is

(a) $A\overline{B}$

(b) $\overline{A} + \overline{B}$

(c) $\overline{A\overline{B}}$

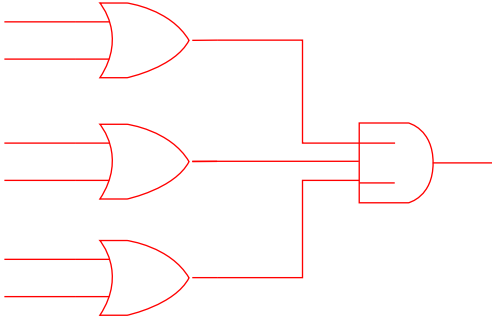
(d) $A + \overline{B}$



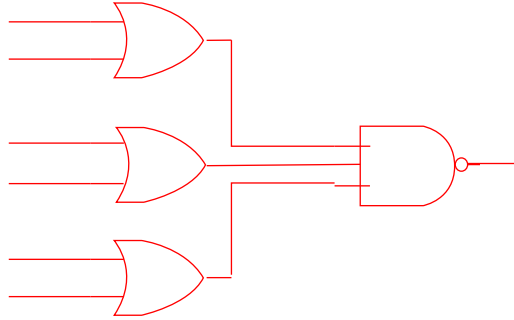
Use the Code :BVREDDY, to get the Maximum discount

50. The circuit shown in figure is equivalent to:

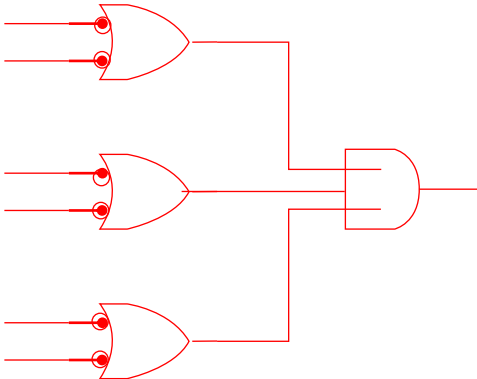
(a)



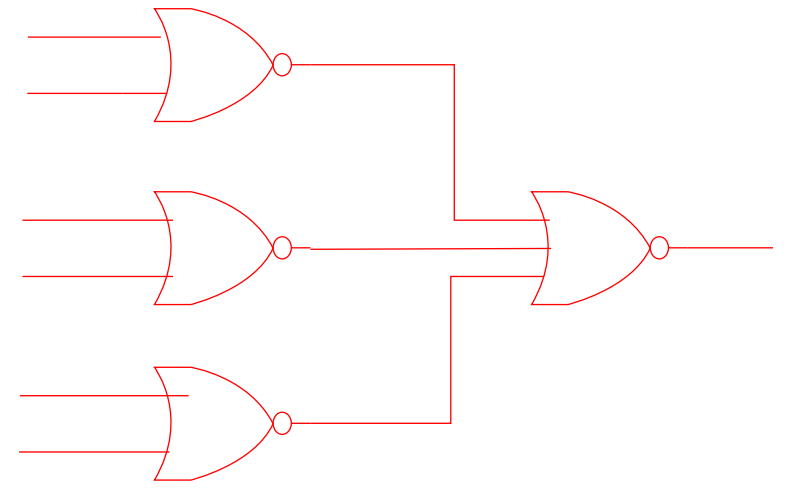
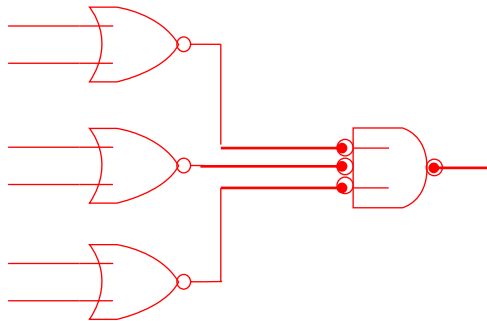
(b)



(c)



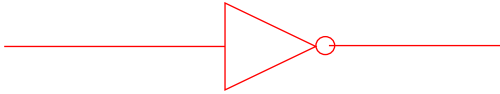
(d)



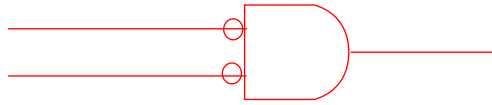
Use the Code :BVREDDY, to get the Maximum discount

51. The logic circuit shown in the figure can be Minimized to

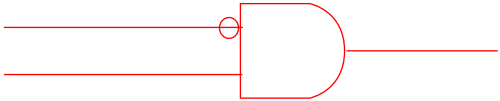
(a)



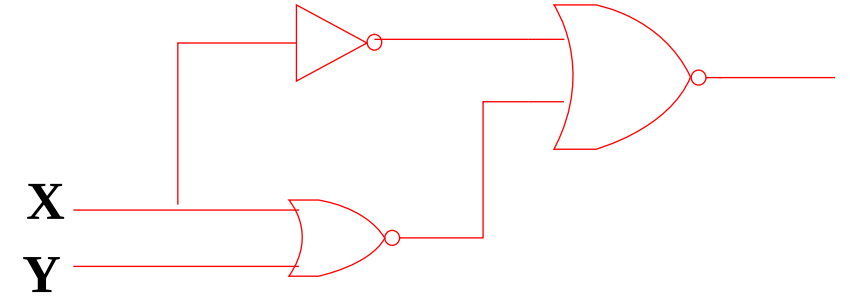
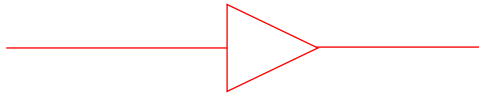
(b)



(c)



(d)



Use the Code :BVREDDY, to get the Maximum discount

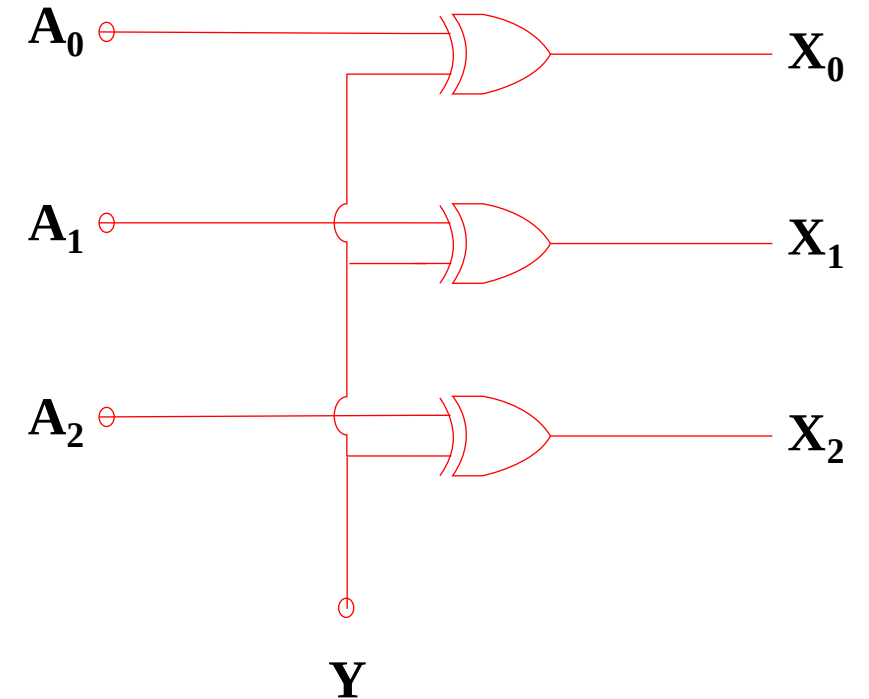
52. In the figure shown, $X_2 X_1 X_0$ will be 1's complement of $A_2 A_1 A_0$ if

(a) $Y = 0$

(b) $Y = 1$

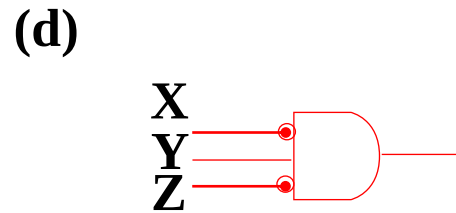
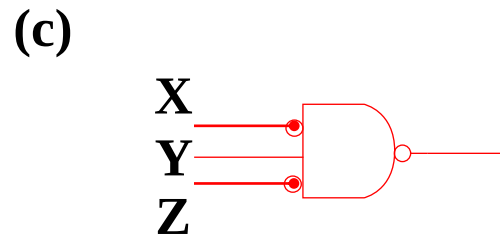
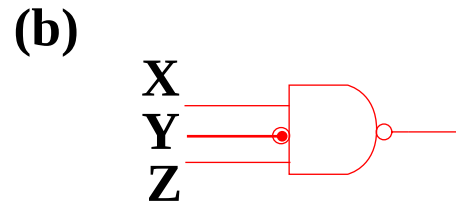
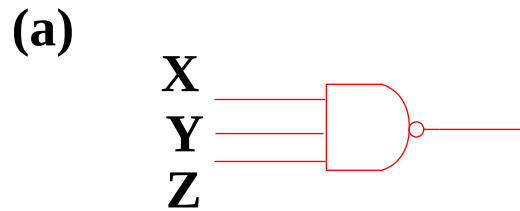
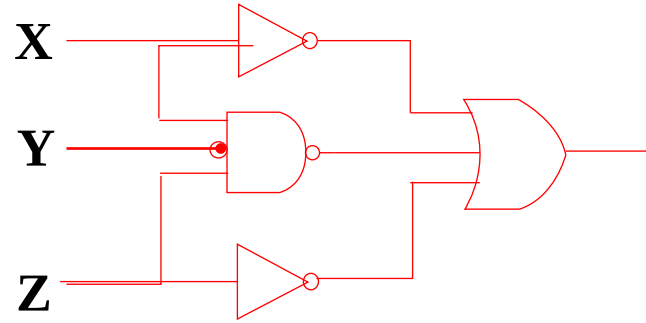
(c) $Y = \bar{A}_0 = \bar{A}_1 = \bar{A}_2$

(d) $Y = A_0 = A_1 = A_2$



Use the Code :BVREDDY, to get the Maximum discount

53. The minimized logic circuit for the circuit shown in figure is



Use the Code :BVREDDY, to get the Maximum discount

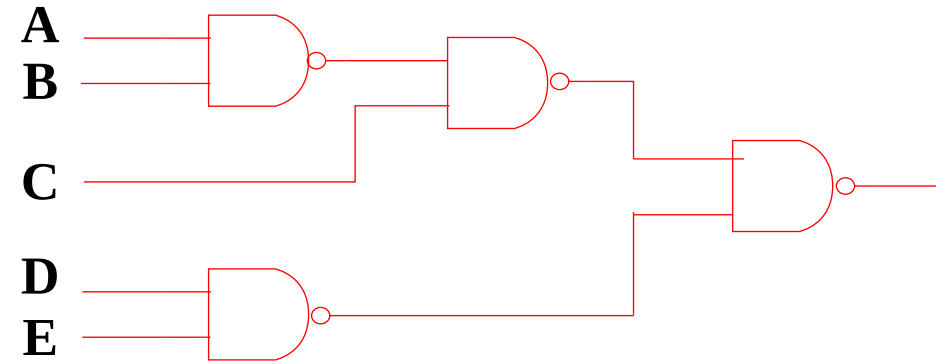
54. The circuit shown in the following figure realizes the function.

(a) $(\bar{A} + \bar{B})C + DE$

(b) $(A + B)C + D + E$

(c) $AB + C + DE$

(d) $AB + C(D + E)$



Use the Code :BVREDDY, to get the Maximum discount

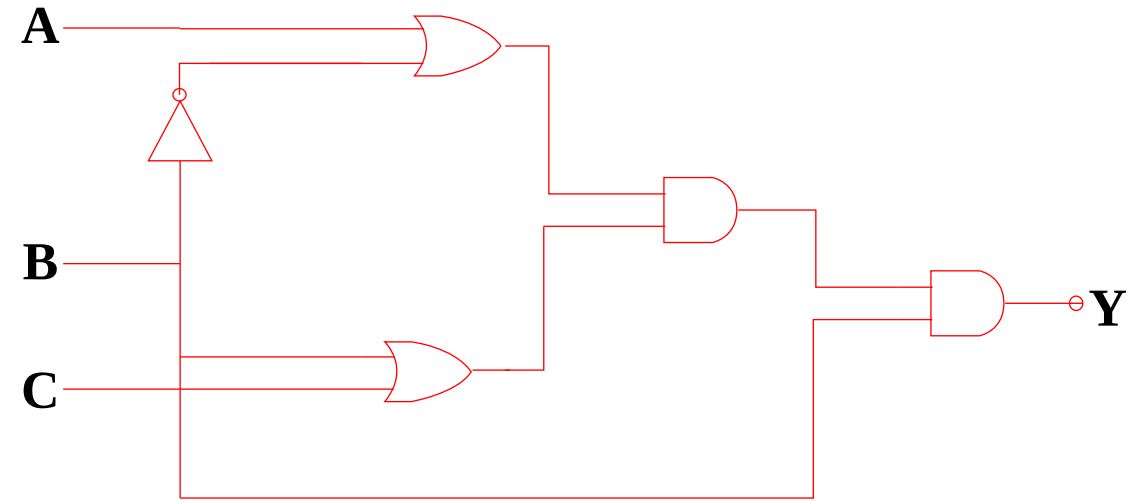
55. The output Y of the logic circuit shown in the figure is

(a) $A + BC$

(b) BC

(c) AB

(d) $AB + C$



Use the Code :BVREDDY, to get the Maximum discount

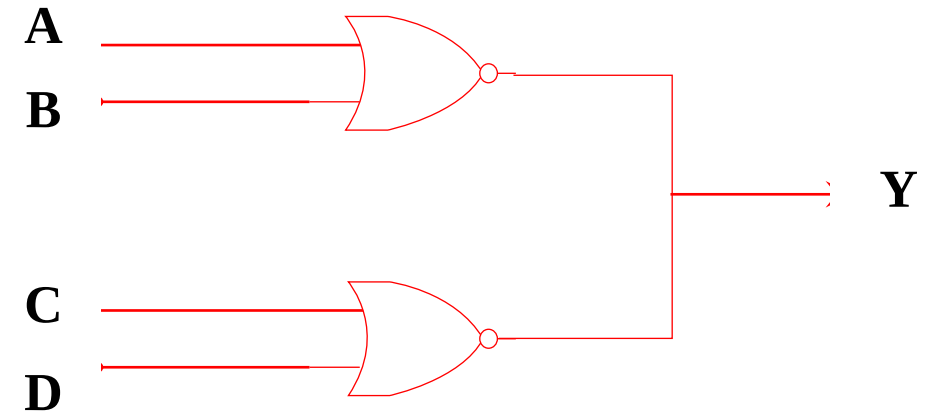
56. When two gates with open collector outputs are tied together as shown in the figure, the output obtained will be

(a) $\overline{A} + \overline{B} + \overline{C} + \overline{D}$

(b) $\overline{A} \overline{B} \overline{C} \overline{D}$

(c) $(\overline{A} + \overline{B}) (\overline{C} + \overline{D})$

(d) $\overline{A + B + C + D}$



Use the Code :BVREDDY, to get the Maximum discount

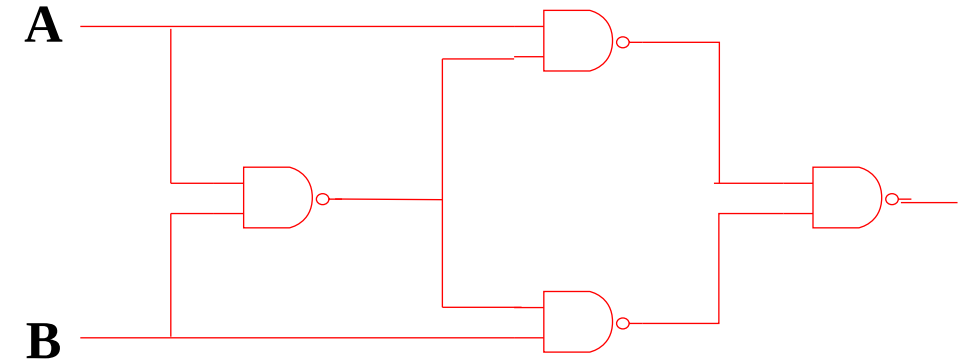
57. The circuit shown in the figure is functionally equivalent to

(a) NOR gate

(b) OR gate

(c) EX-OR gate

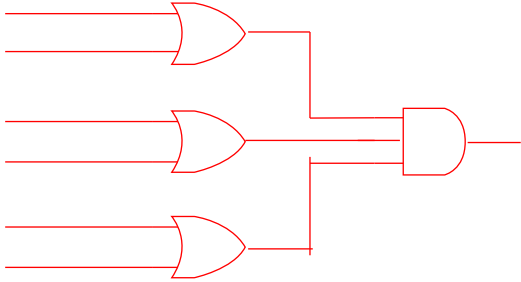
(d) NAND gate



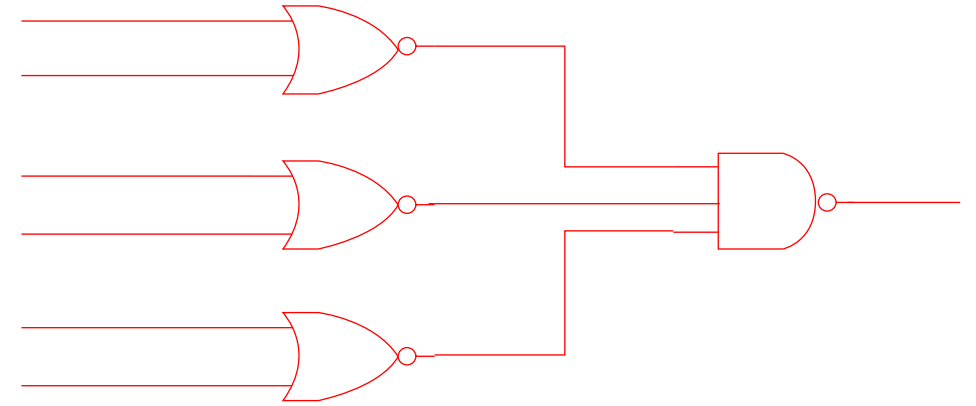
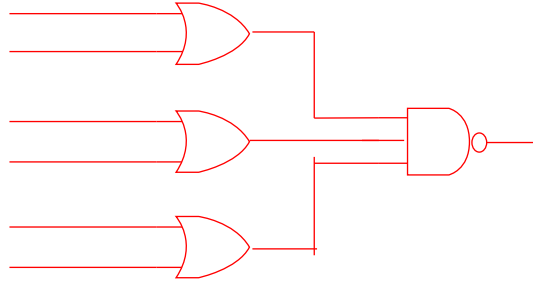
Use the Code :BVREDDY, to get the Maximum discount

58. The circuit shown in fig is equivalent to

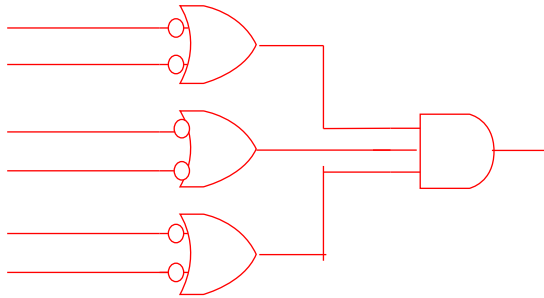
(a)



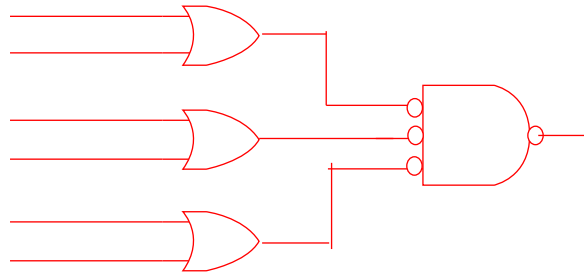
(b)



(c)



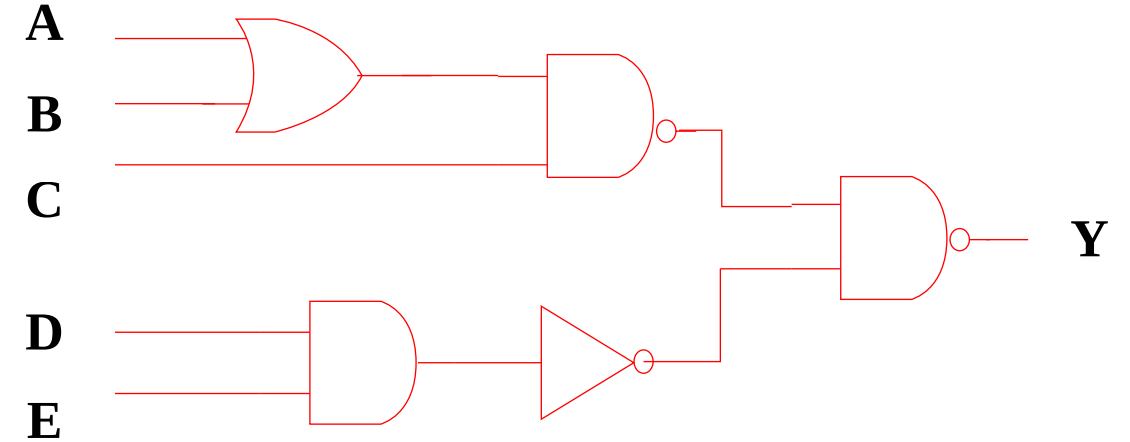
(d)



Use the Code :BVREDDY, to get the Maximum discount

59. The output Y of the circuit shown in the figure :-

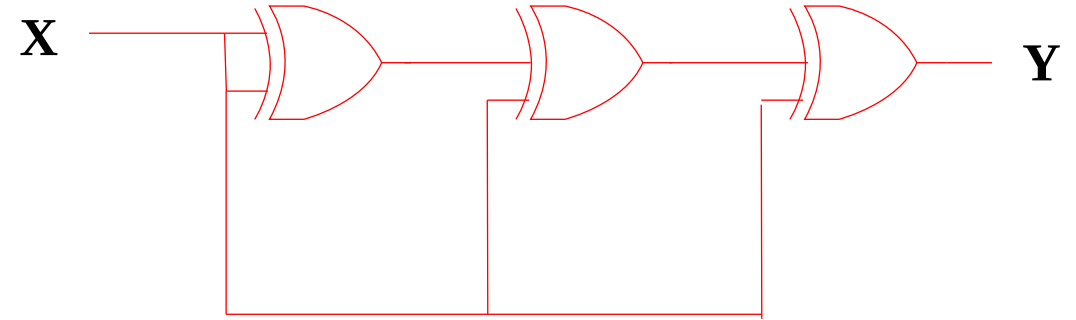
- (a) $(A + B) C + DE$
- (b) $AB + C (D + E)$
- (c) $(A + B) C + D + E$
- (d) $(AB + C) DE$



Use the Code :BVREDDY, to get the Maximum discount

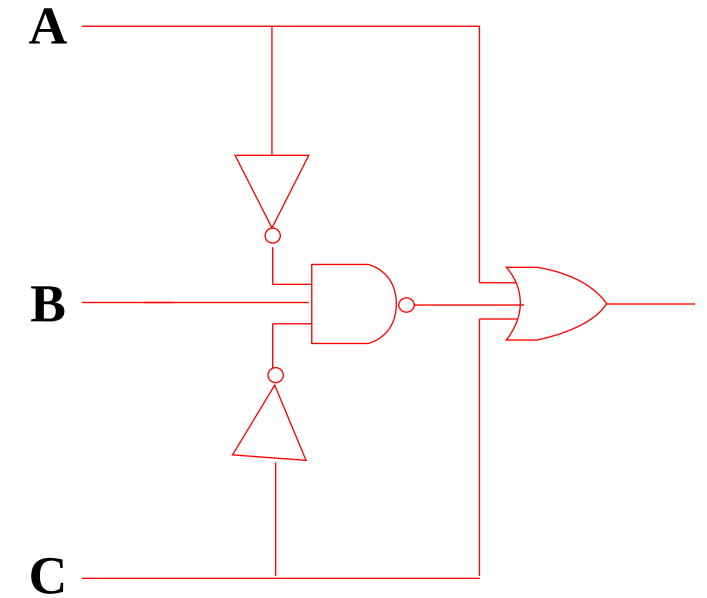
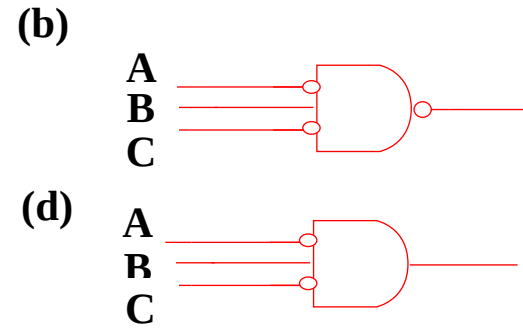
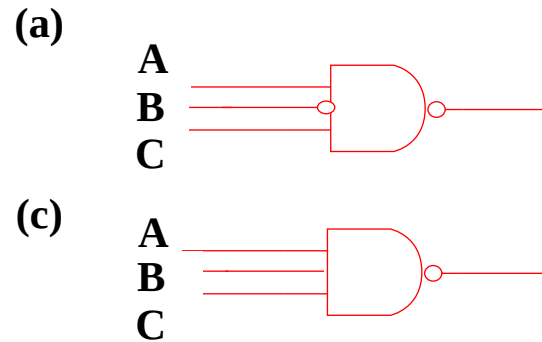
60. The output Y of the given circuit is.

- (a) 1
- (b) ZERO
- (c) X
- (d) $\bar{X}$



Use the Code :BVREDDY, to get the Maximum discount

61. Which one of the following circuit is the Minimized logic circuit for the circuit shown in the figure?



Use the Code :BVREDDY, to get the Maximum discount

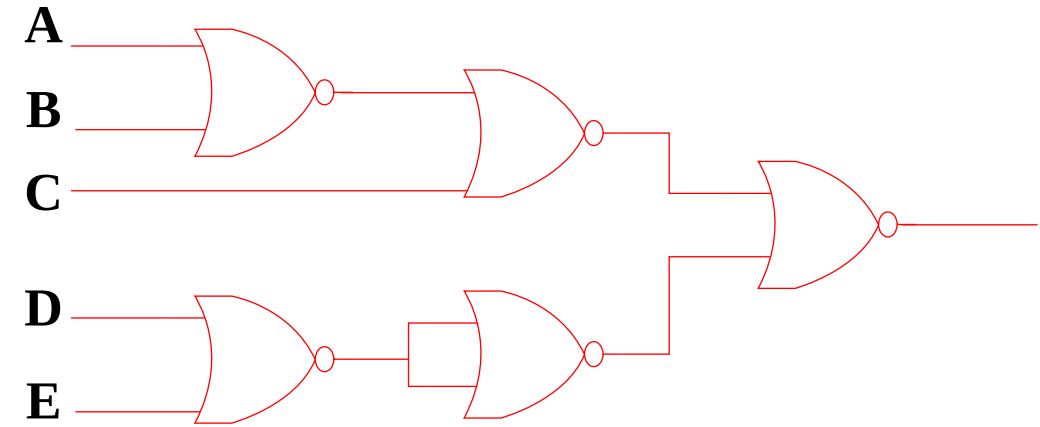
62. The circuit shown in the given figure realizes the function.

(a) $(\overline{A + B + C}) \overline{D} \overline{E}$

(b) $(\overline{A + B + C}) D \overline{E}$

(c) $(A + \overline{B + C}) \overline{D} E$

(d) $(A + B + \overline{C}) \overline{D} \overline{E}$



Use the Code :BVREDDY, to get the Maximum discount

63. The logic operations of two combinational circuits in figure-I and figure-II are.

- | | |
|------------------------|---------------|
| (a) Entirely different | (b) identical |
| (c) Complementary | (d) dual |

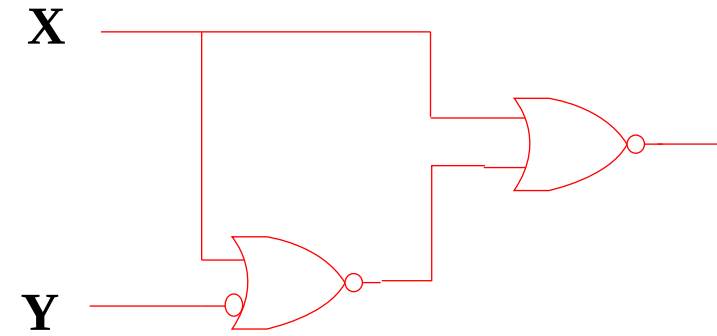


figure I

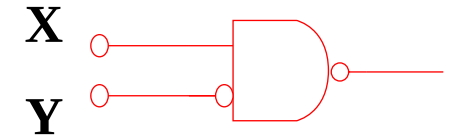


figure II

Use the Code :BVREDDY, to get the Maximum discount

64. If the output of a logic gate is '1' when all its inputs are at logic '0'. The gate is either

(a) An NAND or NOR

(b) An AND or an EX-NOR

(c) An OR or a NAND

(d) An EX-OR or an EX-NOR

Use the Code :BVREDDY, to get the Maximum discount

65. How is inversion achieved using EX-OR gate?

- (a) Giving input signal to the two input lines of the gate tied together
- (b) Giving input to one input line and logic zero to the other line.
- (c) Giving input to one input line and logic one to the other line.
- (d) Inversion cannot be achieved using EX-OR gate.

Use the Code :BVREDDY, to get the Maximum discount

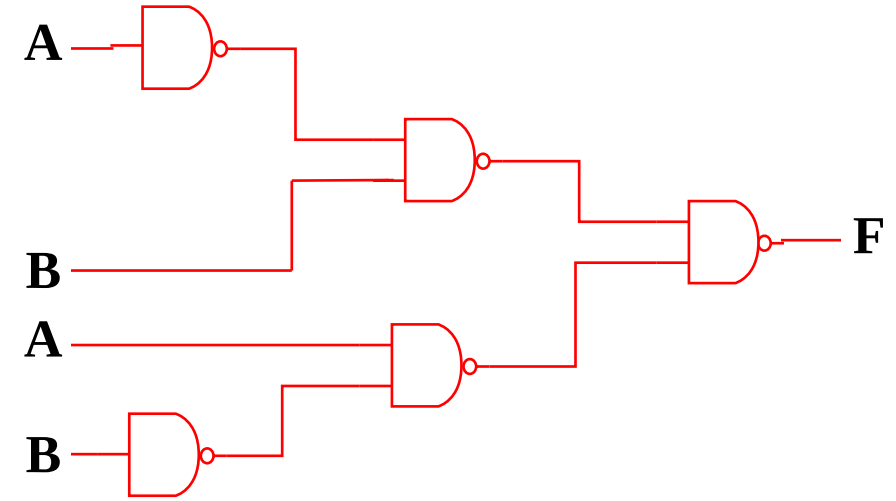
66. The circuit shown below is functionally equivalent to

(a) NOR gate

(b) OR gate

(c) EX-OR gate

(d) NAND gate



Use the Code :BVREDDY, to get the Maximum discount

67. Consider the following logic circuit:

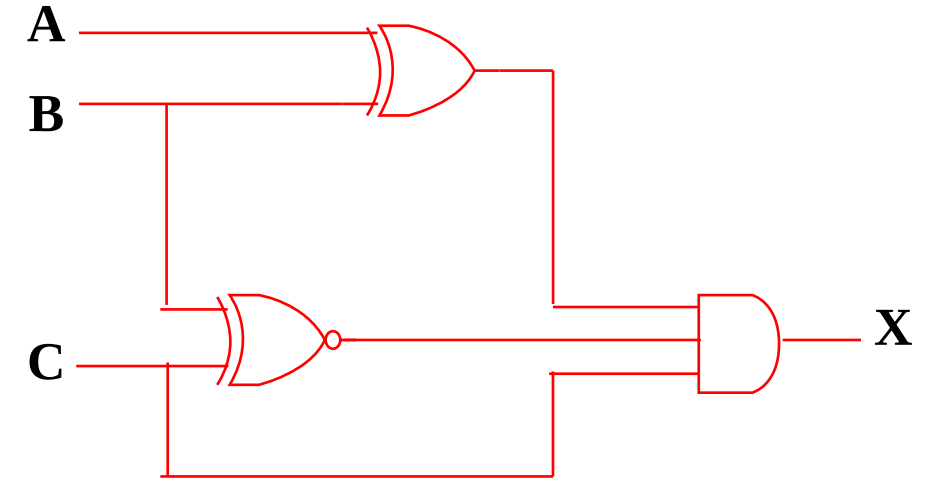
What is the required input condition (A, B, C) to make the output $X=1$, for the below logic circuit?

(a) (1, 0, 1)

(b) (0, 0, 1)

(c) (1, 1, 1)

(d) (0, 1, 1)



Use the Code :BVREDDY, to get the Maximum discount

68. The circuit shown below is to be used to implement the function $Z = f(A, B) = \bar{A} + B$

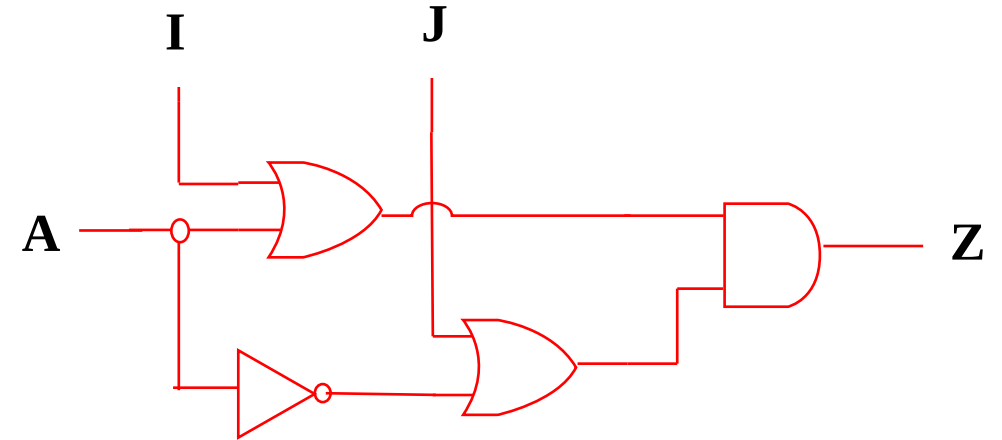
What values are to be selected for I and J?

(a) $I=0, J=B$

(b) $I=A, J=B$

(c) $I=1, J=B$

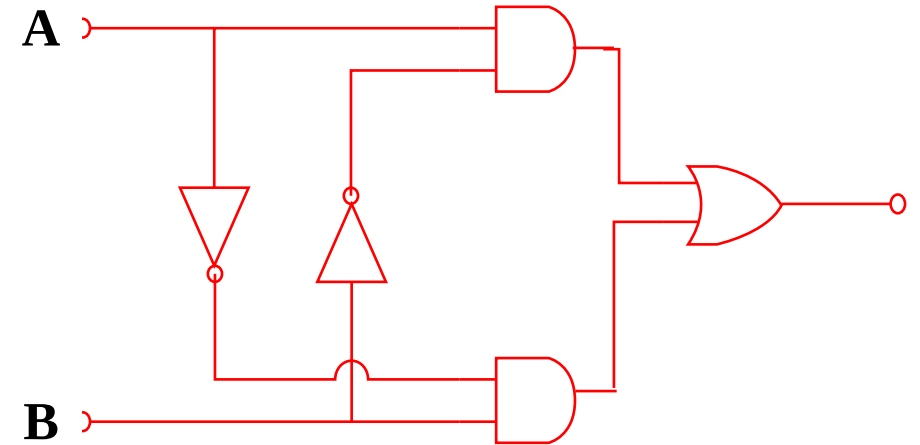
(d) $I=\bar{B}, J=0$



Use the Code :BVREDDY, to get the Maximum discount

69. Which one of the following logical operations is performed by the digital circuit shown below?

- (a) NOR
- (b) NAND
- (c) EX-OR
- (d) OR



Use the Code :BVREDDY, to get the Maximum discount

70. What is the Boolean expression $A \oplus B$ equivalent to?

(a) $A B + \bar{A} \bar{B}$

(b) $\bar{A} B + A \bar{B}$

(c) B

(d) $\bar{A}$

Use the Code :BVREDDY, to get the Maximum discount

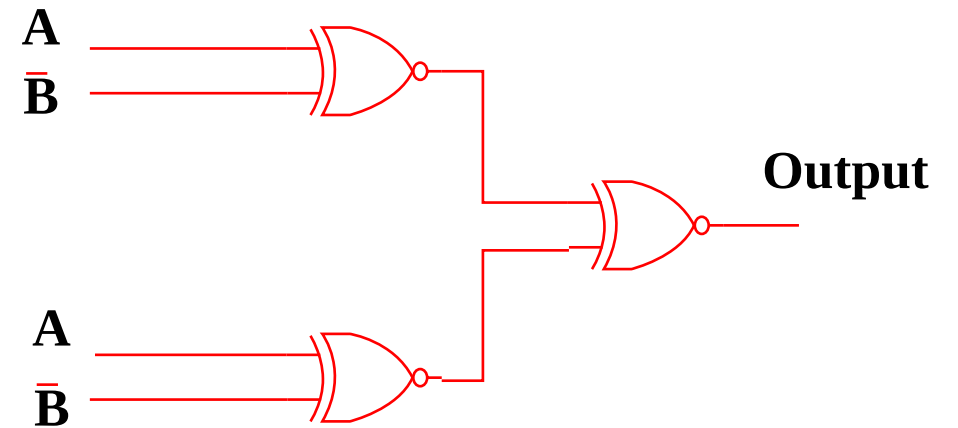
71. The output of the circuit shown in the figure is equal to

(a) 0

(b) 1

(c) $\bar{A} B + A \bar{B}$

(d) $(\overline{A * B}) * (\overline{A * B})$



Use the Code :BVREDDY, to get the Maximum discount

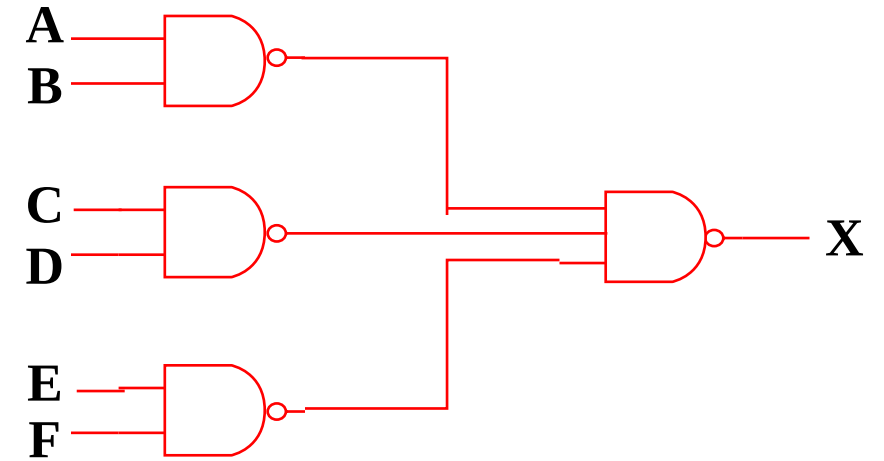
72. The output X of the above logic circuit is:

(a) $AB + CD + EF$

(b) $\overline{AB} + \overline{CD} + \overline{EF}$

(c) $(A + B)(C + D)(E + F)$

(d) $\overline{(A + B)}\overline{(C + D)}\overline{(E + F)}$



Use the Code :BVREDDY, to get the Maximum discount

73. Consider the following gate network:

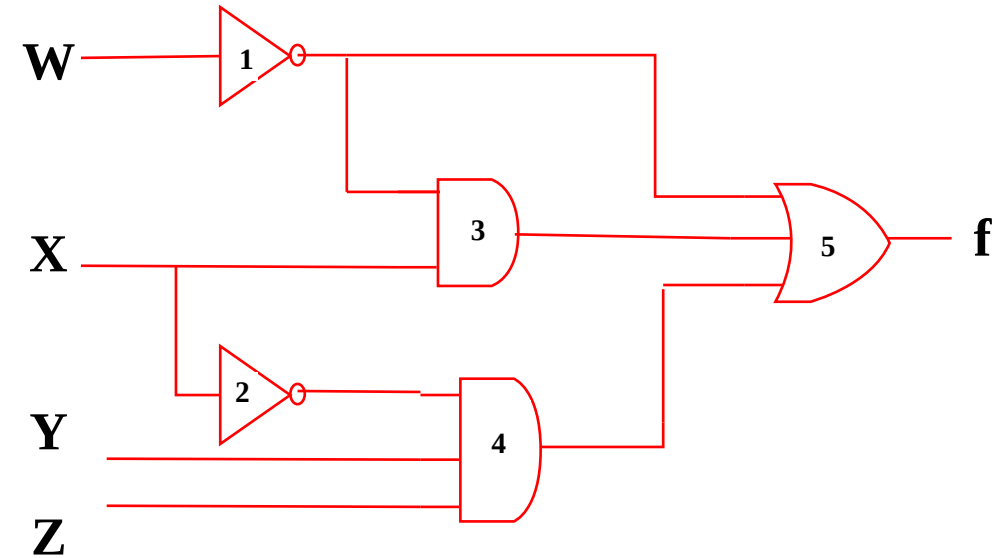
Which one of the following gates is redundant?

(a) Gate No. 1

(b) Gate No. 2

(c) Gate No. 3

(d) Gate No. 4



Use the Code :BVREDDY, to get the Maximum discount

74. The Boolean expression $X(P, Q, R) = \prod(0, 5)$ is to be realized using only two 2-input gates. Which are these gates?

- (a) AND and OR
- (b) NAND and OR
- (c) AND and XOR
- (d) OR and XOR

Use the Code :BVREDDY, to get the Maximum discount

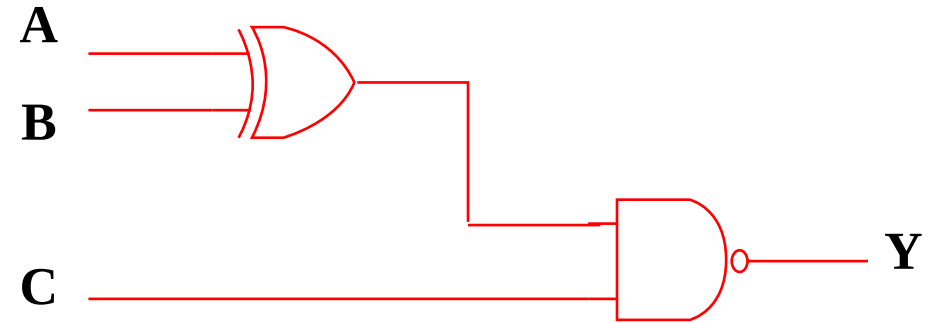
75. The Boolean expression for the output of the below logic circuit is: -

(a) $Y = \bar{A} \bar{B} + AB + \bar{C}$

(b) $Y = \bar{A} B + A\bar{B} + \bar{C}$

(c) $Y = A \oplus B + \bar{C}$

(d) $Y = AB + \bar{C}$



Use the Code :BVREDDY, to get the Maximum discount

76. According to De-Morgan's 2nd theorem

- (a) a NAND gate is always complimentary to an AND gate
- (b) a NAND gate equivalent to a bubbled NAND gate
- (c) a NAND gate equivalent to a bubbled AND gate
- (d) a NAND gate equivalent to a bubbled OR gate

Use the Code :BVREDDY, to get the Maximum discount

77. If $\bar{x}\bar{y} = 1$, then which one of the following is true?

(a) $\bar{x}y + \bar{y}x + xz = x\bar{y} + yz$

(b) $\bar{x}\bar{y}\bar{z} + xyz = xy\bar{z} + \bar{x}\bar{y}\bar{z}$

(c) $\bar{x}y + \bar{y}x = xy + \bar{x}\bar{y}$

(d) $\bar{x}yx = 1$

Use the Code :BVREDDY, to get the Maximum discount

78. There are four Boolean variables x_1, x_2, x_3, x_4 following functions are defined on sets of them:

$$f(x_3, x_2, x_1) = \sum(3, 4, 5),$$

$$g(x_4, x_3, x_2) = \sum(1, 6, 7), \quad h(x_4, x_3, x_2, x_1) = fg. \text{ then } h(x_4, x_3, x_2, x_1) \text{ is :}$$

(a) zero

(b) $\sum(3, 12, 13)$

(c) $\sum(3, 4, 5, 1, 6, 7)$

(d) $\sum(3, 12, 15),$

Use the Code :BVREDDY, to get the Maximum discount

79. The circuit given in the figure is to be used to implement the function $Z = f(A, B) = \bar{A} + B$.

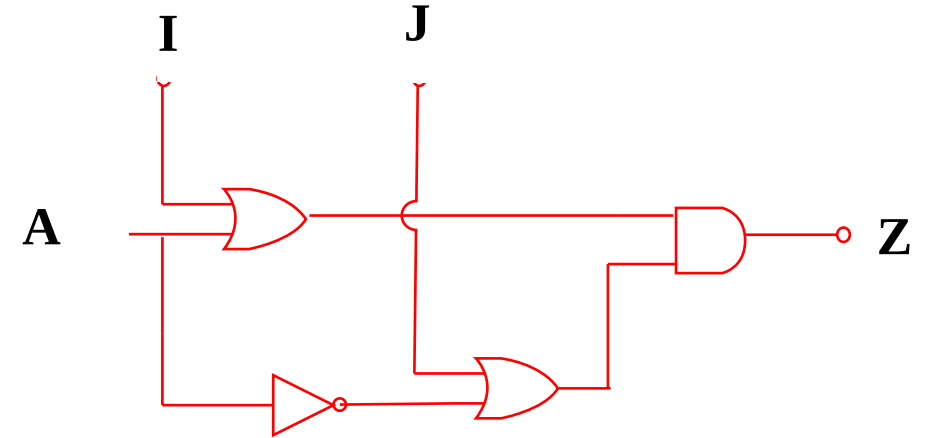
What values should be selected for I and J?

(a) $I = 0$; $J = B$

(b) $I = 1$; $J = B$

(c) $I = 1$; $J = B$

(d) $I = B$; $J = 0$



Use the Code :BVREDDY, to get the Maximum discount

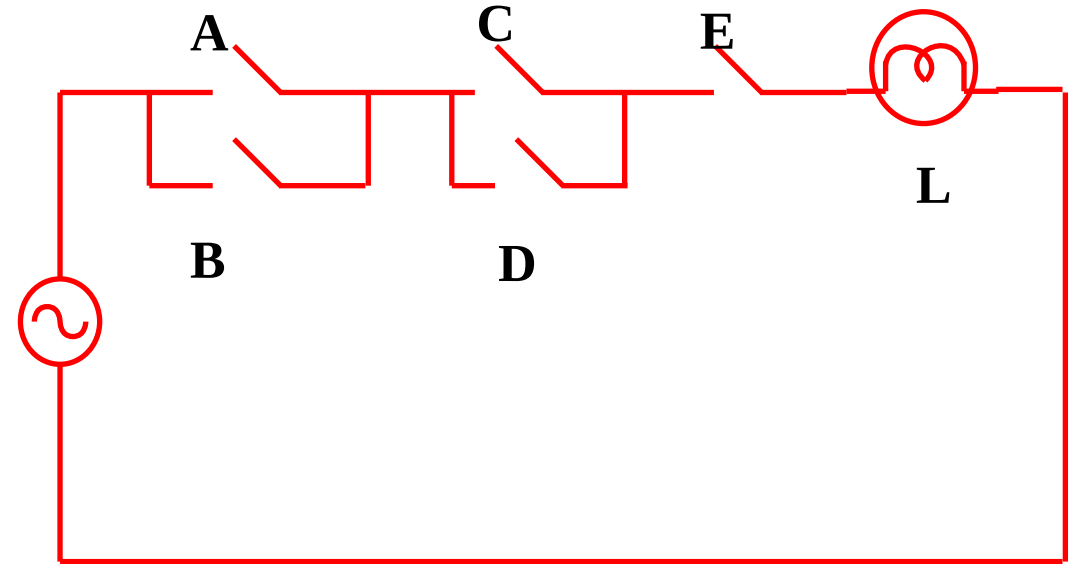
80. The switching circuit given in the figure can be expressed in binary logic notation as

(a) $L = (A + B) (C + D) E$

(b) $L = AB + CD + E$

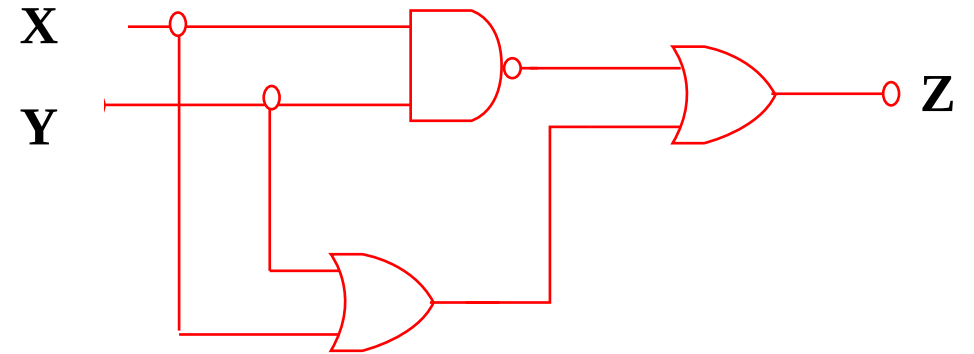
(c) $L = E + (A+B) (C+D)$

(d) $L = (AB + CD) E$



Use the Code :BVREDDY, to get the Maximum discount

81. Which of the following is the truth table of the given logic circuit.



(a)

X	Y	Z
0	0	1
0	1	1
1	0	1
1	1	1

(b)

X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	1

(c)

X	Y	Z
0	0	1
0	1	0
1	0	0
1	1	0

(d)

X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	0

82. The output of an EX-OR gate with A and B as inputs will be.

(a) $AB + \overline{A}\overline{B}$

(b) $(A + B) \overline{(A + B)}$

(c) $(A + B) \overline{AB}$

(d) $\overline{A + B} + AB$

Use the Code :BVREDDY, to get the Maximum discount

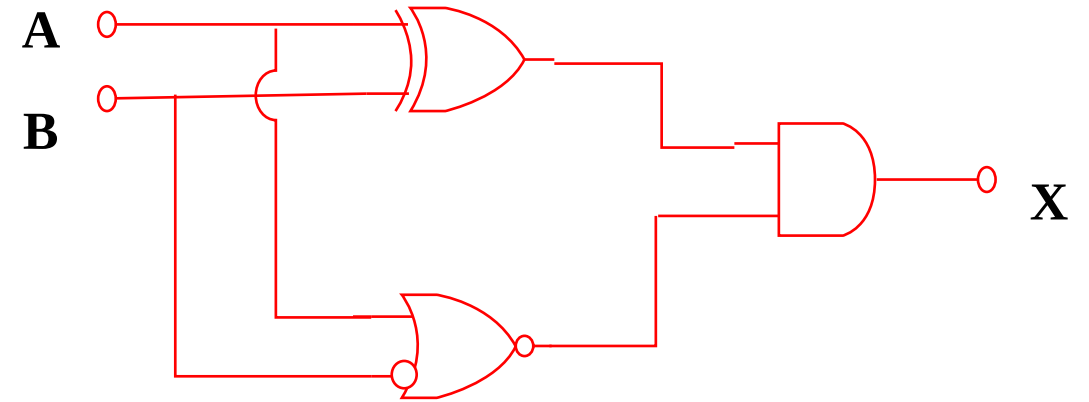
83. The output X of the circuit shown in the figure will be.

(a) AB

(b) $\bar{A}B$

(c) $A\bar{B}$

(d) $\bar{A}\bar{B}$



Use the Code :BVREDDY, to get the Maximum discount

84. Consider the following statements.

- (a) A NAND gate is equivalent to an OR gate with its inputs inverted
- (b) A NOR gate is equivalent to an AND gate with its inputs inverted
- (c) A NAND gate is equivalent to an OR gate with its output inverted
- (d) A NOR gate is equivalent to an AND gate with its output inverted

Which of these statements are correct?

- (a) 1 and 2
- (b) 2 and 3
- (c) 3 and 4
- (d) 1 and 4

Use the Code :BVREDDY, to get the Maximum discount

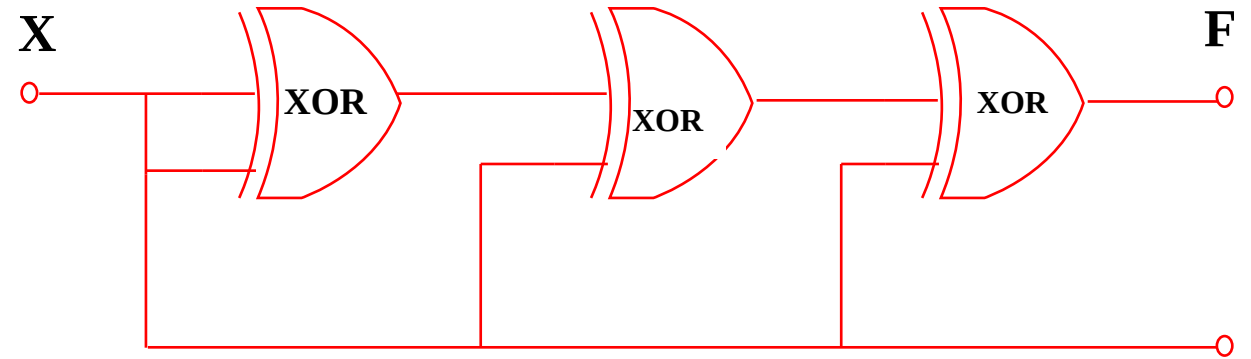
85. For the circuit shown in the given figure the output F will be.

(a) 1

(b) 0

(c) X

(d) $\overline{X}$



Use the Code :BVREDDY, to get the Maximum discount

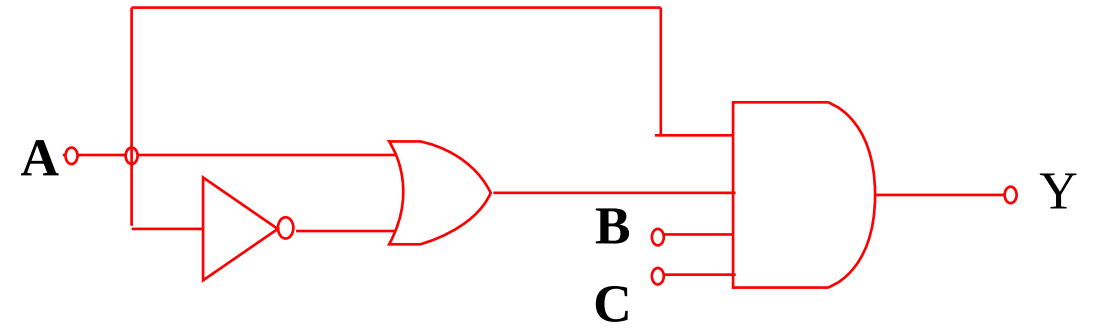
86. The Boolean expression for the output Y in the logic circuit is.

(a) $A \overline{B} C$

(b) ABC

(c) $\overline{A}BC$

(d) $\overline{A} \overline{B} \overline{C}$



Use the Code :BVREDDY, to get the Maximum discount

87. Consider the following:

Any combinational circuit can be built using.

1. NAND gate
2. NOR gate
3. EX-OR gate
4. Multiplexers

Which of these are correct?

- | | |
|----------------|----------------|
| (a) 1, 2 and 3 | (b) 1, 3 and 4 |
| (c) 2, 3 and 4 | (d) 1, 2 and 4 |

Use the Code :BVREDDY, to get the Maximum discount

88. The AND function can be realized by using only n number of NOR gates. What is n =?

(a) 2

(b) 3

(c) 4

(d) 5

Use the Code :BVREDDY, to get the Maximum discount

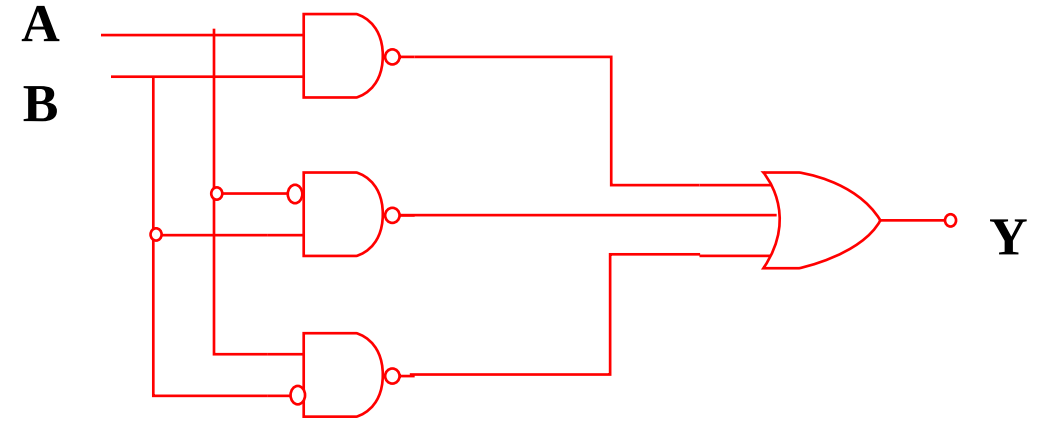
89. In the given circuit, the output Y equals which one of the following?

(a) $A+B$

(b) $\overline{A}B + A\overline{B}$

(c) AB

(d) 1



Use the Code :BVREDDY, to get the Maximum discount

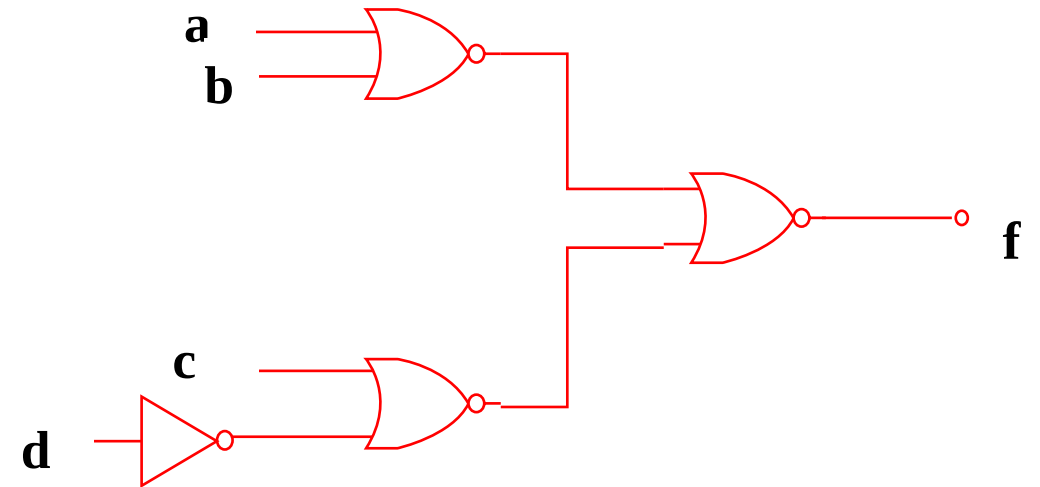
90. Which one of the following is the correct output (f) of the below circuit?

(a) $(a + b)(c + \bar{d})$

(b) $(\bar{a} + \bar{b})(c + \bar{d})$

(c) $(a + \bar{b})(c + \bar{d})$

(d) $(a + b)(\bar{c} + \bar{d})$



Use the Code :BVREDDY, to get the Maximum discount

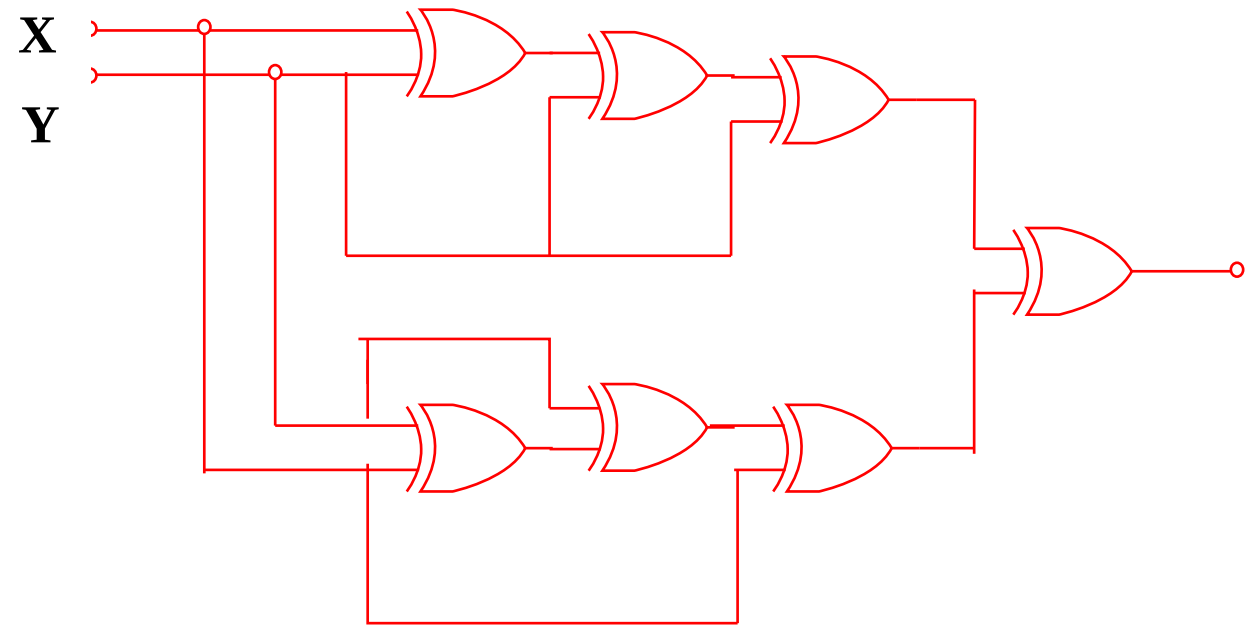
91. The circuit shown in the figure below generates the function of

(a) $X \oplus Y$

(b) 0

(c) $X\bar{Y} + YX + \bar{Y}X$

(d) $X.\bar{Y}$

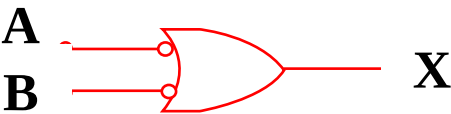


Use the Code :BVREDDY, to get the Maximum discount

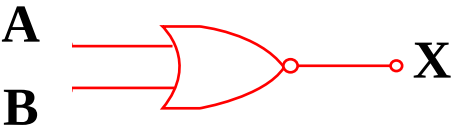
92. Match List-I with List-II and select the correct answer using the codes given below the L

List I

(a)



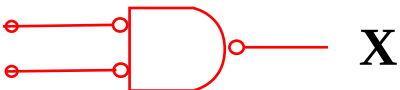
(b)



(c)



(d)



List-II

1. AB

2. $\overline{AB}$

3. $A+B$

4. $\overline{A+B}$

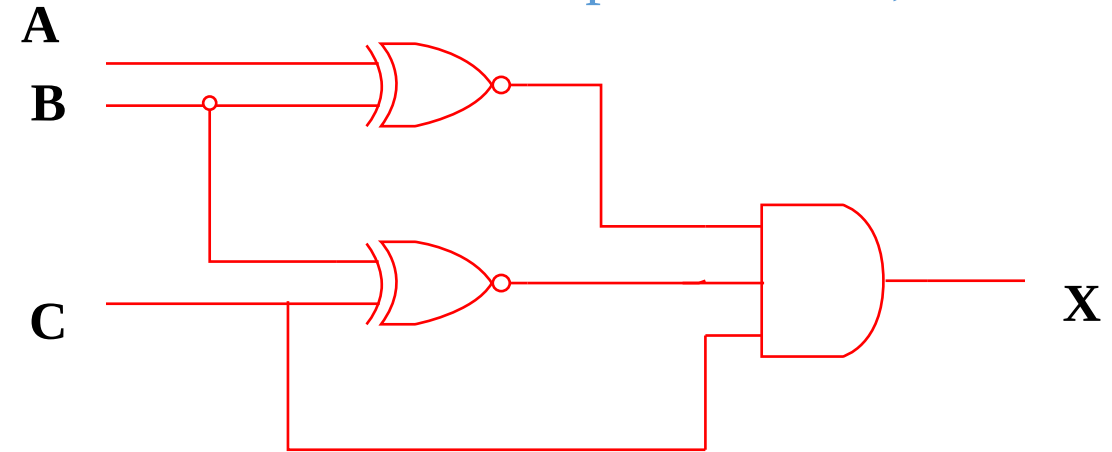
Codes

	A	B	C	D
(a) 3	1	4	2	
(b) 2	1	4	3	
(c) 3	4	1	3	
(d) 2	4	1	3	

Use the Code :BVREDDY, to get the Maximum discount

93. For logic circuits shown, the required inputs A, B and C to make the output X=1 are, respectively

- | | |
|----------------|----------------|
| (a) 1, 0 and 1 | (b) 0, 0 and 1 |
| (c) 1, 1 and 1 | (d) 0, 1 and 1 |



Use the Code :BVREDDY, to get the Maximum discount

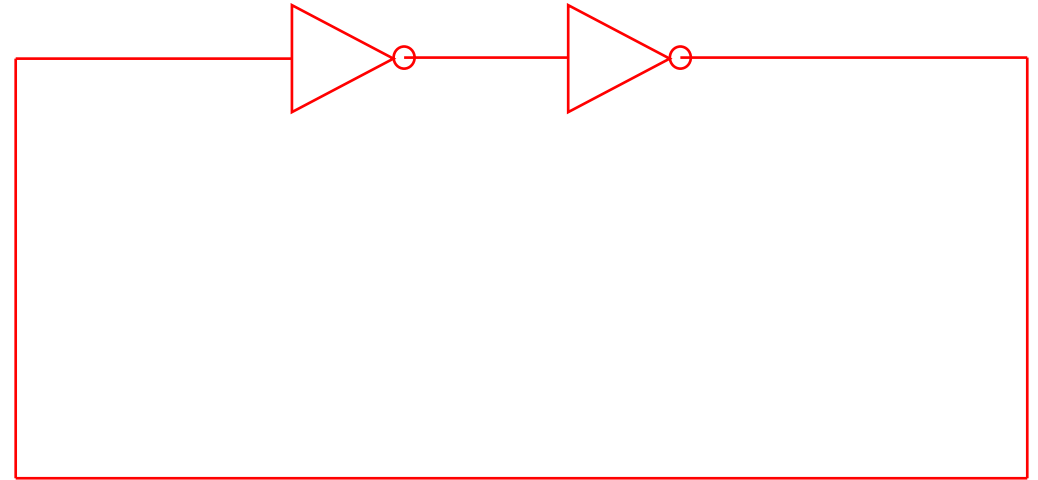
94. If the output of a logic Gate is '1' When all its inputs are at logic '0', the gate is either

- (a) A NAND and a NOR (b) An AND and an EX-NOR
(c) An OR and a NAND (d) An EX-OR and an EX-NOR

Use the Code :BVREDDY, to get the Maximum discount

95. The digital circuit using two invertors as shown in the below figure acts as?

- (a) a Bi-stable Multivibrator
- (b) an Astable Multivibrator
- (c) a Monostable Multivibrator
- (d) an oscillator spectrum



Use the Code :BVREDDY, to get the Maximum discount

96. (a) Let $*$ be a Boolean operation defined as

$$A * B = AB + \overline{A} \overline{B}$$

If $C = A * B$ then evaluate and fill in the blanks:

i. $A * A = \underline{\hspace{2cm}}$

ii. $C * A = \underline{\hspace{2cm}}$

(b) Solve the following Boolean equations for the values of A, B and C:

$$AB + \overline{A}C = 1$$

$$AC + B = 0$$

Use the Code :BVREDDY, to get the Maximum discount

97. What values of A, B, C and D satisfy the following simultaneous Boolean equations?

$$\overline{A} + AB = 0, AB = AC, AB + A\overline{C} + CD = \overline{CD}$$

(A) A=1, B=0, C=0, D=1

(B) A=1, B=1, C=0, D=0

(C) A=1, B=0, C=1, D=1

(D) A=0, B=0, C=0, D=0

Use the Code :BVREDDY, to get the Maximum discount

98. Consider the logic circuit shown in the figure below. The function f_1 , f_2 and f
(In canonical sum of products form in decimal notation) are

$$f_1(w, x, y, z) = \sum 8, 9, 10$$

$$f_2(w, x, y, z) = \sum 7, 8, 12, 13, 14, 15$$

$$f(w, x, y, z) = \sum 8, 9$$

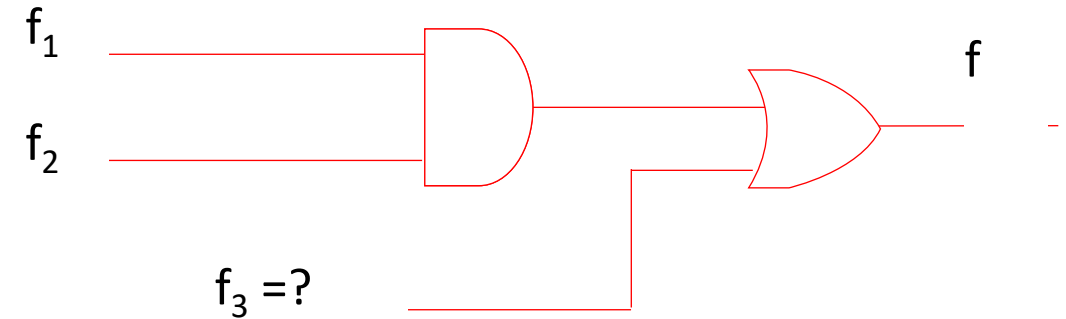
The function f_3 is

(A) $\sum 9, 10$

(B) $\sum 9$

(C) $\sum 1, 8, 9$

(D) $\sum 8, 10, 15$



Use the Code :BVREDDY, to get the Maximum discount

99. What happens when a bit-string is XOR' ed with itself n-times as shown:

$$[B \oplus (B \oplus (B \oplus (B \dots \dots \dots n \text{ times})]$$

- (A) complements when n is even
- (B) complements when n is odd
- (C) divides by 2^n always
- (D) remains unchanged when n is odd

Use the Code :BVREDDY, to get the Maximum discount

100. The simultaneous equations on the Boolean variables x, y, z and w, have the following solution for x, y, z and w, respectively:

$$x + y + z = 1$$

$$xy = 0$$

$$xz + w = 1$$

$$xy + \bar{z}\bar{w} = 0$$

(A) 0 1 0 0

(B) 1 1 0 1

(C) 1 0 1 1

(D) 1 0 0 0

Use the Code :BVREDDY, to get the Maximum discount

101. Define the connective $*$ for the Boolean variables X and Y as: $X * Y = XY + \bar{X}\bar{Y}$. Let $Z = X * Y$. Consider the following expressions P , Q and R .

$$P : X = Y * Z$$

$$Q : Y = X * Z$$

$$R : X * Y * Z = 1$$

Which of the following is TRUE?

- (A) Only P and Q are valid.
- (B) Only Q and R are valid.
- (C) Only P and R are valid.
- (D) All P , Q , R are valid.

Use the Code :BVREDDY, to get the Maximum discount

102. Let # be a binary operator defined as $X\#Y = X' + Y'$ where X and Y are Boolean variables.

Consider the following two statements.

$$(S1) \quad (P\#Q)\#R = P\#(Q\#R)$$

$$(S2) \quad Q\#R = R\#Q$$

Which of the following is/are true for the Boolean variables P, Q and R?

(A) Only S1 is true

(B) Only S2 is true

(C) Both S1 and S2 are true

(D) Neither S1 nor S2 are true

Use the Code :BVREDDY, to get the Maximum discount

103. Let, $x_1 \oplus x_2 \oplus x_3 \oplus x_4 = 0$ where x_1, x_2, x_3, x_4 are Boolean variables, and $\oplus$ is the XOR operator. Which one of the following must always be TRUE?

(A) $x_1 x_2 x_3 x_4 = 0$

(B) $x_1 x_3 + x_2 = 0$

(C) $\bar{x}_1 \oplus \bar{x}_3 = \bar{x}_2 \oplus \bar{x}_4$

(D) $x_1 + x_2 + x_3 + x_4 = 0$

Use the Code :BVREDDY, to get the Maximum discount

104. Let $\oplus$ and $\odot$ denote the Exclusive OR and Exclusive NOR operations, respectively. Which of the following is NOT CORRECT?

(A) $\overline{P \oplus Q} = P \odot Q$

(B) $\overline{P} \oplus Q = P \odot Q$

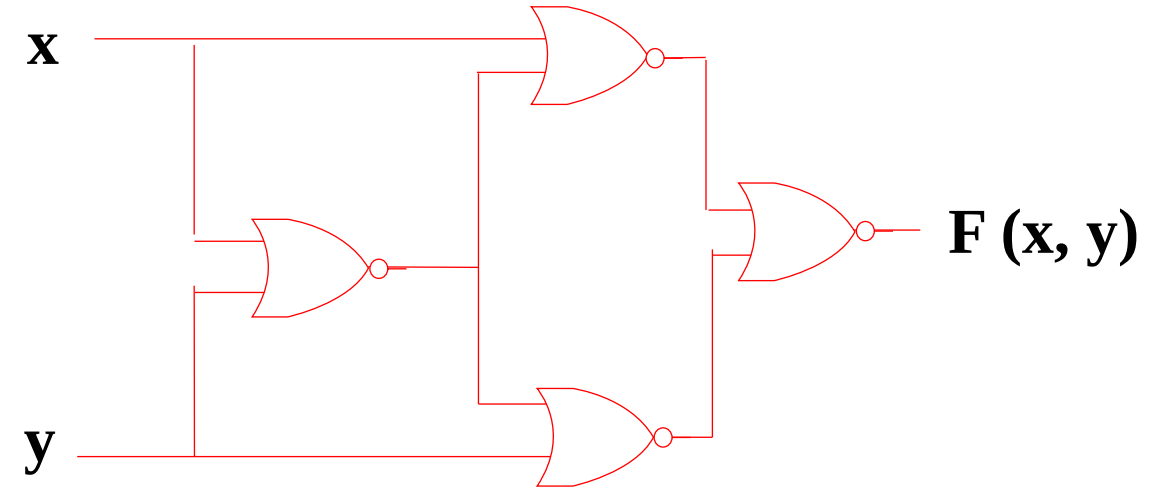
(C) $\overline{P} \oplus \overline{Q} = P \odot Q$

(D) $(P \oplus \overline{P}) \oplus Q = (P \odot \overline{P}) \odot Q$

Use the Code :BVREDDY, to get the Maximum discount

105. Identify the logic function performed by the circuit shown

- (a) exclusive OR
- (b) exclusive NOR
- (c) NAND
- (d) NOR



Use the Code :BVREDDY, to get the Maximum discount

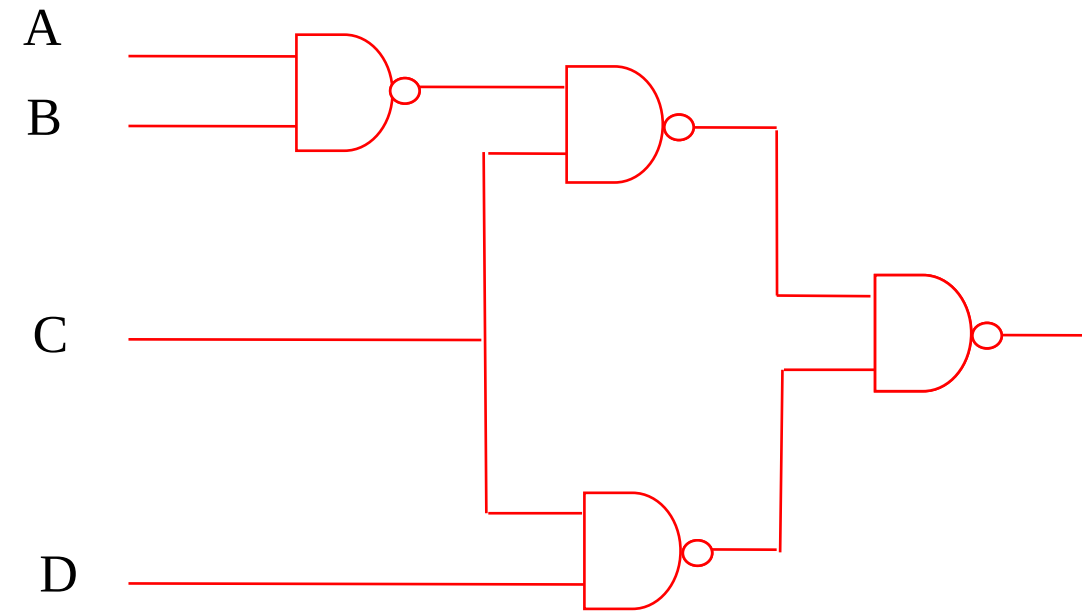
106. The logic expression for the output of the circuit shown in figure below is:

(a) $\bar{A}C + \bar{B}C + CD$

(b) $A\bar{C} + B\bar{C} + \bar{C}D$

(c) $ABC + \bar{C}\bar{D}$

(d) $\bar{A}\bar{B} + \bar{B}\bar{C} + \bar{C}\bar{D}$



Use the Code :BVREDDY, to get the Maximum discount

107. Which of the following operations is commutative but not associative?

(a) AND

(b) OR

(c) NAND

(d) EX-OR

Use the Code :BVREDDY, to get the Maximum discount

108. Which of the following expression is not equivalent to $\overline{x}$?

(a) $x \text{ NAND } x$

(b) $x \text{ NOR } x$

(c) $x \text{ NAND } 1$

(d) $x \text{ NOR } 1$

Use the Code :BVREDDY, to get the Maximum discount

109. Consider the following logic circuit whose inputs are functions f_1 , f_2 , f_3 and output is f .

Given that

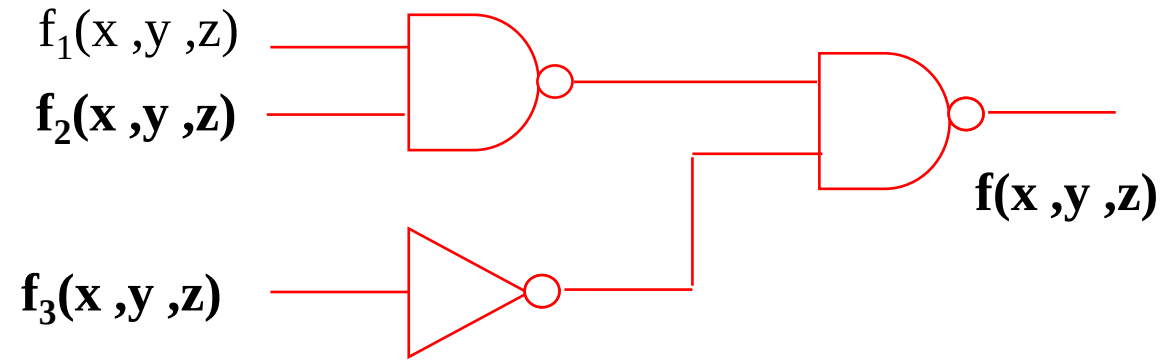
$$f_1(x, y, z) = \sum(0, 1, 3, 5)$$

$$f_2(x, y, z) = \sum(6, 7)$$

$$f(x, y, z) = \sum(1, 4, 5)$$

f_3 is

- (a) $\sum(1, 4, 5)$ (b) $\sum(6, 7)$
(c) $\sum(0, 1, 3, 5)$ (d) None of the above

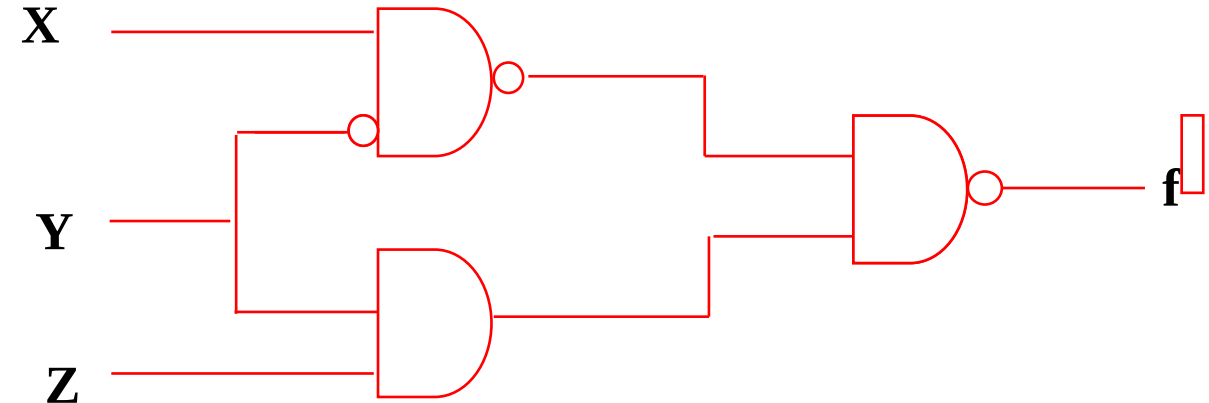


Use the Code :BVREDDY, to get the Maximum discount

110. Consider the following circuit.

Which one of the following is TRUE?

- (a) f is independent of X
- (b) f is independent of Y
- (c) f is independent of Z
- (d) None of X , Y , Z is redundant



Use the Code :BVREDDY, to get the Maximum discount

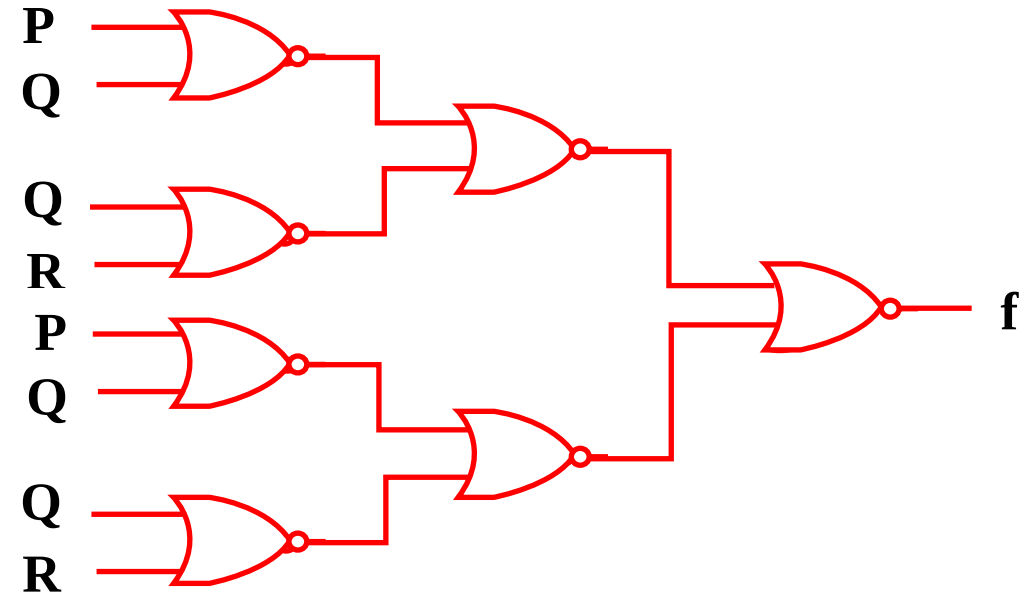
111. What is the Boolean expression for the output f of the combinational logic circuit of NOR gates given below?

(a) $\overline{Q + R}$

(b) $\overline{P + Q}$

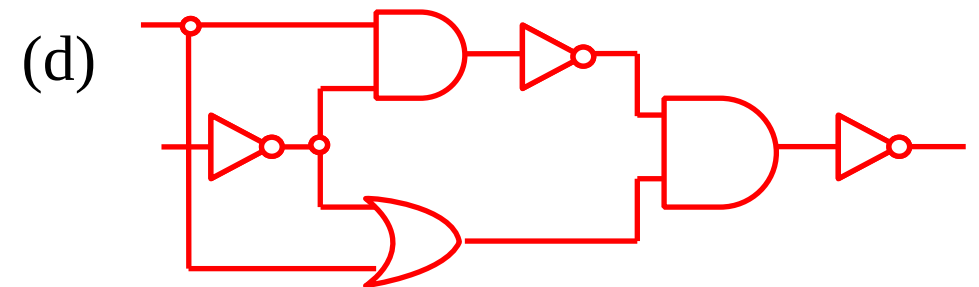
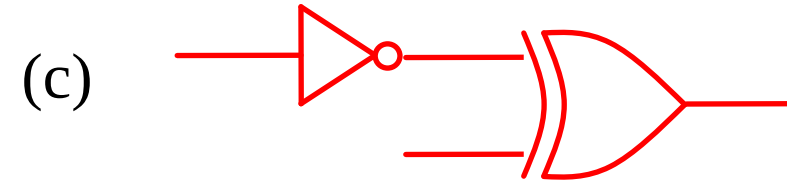
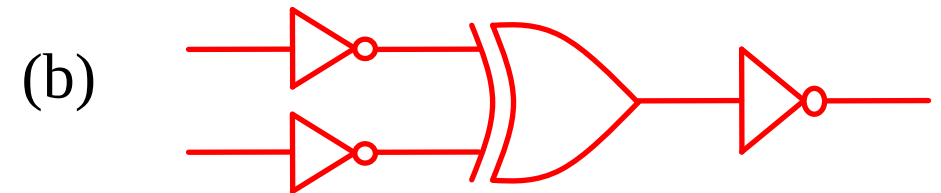
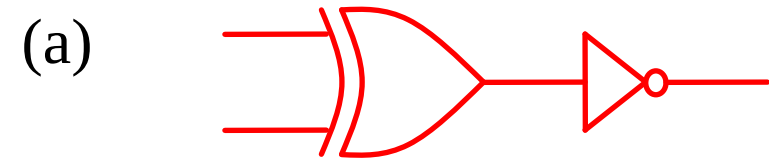
(c) $\overline{P + R}$

(d) $\overline{PR + Q}$



Use the Code :BVREDDY, to get the Maximum discount

112. Which one of the following circuits is NOT equivalent to a 2-input XNOR (exclusive NOR) gate?



Use the Code :BVREDDY, to get the Maximum discount

113. Which one of the following expressions does **NOT** represent exclusive NOR of x and y ?

(a) $xy + x'y'$

(b) $x \oplus y'$

(c) $x' \oplus y$

(d) $x' \oplus y'$

Use the Code :BVREDDY, to get the Maximum discount

114. Consider three 4-variable functions f_1 , f_2 and f_3 which are expressed in sum- of- minterms as

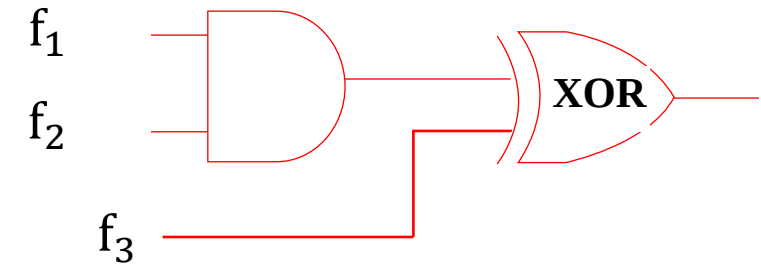
$$f_1 = \sum(0,2,5,8,14)$$

$$f_2 = \sum(2,3,6,8,14,15)$$

$$f_3 = \sum(2,7,11,14)$$

For the following circuit with one AND

gate and one XOR gate, the output function f can be expressed as:



(a) $\sum(7,8,11)$

(b) $\sum(2,14)$

(c) $\sum(2,7,8,11,14)$

(d) $\sum(0,2,3,5,6,7,8,11,14,15)$

Use the Code :BVREDDY, to get the Maximum discount

115. A 3-input majority gate is defined by the logic function $M(a, b, c) = ab + bc + ca$. Which one of the following gates is represented by the function $M(\overline{M(a, b, c)}, M(a, b, \bar{c}), c)$?

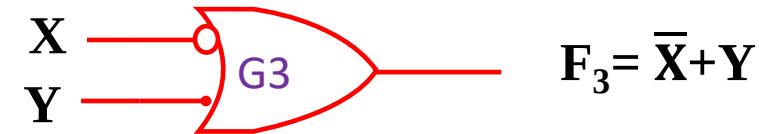
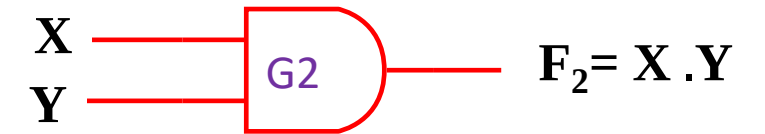
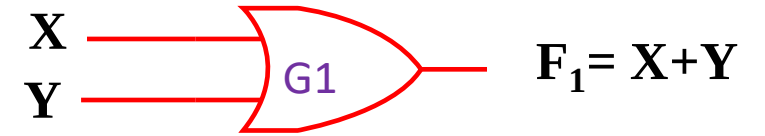
- | | |
|-----------------------|-----------------------|
| (a) 3-input NAND gate | (b) 3-input XOR gate |
| (c) 3-input NOR gate | (d) 3-input XNOR gate |

Use the Code :BVREDDY, to get the Maximum discount

116. A universal logic gate can implement any Boolean function by connecting sufficient number of them appropriately. Three gates are shown.

Which one of the following statements is TRUE?

- (a) Gate 1 is a universal gate
- (b) Gate 2 is a universal gate
- (c) Gate 3 is a universal gate
- (d) None of the gates shown is a universal gate



Use the Code :BVREDDY, to get the Maximum discount

117. A Boolean function f of two variables x and y is defined as follows:

$$f(0,0) = f(0,1) = f(1,1) = 1; f(1,0) = 0$$

Assuming complements of x and y are not available, a minimum cost solution for realizing f using only 2-input NOR gates and 2-input OR gates (each having unit cost) would have a total cost of

(a) 1 unit

(b) 4 units

(c) 3 units

(d) 2 units

Use the Code :BVREDDY, to get the Maximum discount

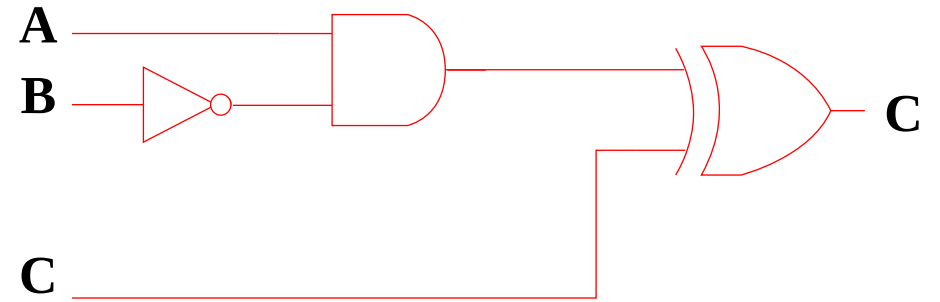
118. The complete set of only those Logic Gates designated as Universal Gates is

- (a) NOT, OR and AND gate
- (b) XNOR, NOR and NAND gate
- (c) NOR and NAND gate
- (d) XOR, NOR and NAND gate

Use the Code :BVREDDY, to get the Maximum discount

119. All the logic gates shown in the figure have a propagation delay of 20 ns.

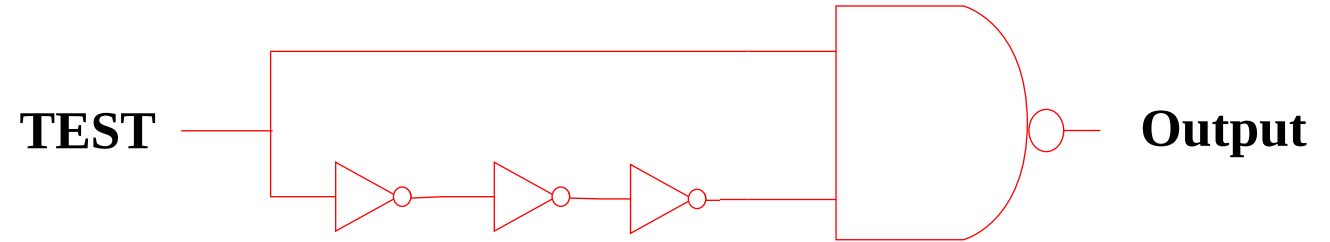
Let $A = C = 0$ and $B=1$ until time $t = 0$. At $t= 0$, all the inputs flip (i.e., $A = C = 1$ and $B = 0$) and remain in that state. For $t >0$, output $Z= 1$ for a duration (in ns) of



Use the Code :BVREDDY, to get the Maximum discount

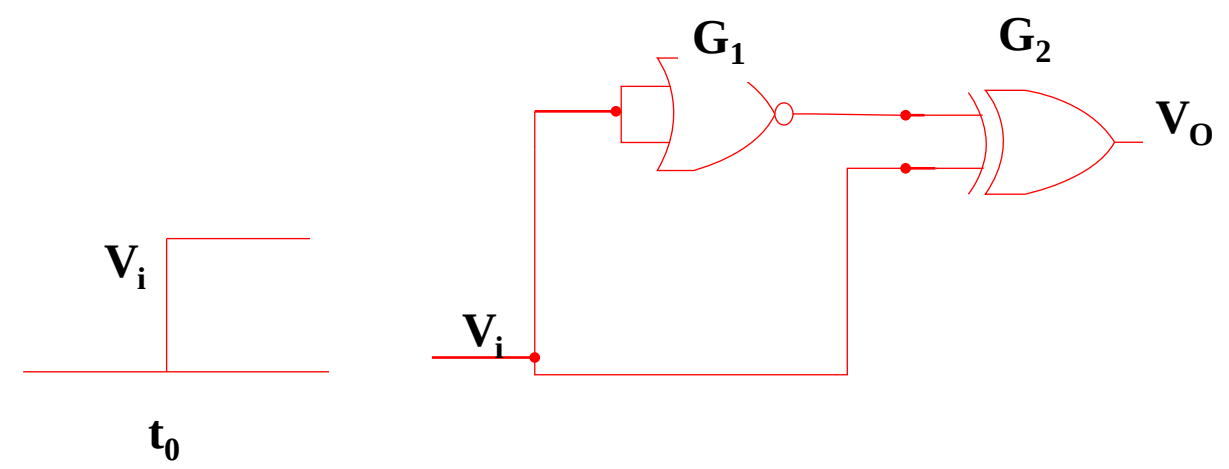
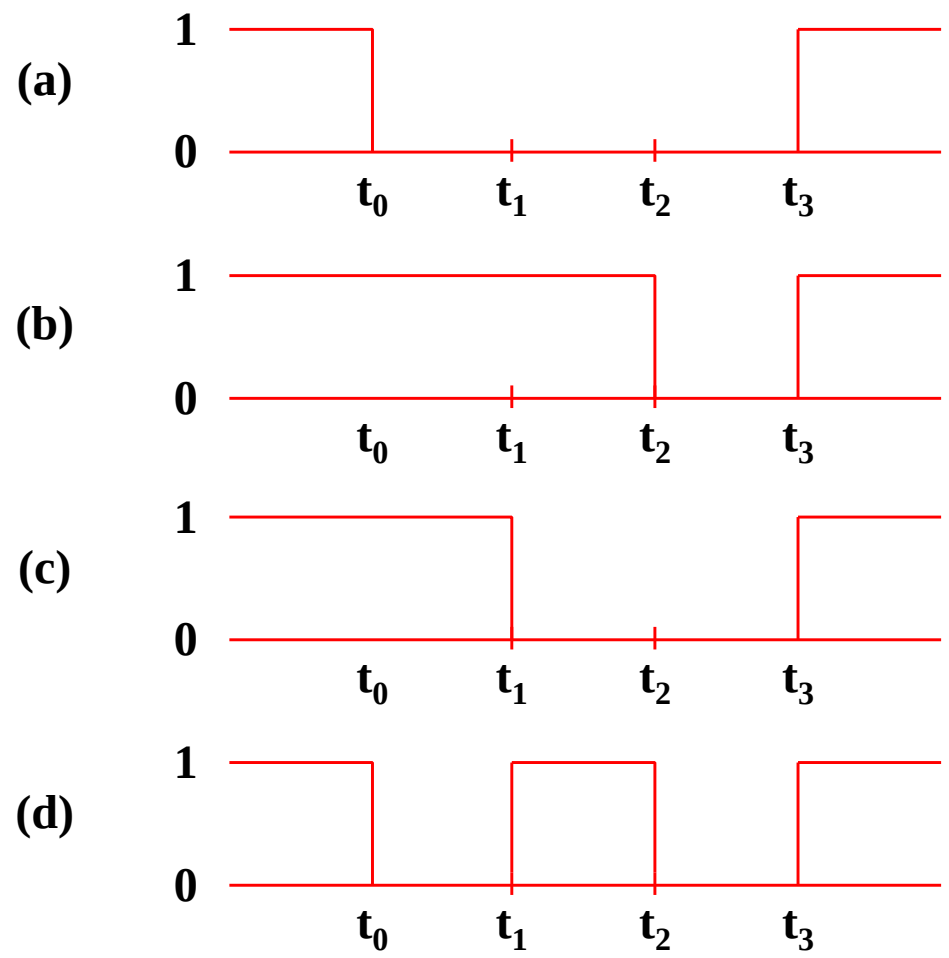
120. Consider the logic circuit with input signal TEST shown in the figure. All gates in the figure shown have identical non-zero delay. The signal TEST which was at logic LOW is switched to logic HIGH and maintained at logic HIGH. The output

- (a) stays HIGH throughout
- (b) stays LOW throughout
- (c) pulses from LOW to HIGH to LOW
- (d) pulses from HIGH to LOW to HIGH



Use the Code :BVREDDY, to get the Maximum discount

121. The gates G_1 and G_2 in figure have propagation delays of 10nsec and 20nsec respectively. If the input V_i makes an abrupt change from logic 0 to 1 at time $t = t_0$ then the output waveform V_o is



122. Which one of the following is equivalent to AND – OR realization

(a) NAND – NOR realization

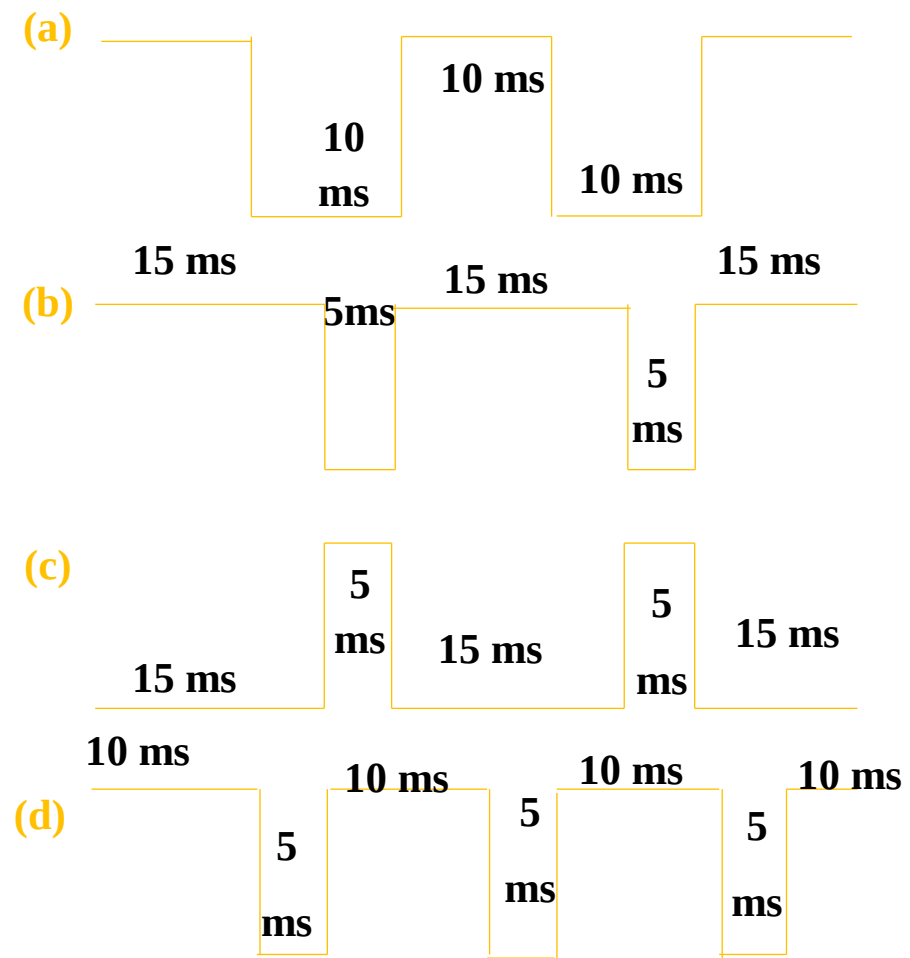
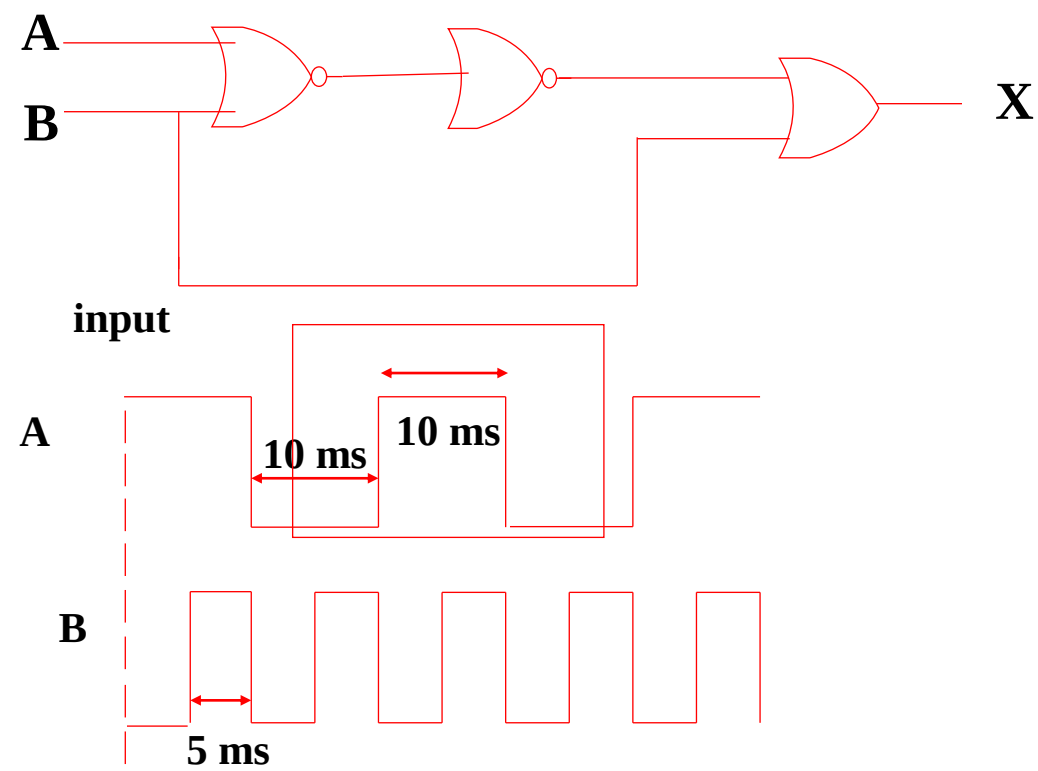
(b) NOR – NOR realization

(c) NOR – NAND realization

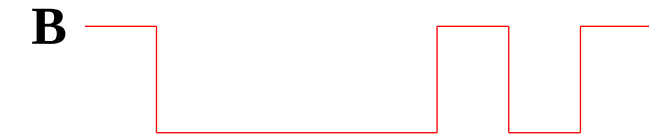
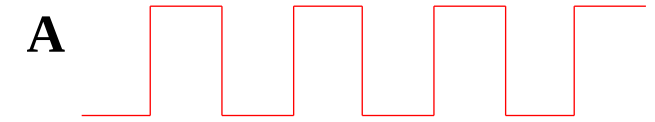
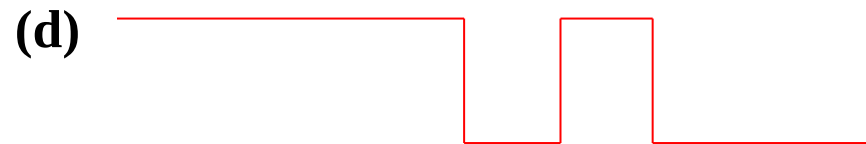
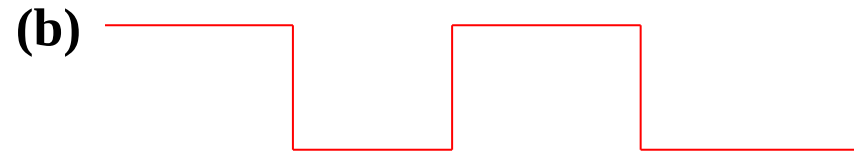
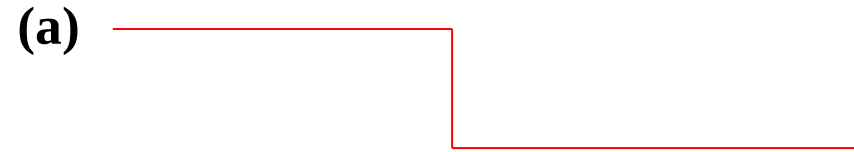
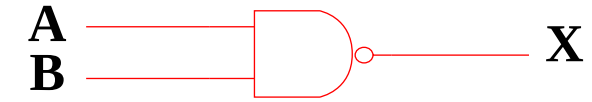
(d) NAND – NAND realization

Use the Code :BVREDDY, to get the Maximum discount

123. The output (x) waveform for the below combinational circuit for the inputs at A and B (waveform shown in the figure) will be.



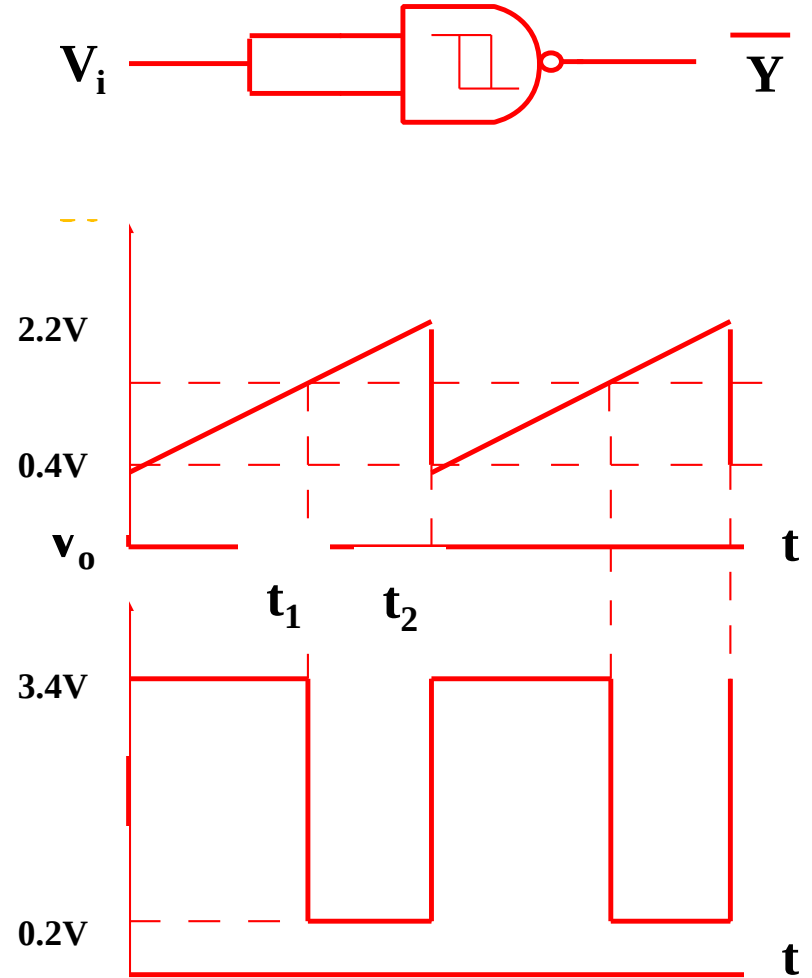
124. The given figure shows a NAND gate with input waveforms A and B. The correct output waveform X of the gate is.



Use the Code :BVREDDY, to get the Maximum discount

125. The input waveform V_i and the output waveform V_o of a Schmitt NAND as shown in the given figures. The duty cycle of the output waveform will be

- (a) 100% (b) 85.5%
(c) 72.2% (d) 25%



Use the Code :BVREDDY, to get the Maximum discount

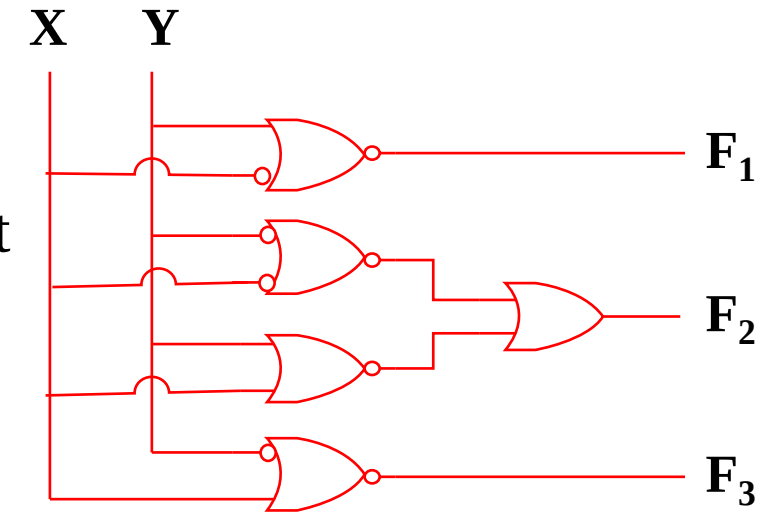
126. The circuit shown in the given figure is

(a) An Adder circuit

(b) A subtractor circuit

(c) A Comparator circuit

(d) A Parity generator circuit



Use the Code :BVREDDY, to get the Maximum discount

127. Assume that only x and y logic inputs are available, and their complements $\bar{x}$ and $\bar{y}$ are not available. What is the minimum number of 2-input NAND gates required to implement $x \oplus y$?

(a) 2

(b) 3

(c) 4

(d) 5

Use the Code :BVREDDY, to get the Maximum discount

128. The output of a two level AND-OR gate network is F. What is the output when all the gates are replaced by NOR gates?

- (a) F
- (b) $\overline{F}$
- (c) F^D
- (d) $\overline{F}^D$

Use the Code :BVREDDY, to get the Maximum discount

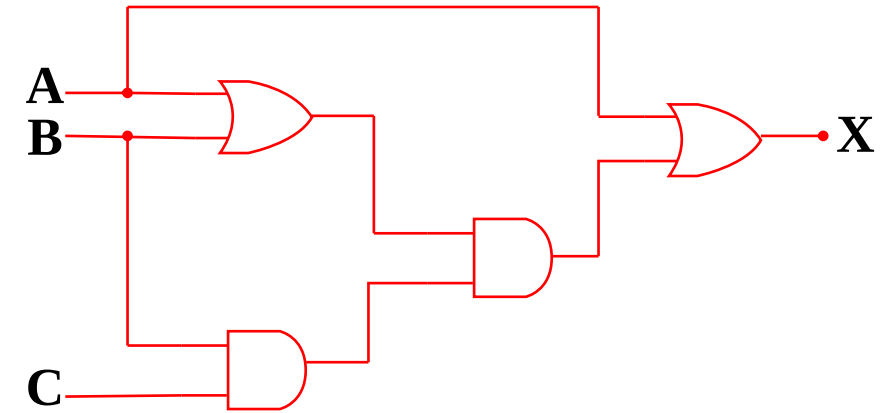
129. For the logic circuit given below, what is the simplified Boolean function?

(a) $X = AB + C$

(b) $X = BC + A$

(c) $X = AB + AC$

(d) $X = AC + B$



Use the Code :BVREDDY, to get the Maximum discount

130. Which of the following are universal gates?

1. NAND 2. NOR 3. XOR

Select the correct answer from the codes given below:

(a) 1 and 2 only

(b) 1 and 3 only

(c) 2 and 3 only

(d) 1, 2 and 3

Use the Code :BVREDDY, to get the Maximum discount

131. In NOR-NOR Configuration, the minimum number of NOR gates needed to implement the switching function $X + X \bar{Y} + X \bar{Y} Z$ is.

(a) 5

(b) 3

(c) 2

(d) 0

Use the Code :BVREDDY, to get the Maximum discount

132. Which of the following are universal gates?

1. AND 2. NAND 3. OR 4. NOR 5. NOT

- (a) 1, 2, 3, 4 and 5 (b) 1, 3 and 4 only
(c) 2, 3 and 5 only (d) 2 and 4 only

Use the Code :BVREDDY, to get the Maximum discount

133. The logic function; $\text{Out} = ab + bc + ca$ defines:

- 1.The output of a 3-input XOR gate
- 2.The output of a 3-input majority gate
- 3.The sum output of a full adder
- 4.The carry output of a full adder

- | | |
|-------------|-------------|
| (a) 1 and 2 | (b) 2 and 3 |
| (c) 3 and 4 | (d) 2 and 4 |

Use the Code :BVREDDY, to get the Maximum discount

134. The minimum number of NAND gates required to implement $A + A\bar{B} + A\bar{B}C$ is

- (a) zero
- (b) 1
- (c) 4
- (d) 7

Use the Code :BVREDDY, to get the Maximum discount

135. The minimum number of gates required to realize the function $A B + \overline{C}$ (using NAND gates only) is

(a) 2

(b) 3

(c) 4

(d) 6

Use the Code :BVREDDY, to get the Maximum discount

136. Match List-I with List-II and select the correct answer using the codes given below the Lists:

List-I (Boolean identity)

List-II (Boolean expression)

A. $Y \cdot (Y+Z)$

1. $(X+Y)(X+Z)$

B. $Y + \bar{Y} \cdot Z$

2. (Y)

C. $Y \oplus Z$

3. $(Y+Z)$

D. $X+Y \cdot Z$

4. $(Y+Z) \cdot (\bar{Y} + \bar{Z})$

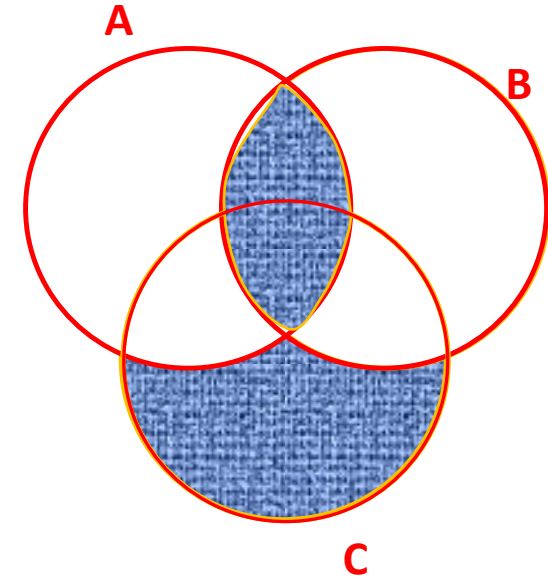
Codes:

	A	B	C	D
(a)	1	2	4	3
(b)	2	3	1	4
(c)	2	3	4	1
(d)	3	2	1	4

Use the Code :BVREDDY, to get the Maximum discount

137. The Boolean expression for the shaded area in the Venn diagram shown is.

- (a) $A + \bar{B} + C$ (b) $AB + \bar{A}BC$
(c) $A\bar{B}C + \bar{A}BC$ (d) $AB + \bar{A}\bar{B}C$



Use the Code :BVREDDY, to get the Maximum discount

138. Consider the following expressions:

1. $Y = f(A, B, C, D) = \sum (1, 2, 4, 7, 8, 11, 13, 14)$

2. $Y = f(A, B, C, D) = \sum (3, 5, 7, 10, 11, 12, 13, 14)$

3. $Y = f(A, B, C, D) = \prod (0, 3, 5, 6, 9, 10, 12, 15)$

4. $Y = f(A, B, C, D) = \prod (1, 2, 4, 6, 8, 9, 15)$

Which of these expressions are equivalents of the expression

$$Y = A \oplus B \oplus C \oplus D$$

(a) 1 and 2

(b) 1 and 4

(c) 2 and 3

(d) 1 and 3

Use the Code :BVREDDY, to get the Maximum discount

139. Consider the operations

$$f(X, Y, Z) = XY'Z + XY' + Y'Z' \text{ and } g(X, Y, Z) = X'YZ + X'YZ' + XY$$

Which one of the following is correct?

- (A) Both {f} and {g} are functionally complete
- (B) Only {f} is functionally complete
- (C) Only {g} is functionally complete
- (D) Neither {f} nor {g} is functionally complete

Use the Code :BVREDDY, to get the Maximum discount

140. Consider the Boolean operator # with the following properties:

$x \# 0 = x, x \# 1 = \bar{x}, x \# x = 0$ and $x \# \bar{x} = 1$. Then $x \# y$ is equivalent to

(A) $x\bar{y} + \bar{x}y$ (B) $x\bar{y} + \bar{x}\bar{y}$

(C) $x\bar{y} + xy$ (d) $xy + \bar{x}\bar{y}$

Use the Code :BVREDDY, to get the Maximum discount

141. Which one of the following is NOT a valid identity?

(A) $(x \oplus y) \oplus z = x \oplus (y \oplus z)$

(B) $x \oplus y = (xy + x'y')'$

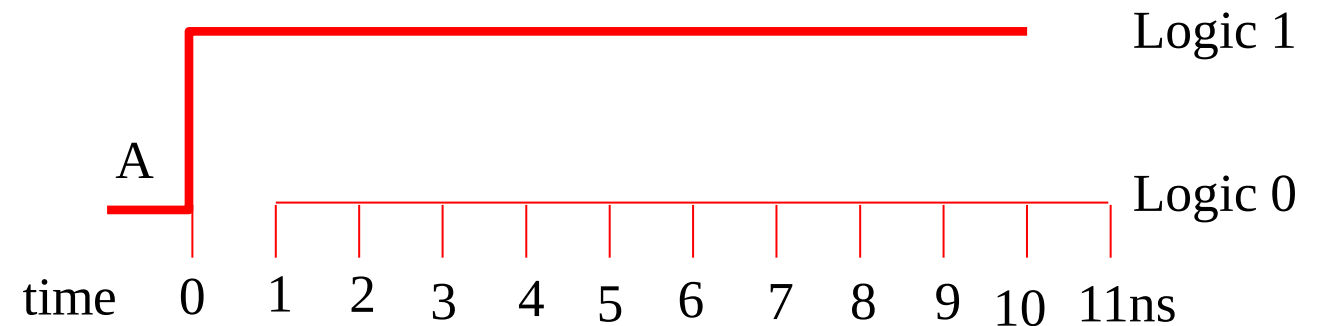
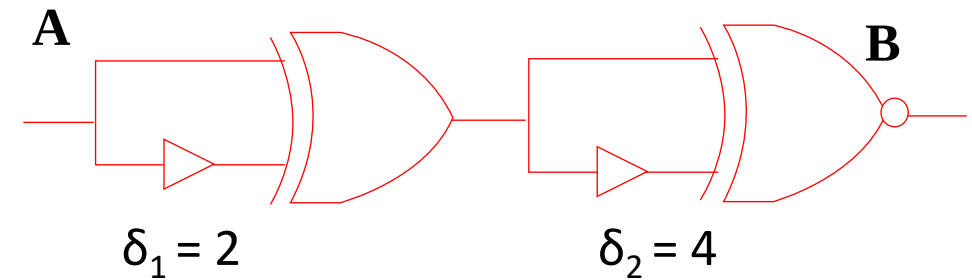
(C) $(x + y) \oplus z = x \oplus (y + z)$

(D) $(x \oplus y) = x + y, \text{ if } xy = 0$

Use the Code :BVREDDY, to get the Maximum discount

142. Consider the following circuit composed of XOR gates and non-inverting buffers. The non-inverting buffers have delays $\delta_1 = 2\text{ns}$ and $\delta_2 = 4\text{ns}$ as shown in the figure. Both XOR gates and all wires have zero delay. Assume that all gate inputs, outputs and wires are stable at logic level 0 at time 0. If the following waveform is applied at input A, how many transition(s) (change of logic levels) occur(s) at B during the interval from 0 to 10 ns?

- (a) 1
- (b) 2
- (c) 3
- (d) 4



Use the Code :BVREDDY, to get the Maximum discount

143. For a digital circuit having 4 inputs , 0 is represented by 0000, 1 by 0001, ..., 9 by 1001.

A combinational circuit is to be designed which takes these 4 bits as input and output is 1 if the digit ≥ 5 , and 0 otherwise. If only AND, OR and NOT gates may be used, what is the minimum number of gates required?

(a) 2

(b) 3

(c) 4

(d) 5

Use the Code :BVREDDY, to get the Maximum discount

144. What is the minimum number of gates required to implement the Boolean function $(AB+C)$ if we have to use only 2-input NOR gates?

(a) 2

(b) 3

(c) 4

(d) 5

Use the Code :BVREDDY, to get the Maximum discount

145. The sum of products form can be realized by using ----- logic

(a) NAND-NAND

(b) AND-OR

(c) Either 1 or 2

(d) None

Use the Code :BVREDDY, to get the Maximum discount

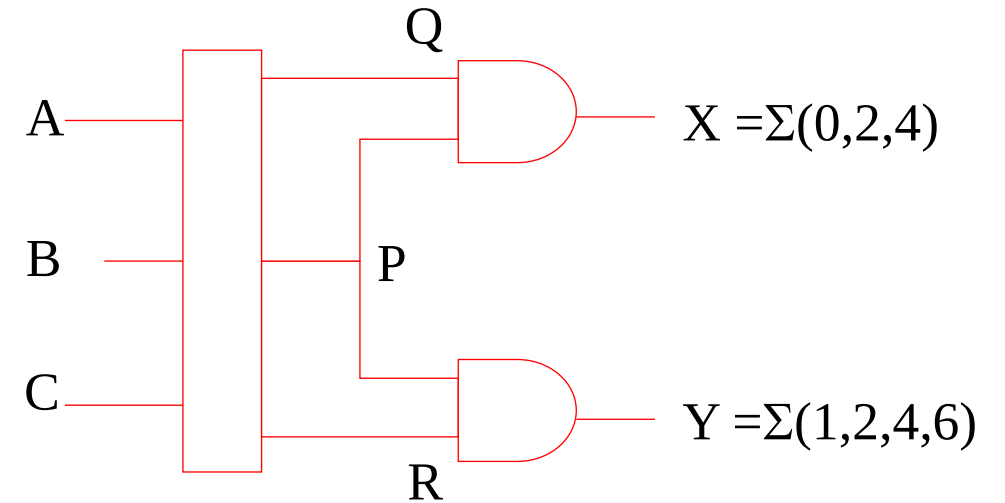
146. Given a combinational network with three inputs A, B and C, three intermediate output, P, Q and R, and two output $X = P.Q = \Sigma(0, 2, 4)$ and $Y = P.R = \Sigma(1, 2, 4, 6)$ as shown below, find the smallest function P (containing minimum number of minterms that can produce the output X and Y).

(a) $\Sigma(2, 4)$

(b) $\Sigma(0, 1, 2, 4, 6)$

(c) $\Sigma(3, 5, 7)$

(d) $\Sigma(1, 2, 6)$



Use the Code :BVREDDY, to get the Maximum discount

147. What is the maximum number of different Boolean functions involving n Boolean variables?

(A) n^2

(B) 2^n

(C) 2^{2^n}

(D) 2^{n^2}

Use the Code :BVREDDY, to get the Maximum discount

148. The dual of a Boolean function $F(X_1, X_2, \dots, X_n)$, written as F^D , is the same Expression as that of F with $+$ and swapped. F is said to be self-dual if $F = F^D$. The number of self-dual functions with n Boolean variables is

(A) $2n$

(B) $2n-1$

(C) 2^{2^n}

(D) $2^{2^{n-1}}$

Use the Code : BVREDDY , to get Maximum Discount

149. The SOP form of logical expression is most suitable for designing logic circuit using only

(a) NOR gates

(b) NAND gates

(c) AND gates

(d) X-OR gates

Use the Code : BVREDDY , to get Maximum Discount

150. The SOP (sum of products) form of a Boolean function is $\sum(0,1,3,7,11)$, where inputs are A, B, C, D (A is MSB, and D is LSB). The equivalent minimized expression of the function is

(a) $(\bar{B} + C)(\bar{A} + C)(\bar{A} + \bar{B})(\bar{C} + D)$

(b) $(\bar{B} + C)(\bar{A} + C)(\bar{A} + \bar{C})(\bar{C} + D)$

(c) $(\bar{B} + C)(\bar{A} + C)(\bar{A} + \bar{C})(\bar{C} + \bar{D})$

(d) $(\bar{B} + C)(A + \bar{B})(\bar{A} + \bar{B})(\bar{C} + D)$

Use the Code :BVREDDY, to get the Maximum discount

151. The POS form of logical expression is most suitable for designing logic circuits using only

- (a) X-OR gates
- (b) AND gates
- (c) NAND gates
- (d) NOR gates

Use the Code : BVREDDY , to get Maximum Discount

152. The sum of products form can be realized by using ----- logic

- | | |
|-------------------|------------|
| (a) NAND-NAND | (b) AND-OR |
| (c) Either 1 or 2 | (d) None |

Use the Code : BVREDDY , to get Maximum Discount

153. The Maximum number of different Boolean functions that can be implemented with '4' variable is

(a) 216

(b) 162

(c) 42

(d) 65536

Use the Code : BVREDDY , to get Maximum Discount

154. The Number of switching functions of 3 variables is

- (a) 8
- (b) 64
- (c) 128
- (d) 256

Use the Code :BVREDDY, to get the Maximum discount

155. For the Boolean expression $f = \bar{a}\bar{b}\bar{c} + \bar{a}b\bar{c} + a\bar{b}\bar{c} + abc + ab\bar{c}$ the minimized Product of Sum (POS) expression is

(a) $f = (b + \bar{c}).(a + \bar{c})$

(b) $f = (\bar{b} + c).(\bar{a} + c)$

(c) $f = (\bar{b} + c).(a + \bar{c})$

(d) $f = \bar{c} + abc$

Use the Code :BVREDDY, to get the Maximum discount

156. The Boolean expression

$$Y = \bar{A}\bar{B}\bar{C}D + \bar{A}BC\bar{D} + A\bar{B}\bar{C}D + AB\bar{C}\bar{D}$$

can be minimized to

(a) $Y = \bar{A}\bar{B}\bar{C}D + \bar{A}B\bar{C} + A\bar{C}D$

(b) $Y = \bar{A}\bar{B}\bar{C}D + BC\bar{D} + A\bar{B}\bar{C}D$

(c) $Y = \bar{A}BC\bar{D} + \bar{B}\bar{C}D + A\bar{B}\bar{C}D$

(d) $Y = \bar{A}BC\bar{D} + \bar{B}\bar{C}D + AB\bar{C}\bar{D}$

Use the Code :BVREDDY, to get the Maximum discount

158. The number of distinct Boolean expression of 4 variables is

- (a) 16
- (b) 256
- (c) 1024
- (d) 65536

Use the Code :BVREDDY, to get the Maximum discount

158. $Y = f(A, B) = \prod M(0, 1, 2, 3)$ represents (M is max term)

(a) NOR Gate

(b) NAND Gate

(c) OR Gate

(d) a situation where output is independent of input.

Use the Code :BVREDDY, to get the Maximum discount

159. With four Boolean variables how many Boolean expressions can be formed.

(a) 16

(b) 256

(c) 1024 (1k)

(d) 64k (64×1024)

Use the Code :BVREDDY, to get the Maximum discount

160. Match List-I (Boolean Logic Function) with List-II (Inverse of Function) and select the correct answer using the codes given below the lists:

List-I

A. $ab + bc + ca + abc$

B. $ab + \bar{a}\bar{b} + \bar{c}$

C. $a + bc$

D. $(\bar{a} + \bar{b} + \bar{c})(a + \bar{b} + \bar{c})(\bar{a} + \bar{b} + c)$

List-II

1. $\bar{a}(\bar{b} + \bar{c})$

2. $\bar{a}\bar{b} + \bar{b}\bar{c} + \bar{c}\bar{a}$

3. $(a \oplus b)c$

4. $abc + \bar{a}bc + ab\bar{c}$

Codes:

	A	B	C	D
(a)	3	2	1	4
(b)	2	3	1	4
(c)	3	2	4	1
(d)	2	3	4	1

Use the Code :BVREDDY, to get the Maximum discount

161. The output of a two level AND-OR gate network is F. What is the output when all the gates are replaced by NOR gates?

- (a) F
- (b) $\overline{F}$
- (c) F^D
- (d) $\overline{F}^D$

Use the Code :BVREDDY, to get the Maximum discount

162. The number of Boolean functions which can be generated with four variables is?

(a) 4

(b) 16

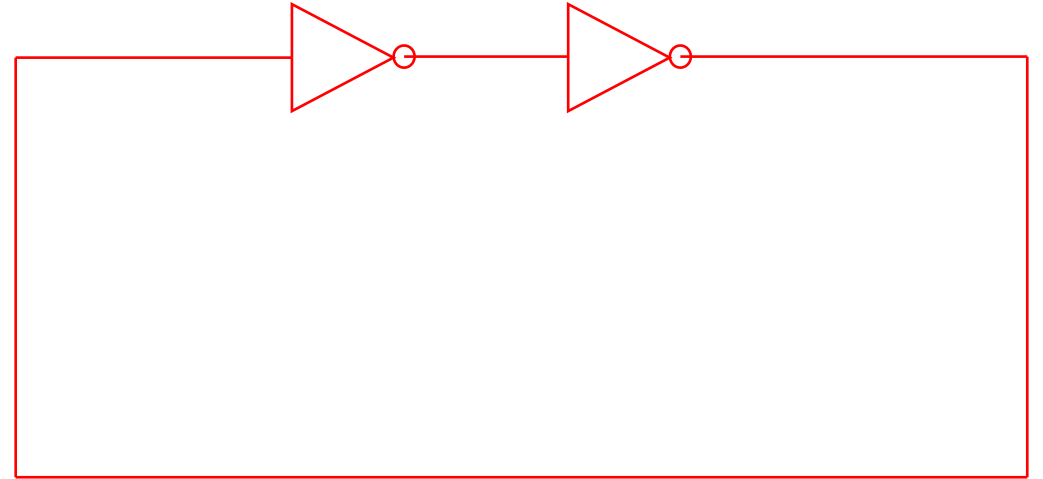
(c) 256

(d) 65,536

Use the Code :BVREDDY, to get the Maximum discount

163. The digital circuit using two invertors as shown in the below figure acts as?

- (a) a Bi-stable Multivibrator
- (b) an Astable Multivibrator
- (c) a Monostable Multivibrator
- (d) an oscillator spectrum



Use the Code :BVREDDY, to get the Maximum discount

K- Map's

1. The output expression for the Karnaugh map shown below is

(a) $B\bar{D} + BCD$

(b) $B\bar{D} + AB$

(c) $\bar{B}D + ABC$

(d) $B\bar{D} + ABC$

		C D			
		00	01	11	10
A B	00	0	0	0	0
	01	1	0	0	1
	11	1	0	1	1
	10	0	0	0	0

Use the Code : BVREDDY , to get Maximum Discount

2. Karnaugh map is used to

- (a) minimize the number of flip-flops in a digital circuit
- (b) minimize the number of gates only in a digital circuit
- (c) minimize the number of gates and fan in a digital circuit
- (d) Design gates

Use the Code : BVREDDY , to get Maximum Discount

3. The output expression for the Karnaugh map shown below is

(a) $A + \bar{B}$

(b) $A + \bar{C}$

(c) $\bar{A} + \bar{C}$

(d) $\bar{A} + C$

A \ B C	B C			
	00	01	11	10
00	1	0	0	1
01	1	1	1	1

Use the Code : BVREDDY , to get Maximum Discount

4. If $F(A, B, C, D) = \prod M(0, 1, 3, 4, 5, 7, 9, 11, 12, 13, 14, 15)$ is a maxterm representation of a Boolean function $f(A, B, C, D)$ where A is the MSB and D is the LSB. The equivalent minimized representation of this function is

(a) $(A + \bar{C} + D)(\bar{A} + B + D)$

(b) $A\bar{C}D + \bar{A}BD$

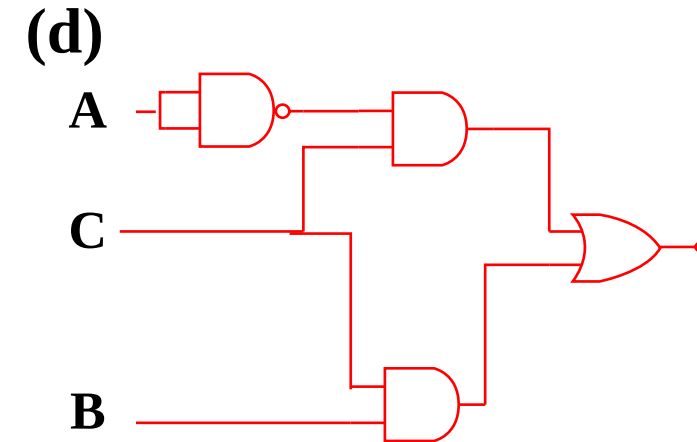
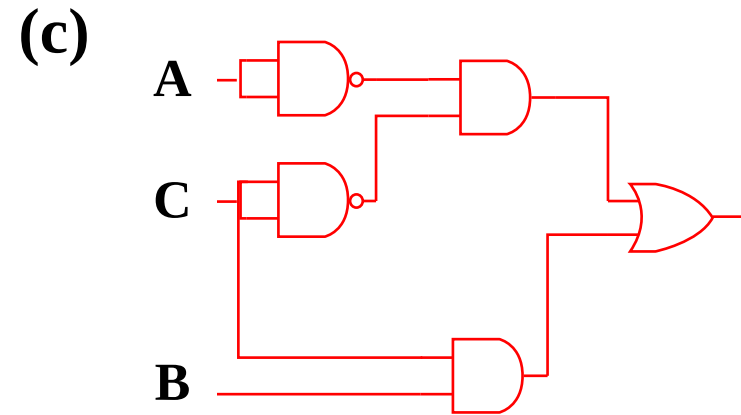
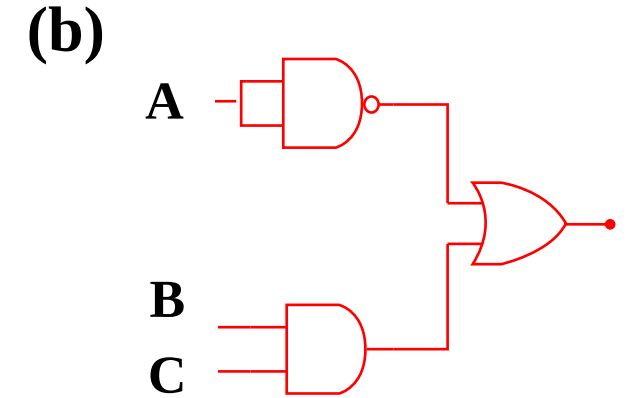
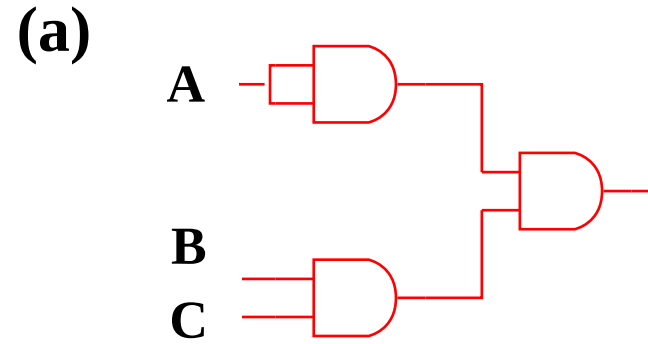
(c) $\bar{A}C\bar{D} + A\bar{B}\bar{D}$

(d) $(B + \bar{C} + D)(A + \bar{B} + \bar{C} + D)(\bar{A} + B + C + D)$

Use the Code : BVREDDY , to get Maximum Discount

5. Which of the following logic circuits is a realization of the function F whose Karnaugh map is shown in figure?

A B					
C		00	01	11	10
	0	1	1		
	1		1	1	



Use the Code : BVREDDY , to get Maximum Discount

6. Statement for Linked Answer Questions

The following Karnaugh map represents a function F

[i] A minimized form of the function F is

(a) $F = \bar{X}Y + YZ$ (b) $F = \bar{X}\bar{Y} + YZ$

(c) $F = \bar{X}\bar{Y} + Y\bar{Z}$ (d) $F = \bar{X}\bar{Y} + \bar{Y}Z$

X \ Y Z	Y Z			
	00	01	11	10
0	1	1	1	0
1	0	0	1	0

Use the Code : BVREDDY , to get Maximum Discount

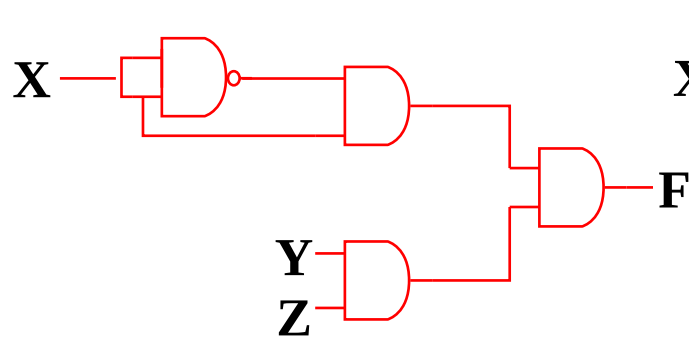
7. Statement for Linked Answer Questions

The following Karnaugh map represents a function F

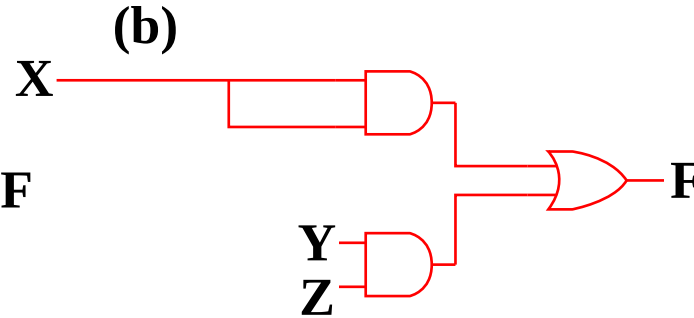
[ii] Which of the following circuits is a realization of the above function F?

		Y Z			
		00	01	11	10
X	0	1	1	1	0
	1	0	0	1	0

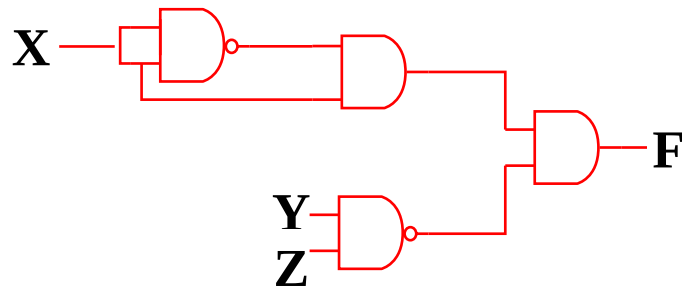
(a)



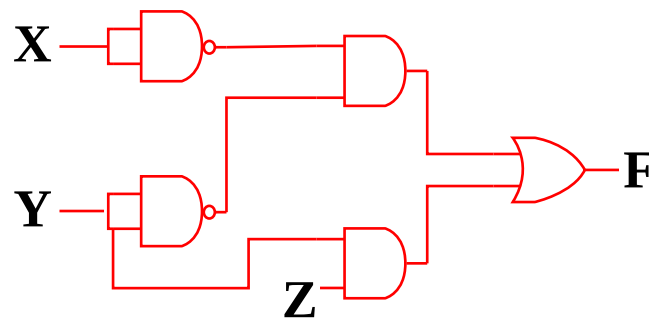
(b)



(c)



(d)



Use the Code : BVREDDY , to get Maximum Discount

8. The Boolean expression for the truth table shown is:

- (a) $B(A + C)(\bar{A} + \bar{C})$
- (b) $B(A + \bar{C})(\bar{A} + C)$
- (c) $\bar{B}(A + \bar{C})(\bar{A} + C)$
- (d) $B(A + C)(\bar{A} + \bar{C})$

A	B	C	D
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Use the Code : BVREDDY , to get Maximum Discount

9. The Karnaugh map for a four variable Boolean function is given in figure. The correct Boolean sum of product is

- (a) $PQRS + \bar{Q}S$ (b) $\bar{P}QR\bar{S} + \bar{Q}S$
 (c) $PQR + Q\bar{S}$ (d) $PQRS + \bar{Q}$

		P Q			
		00	01	11	10
R S	00	0	0	0	0
	01	1	0	0	1
	11	1	0	0	1
	10	0	1	0	0

Use the Code : BVREDDY , to get Maximum Discount

10. The minimal sum-of-products expression for the logic function f represented by the given Karnaugh map is.

(a) $QS + \overline{P}RS + PQR + \overline{P}RS + \overline{P}Q\overline{R}$

(b) $\overline{Q}\overline{S} + \overline{P}R\overline{S} + \overline{P}\overline{Q}R + \overline{P}R\overline{S} + P\overline{Q}R$

(c) $\overline{P}R\overline{S} + \overline{P}\overline{Q}\overline{R} + \overline{P}R\overline{S} + P\overline{Q}R$

(d) $\overline{P}R\overline{S} + PQR + \overline{P}RS + \overline{P}QR$

		PQ			
		00	01	11	10
RS	00	0	1	0	0
	01	0	1	1	1
	11	1	1	1	0
	10	0	0	1	0

Use the Code : BVREDDY , to get Maximum Discount

11. The minimal sum of products form of the Boolean expression

$$Y = \overline{P}\overline{Q}\overline{R}\overline{S} + P\overline{Q}\overline{R}\overline{S} + P\overline{Q}\overline{R}S + P\overline{Q}R\overline{S} + P\overline{Q}RS + \overline{P}\overline{Q}R\overline{S}$$

(a) $Y = P\overline{Q} + \overline{Q}\overline{S}$

(b) $Y = P\overline{Q} + \overline{Q}R\overline{S}$

(c) $Y = PQ + \overline{Q}\overline{R}\overline{S}$

(d) $Y = \overline{Q}\overline{S} + P\overline{Q}R$

Use the Code : BVREDDY , to get Maximum Discount

12. Min-term (Sum of Products) expression for a Boolean function is given as follows

$$f(A, B, C) = \sum m(0,1,2,3,5,6)$$

(a) $A + (B \oplus C)$

(b) $A \oplus (B + C)$

(c) $\bar{A} + (B \oplus C)$

(d) $\overline{ABC}$

Use the Code : BVREDDY , to get Maximum Discount

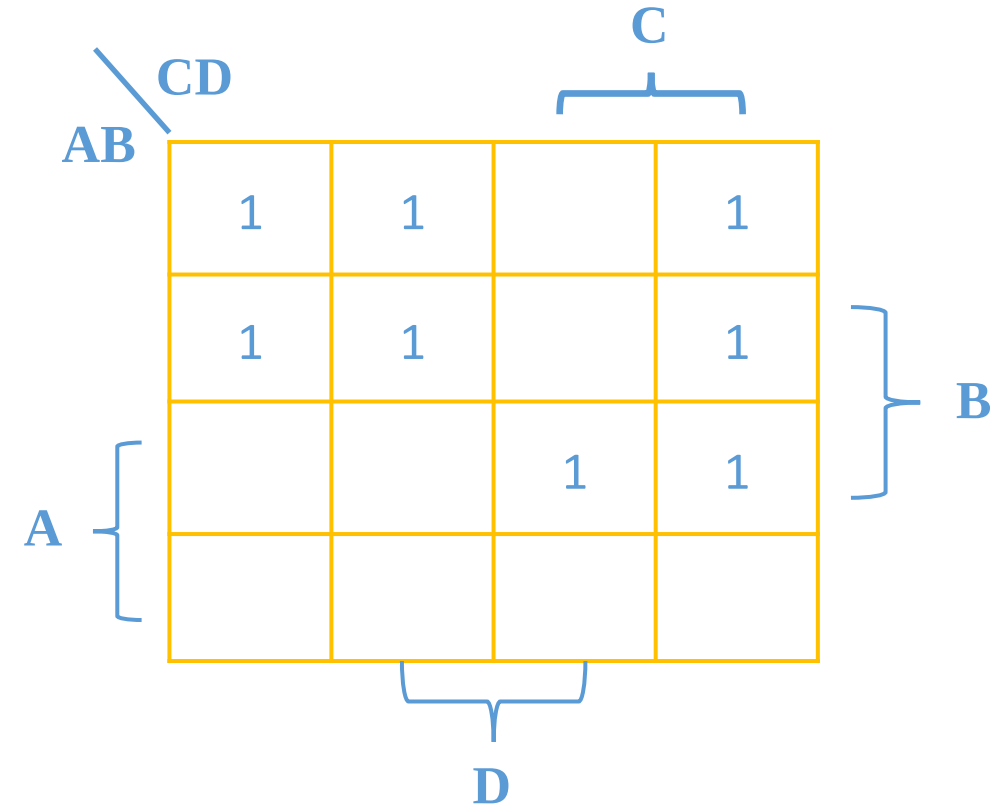
13. The simplified Boolean expression from the Karnaugh map given in the figure below is?

(a) $\bar{A}\bar{C} + \bar{A}\bar{D} + ABC$

(b) $\bar{A}\bar{B} + \bar{A}\bar{D} + ABC$

(c) $AC + ACD + ABC + BCD$

(d) $\bar{A}\bar{B} + \bar{C}\bar{D} + \bar{A}\bar{D}$



Use the Code : BVREDDY , to get Maximum Discount

14. How many min terms (excluding redundant terms) does the minimal switching function

$f(v, w, x, y, z) = x + \bar{y}z$ originally have

(a) 16

(b) 20

(c) 24

(d) 32

Use the Code : BVREDDY , to get Maximum Discount

15. For the Karnaugh map shown in the figure below, the minimal Boolean function is

(a) $x^1 + z^1 + y$

(b) $xz^1 + z + yz^1$

(c) $xy + z + y^1z$

(d) $x^1z^1 + yz$

X \ YZ	0	1
00	1	1
01	1	
11	1	1
10	1	1

Use the Code : BVREDDY , to get Maximum Discount

16. Consider the following circuits (Assume all gates to have a finite propagation delay).

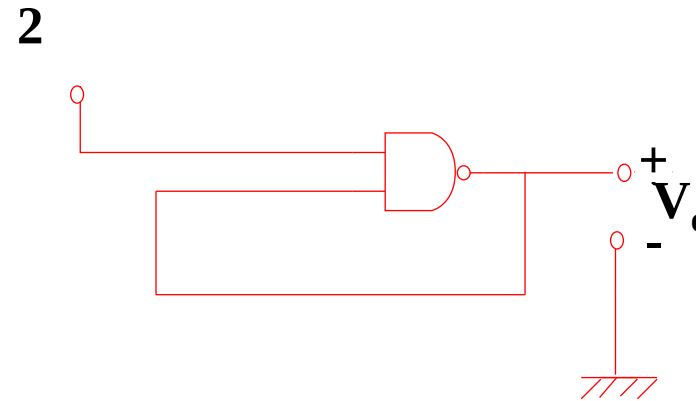
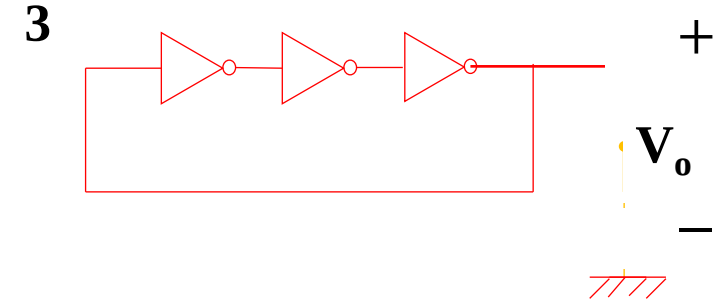
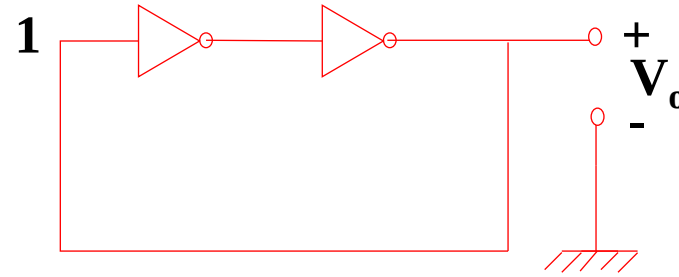
Which of these circuits generate a periodic square wave output?

(a) 1 and 2

(b) 3

(c) 2, 3

(d) 1, 2, 3



Use the Code : BVREDDY , to get Maximum Discount

17. For a function F, the Karnaugh map is shown in the figure below. Then minimal representation of F is

(a) $AB + \bar{C}$

(b) $C + \bar{A}B$

(c) $A + B + C$

(d) $A + \bar{B}C$

		BC			
		$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
A	$\bar{A}$		1	1	1
A	A		1	1	

A

Use the Code : BVREDDY , to get Maximum Discount

18. For a four variable K-Map, if each cell is assigned one integer value in range 0-15 then which is the cells adjacent cells adjacent to the cell corresponding to decimal value 7?

- (a) 3, 5, 6 and 18 (b) 3, 5, 10 and 11
(c) 3, 5, 6 and 15 (d) 4, 6, 8 and 15

Use the Code : BVREDDY , to get Maximum Discount

19. While obtaining minimal sum of products expression:

- (a) all don't-cares are ignored
- (b) all don't-cares are treated as logic ones
- (c) all don't-cares are treated as logic zeroes
- (d) only such don't-cares that aid minimization are treated as logic ones

Use the Code : BVREDDY , to get Maximum Discount

20. K-map method of simplification can be applied when the given function is in

- (a) Product of sum form
- (b) Sum of Product form
- (c) Canonical form
- (d) Any form

Use the Code : BVREDDY , to get Maximum Discount

21. Let $f(x, y, z) = \bar{x} + \bar{y}x + xz$ be a switching function. Which one of the following is valid?

- (A) $\bar{y}x$ is a prime implicant of f
- (B) xz is a min-term of f
- (C) xz is an implicant of f
- (D) y is a prime implicant of f

Use the Code : BVREDDY , to get Maximum Discount

22. The function represented by the Karnaugh map given below is:

(A) $A.B$

(B) $AB + BC + CD$

(C) $\overline{B \oplus C}$

(D) $A.BC$

A \ BC	BC			
	00	01	10	11
0	1	0	0	1
1	1	0	0	1

Use the Code : BVREDDY , to get Maximum Discount

23. Which of the following function implements the Karnaugh map shown below?

- (A) $\bar{A}B + CD$ (B) $D(C + A)$
(C) $AD + \bar{A}B$ (D) $(C + D)(\bar{C} + D) + (A + B)$

AB \ CD	CD			
	00	01	11	10
00	0	0	1	0
01	X	X	1	X
11	0	1	1	0
10	0	1	1	0

Use the Code : BVREDDY , to get Maximum Discount

24. Which functions does NOT implement the Karnaugh map given below?

(A) $(w + x) y$

(B) $xy + yw$

(C) $(w + x) (\overline{w} + y) (\overline{x} + y)$

(D) None of the above

xy \ zw	zw			
	00	01	11	10
00	0	X	0	0
01	0	X	1	1
11	1	1	1	1
10	0	X	0	0

Use the Code : BVREDDY , to get Maximum Discount

25. Given the following Karnaugh map, which one of the following represents the minimal Sum of Products of the map?

(A) $xy + y'z$

(B) $wx'y' + xy + xz$

(C) $w'x + y'z + xy$

(D) $xz + x'y$

yz \ wx	wx			
	00	01	11	10
00	0	X	0	X
01	X	1	X	1
11	0	X	1	0
10	0	1	X	0

Use the Code : BVREDDY , to get Maximum Discount

26. Minimum sum of product expression for $f(w,x,y,z)$ shown in Karnaugh-map below is

(A) $xz + y'z$

(B) $xz' + zx'$

(C) $x'y + zx'$

(D) None of the above

yz \ wx	wx			
	00	01	11	10
00	0	1	1	0
01	x	0	0	1
11	x	0	0	1
10	0	1	1	x

Use the Code : BVREDDY , to get Maximum Discount

27. The literal count of a Boolean expression is the sum of the number of times each literal appears in the expression. For example, the literal count of $(xy + xz)$ is 4. What are the minimum possible literal counts of the product of sum and sum of product representations respectively of the function given by the following Karnaugh map? Here, X denotes “don’t care”.

(A) (11, 9)

(B) (9, 13)

(C) (9, 10)

(D) (11, 11)

WZ xy	00	01	11	10
00	X	1	0	1
01	0	1	X	0
11	1	X	X	0
10	X	0	0	X

Use the Code : BVREDDY , to get Maximum Discount

28. Which are the essential prime implicants of the following Boolean function?

$$f(a, b, c) = a'c + ac' + b'c$$

(A) $a'c$ and ac'

(B) $a'c$ and $b'c$

(C) $a'c$ only

(D) ac' and bc'

Use the Code : BVREDDY , to get Maximum Discount

29. Let $f(w, x, y, z) = \Sigma(0, 4, 5, 7, 8, 9, 13, 15)$.

Which of the following expressions are NOT equivalent to f ?

P. $x'y'z' + w'xy' + wy'z + xz$

Q. $w'y'z' + wx'y' + xz$

P. R. $w'y'z' + wx'y' + xyz + xy'z$

S. $x'y'z' + wx'y' + w'y$

(A) P only

(B) Q and S

(C) R and S

(D) S only

Use the Code : BVREDDY , to get Maximum Discount

30. In the Karnaugh map shown above, X denotes a don't care term. What is the minimal form of the function represented by the Karnaugh map?

- (A) $\bar{b}.\bar{d} + \bar{a}.\bar{d}$ (B) $\bar{a}.\bar{b} + \bar{b}.\bar{d} + \bar{a}.b.\bar{d}$
(C) $\bar{b}.\bar{d} + \bar{a}.b.\bar{d}$ (D) $\bar{a}.\bar{b} + \bar{b}.\bar{d} + \bar{a}.\bar{d}$

		ab			
		00	01	11	10
cd	00	1	1		1
	01	X			
	11	X			
	10	1	1		X

Use the Code : BVREDDY , to get Maximum Discount

31. What is the minimal form of the Karnaugh map shown below? Assume that X denotes a don't care.

(A) $\overline{b}.\overline{d} + \overline{b}.\overline{c}$

(B) $\overline{a}.\overline{b} + \overline{b}.\overline{d} + \overline{a}.b.\overline{d}$

(C) $\overline{b}.\overline{d} + \overline{a}.b.\overline{d}$

(D) $\overline{a}.\overline{b} + \overline{b}.\overline{d} + \overline{a}.\overline{d}$

cd \ ab	ab			
	00	01	11	10
00	1	X	X	1
01	X			1
11				
10	1			X

Use the Code : BVREDDY , to get Maximum Discount

32. Consider the following minterm expression for F.

$$F(P, Q, R, S) = \sum 0, 5, 7, 10, 15$$

The minterms 2, 7, 8 and 13 are 'do not care' terms. The minimal sum of-products form for F is

- (A) $Q\bar{S} + \bar{Q}S$
- (B) $\bar{Q}\bar{S} + QS$
- (C) $\bar{Q}\bar{R}\bar{S} + \bar{Q}R\bar{S} + Q\bar{R}S + QRS$
- (D) $\bar{P}\bar{Q}\bar{S} + \bar{P}QS + PQS + P\bar{Q}\bar{S}$

Use the Code : BVREDDY , to get Maximum Discount

33. The total number of prime implicants of the function

$$f(w, x, y, z) = \sum(0, 2, 4, 5, 6, 10) \text{ is } \underline{\hspace{2cm}}$$

Use the Code : BVREDDY , to get Maximum Discount

34. Consider the Karnaugh map given below, where x represents “don’t care” and blank represents 0. Assume for all inputs (a, b, c, d) the respective complements ($\bar{a}$, $\bar{b}$, $\bar{c}$, $\bar{d}$) are also available. The above logic is implemented in 2-input NOR gates only. The minimum number of gates required is _____.

dc \ ba		ba			
		00	01	11	10
dc	00		x	x	
	01	1			x
	11	1			1
	10		x	x	

Use the Code : BVREDDY , to get Maximum Discount

35. Given $f(w, x, y, z) = \sum m(0, 1, 2, 3, 7, 8, 10) + \sum_d(5, 6, 11, 15)$
where d represents the don't care condition in Karnaugh maps. Which of the
following is a minimal product-of-sums (POS) form of $f(w, x, y, z)$?

(A) $f = (\bar{w} + \bar{z})(\bar{x} + z)$

(B) $f = (\bar{w} + z)(x + z)$

(C) $f = (w + z)(\bar{x} + z)$

(D) $f = (w + \bar{z})(\bar{x} + z)$

Use the Code : BVREDDY , to get Maximum Discount

36. If w, x, y, z are Boolean variables, then which one of the following is INCORRECT?

(A) $wx + w(x + y) + x(x + y) = x + wy$

(B) $\overline{w\bar{x}(y + \bar{z})} + \bar{w}x = \bar{w} + x + \bar{y}z$

(C) $(w\bar{x}(y + \bar{z}) + \bar{w}\bar{x})y = x\bar{y}$

(D) $(w + y)(wxy + wyz) = wxy + wyz$

Use the Code : BVREDDY , to get Maximum Discount

37. Consider the minterm list form of a Boolean function F given below.

$$f(w, x, y, z) = \sum m(0, 2, 5, 7, 9, 11) + \sum d(3, 8, 10, 12, 14)$$

Here, m denotes a minterm and d denotes a don't care term. The number of essential prime implicants of the function F is _____

Use the Code : BVREDDY , to get Maximum Discount

38. What is the minimum number of 2-input NOR gates required to implement a 4-variable function expressed in sum-of -minterms form as $f = \Sigma(0,1,5,7,8,10,13,15)$ Assume that all the inputs and their complements are available.

Use the Code : BVREDDY , to get Maximum Discount

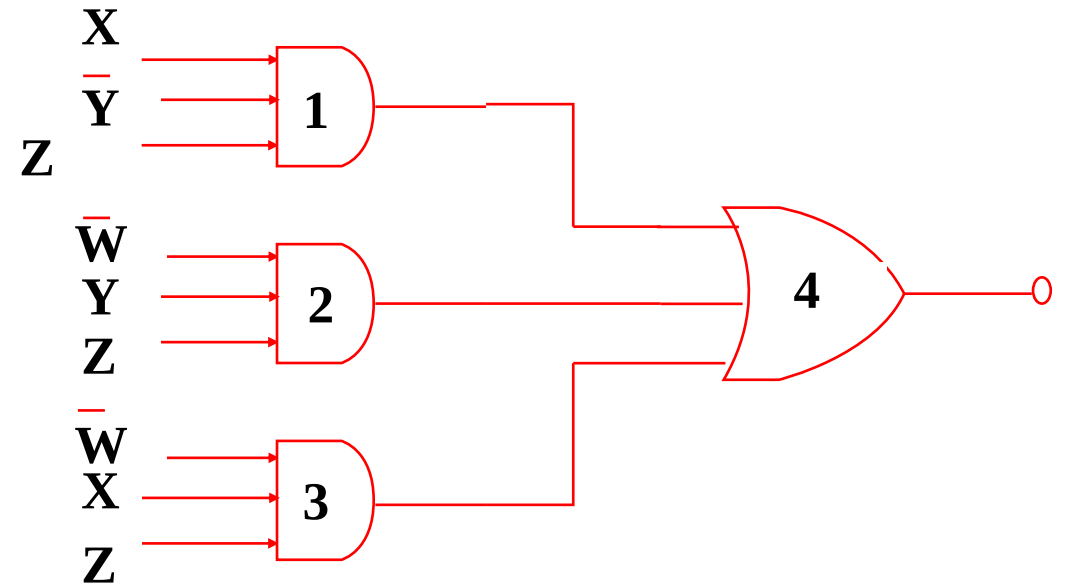
39. Which one of the following gates labeled 1, 2, 3, and 4 in the network shown in the figure is redundant?

(a) 1

(b) 2

(c) 3

(d) 4



Use the Code : BVREDDY , to get Maximum Discount

40. Consider the Boolean function,

$$F(w, x, y, z) = wy + xy + \bar{w}xyz + \bar{w}\bar{x}y + xz + \bar{x}y\bar{z}.$$

Which one of the following is the complete set of essential prime implicants?

- | | |
|--------------------------------|-----------------------------|
| (a) $w, y, xz, \bar{x}\bar{z}$ | (b) w, y, xz |
| (c) $y, \bar{x}\bar{y}\bar{z}$ | (d) $y, xz, \bar{x}\bar{z}$ |

Use the Code : BVREDDY , to get Maximum Discount

41. For an n -variable Boolean function, the maximum number of prime implicants is

(a) $2(n - 1)$

(b) $\frac{n}{2}$

(c) 2^n

(d) $2^{(n-1)}$

Use the Code : BVREDDY , to get Maximum Discount

42. In the sum of product function $f(X, Y, Z) = \sum(2,3,4,5)$ the prime implicants are

(a) $\bar{X}Y, X\bar{Y}$

(b) $\bar{X}Y, X\bar{Y}\bar{Z}, X\bar{Y}Z$

(c) $\bar{X}Y\bar{Z}, \bar{X}YZ, X\bar{Y}$

(d) $\bar{X}Y\bar{Z}, \bar{X}YZ, X\bar{Y}\bar{Z}, X\bar{Y}Z$

Use the Code : BVREDDY , to get Maximum Discount

43. The Minimized expression for the given K-map is (X: don't care)

(a) $CB+BD+CD$

(b) $AB+C\bar{B}+B\bar{C}$

(c) $C\bar{B}+AC+B\bar{C}$

(d) $\bar{C}B+BD+C\bar{B}$

AB CD				
	00	01	11	10
00		1	1	
01		1	X	
11	1	1	X	X
10	1		X	X

Use the Code : BVREDDY , to get Maximum Discount

44. The number of product terms in the minimized sum-of-product expression obtained through the following K-map is (where "d" denotes don't care states)

(a) 2

(b) 3

(c) 4

(d) 5

1	0	0	1
0	d	0	0
0	0	d	1
1	0	0	1

Use the Code : BVREDDY , to get Maximum Discount

45. Consider the Karnaugh map given below:

The function represented by this map can be simplified to the minimal form as

(a) $X_1 \bar{X}_2 \bar{X}_4 + X_2 X_4 + X_1 \bar{X}_3$

(b) $X_1 X_2 X_4 + X_2 X_4 + X_1 \bar{X}_2 \bar{X}_3 \bar{X}_4$

(c) $X_2 X_4 + \bar{X}_2 \bar{X}_4 + X_3 \bar{X}_1$

(d) $X_1 \bar{X}_2 \bar{X}_4 + \bar{X}_1 X_2 \bar{X}_3 X_4 + X_1 X_2$

		X_3X_4			
		X_1X_2	00	01	11
X_3X_4	00	1		d	D
	01		1	d	1
	11		d	1	
	10	1	d		D

46. The Minimized expression for the given K-map (x: don't-care) is.

(a) $A + \overline{B}C$

(b) $B + AC$

(c) $C + AB$

(d) ABC

CD \ AB	AB			
	00	01	11	10
00	0	0	1	1
01	0	X	X	1
11	X	X	1	X
10	1	0	1	1

Use the Code : BVREDDY , to get Maximum Discount

47. By inspecting the Karnaugh map plot of the switching function $F(X_1 X_2 X_3) = \sum (1,3,6,7)$ can say that the redundant prime implicant is

(a) $\bar{X}_1 X_3$

(b) $X_2 X_3$

(c) $X_1 X_2$

(d) X_3

Use the Code : BVREDDY , to get Maximum Discount

48. What is the minimized logic expression corresponding to the given Karnaugh-map?

(a) XZ

(b) $\overline{W} X \overline{Y} + \overline{W} Y Z + W \overline{Y} Z + W X Y$

(c) $\overline{W} X \overline{Y} + \overline{W} Y Z + W \overline{Y} \overline{Z} + W X \overline{Y}$

(d) $XZ + \overline{W} Y Z + \overline{W} X \overline{Y} + W X Y + W \overline{Y} Z$

YZ WX	00	01	11	10
00			1	
01	1	1	1	
11		1	1	1
10		1		

Use the Code : BVREDDY , to get Maximum Discount

49. When the Boolean function $F(X_1 X_2 X_3) = \sum (0,1,2,3) + \sum \emptyset (4,5,6,7)$ is minimised, what does one get?

(a) 1

(b) 0

(c) X_1

(d) X_3

Use the Code : BVREDDY , to get Maximum Discount

50. Consider the following statements:

- 1.Minimization using Karnaugh map may not provide unique solution.
- 2.Redundant grouping in Karnaugh map may result in non-Minimized solution.
- 3.Don't care states if used in Karnaugh map for Minimization, the minimal solution is not obtained.

Which of the statements given above are correct?

- | | |
|------------------|------------------|
| (a) 1, 2 and 3 | (b) 2 and 3 only |
| (c) 1 and 3 only | (d) 1 and 2 only |

Use the Code : BVREDDY , to get Maximum Discount

51. Digital input signals A, B, C with A as the MSB and C as the LSB are used to realize the Boolean function $F = m_0 + m_2 + m_3 + m_5 + m_7$, where m_i denotes the i th minterm. In addition, F has a don't care for m_1 . The simplified expression for F is given by

(a) $\bar{A}\bar{C} + \bar{B}C + AC$

(b) $\bar{A} + C$

(c) $\bar{C} + A$

(d) $\bar{A}C + BC + A\bar{C}$

Use the Code : BVREDDY , to get Maximum Discount